

CAS Machine Learning – Time Series

Forecasting with ARIMA

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Informatik



Outline

- Stationarity
- Differencing
- Autoregressive model
- Moving average model
- ARMA, ARIMA, SARIMA models
- Forecasting
- Evaluation workflow
- Confidence intervals
- Residual analysis
- Information criteria
- Performance metrics

Stationarity

Strict stationarity: the joint distribution of any collection of the TS RVs is identical to that of the shifted RVs:

$$P[X_{t_1} = x_{t_1}, \dots, X_{t_k} = x_{t_k}] = P[X_{t_1+h} = x_{t_1+h}, \dots, X_{t_k+h} = x_{t_k+h}] \quad \forall k \geq 1, \forall t_1, \dots, t_k, \forall h \geq 0$$

(Weak) stationarity: mean and auto-covariance are independent of time (γ only depends on the **shift** $h = |s - t|$):

$$\gamma(t+h, t) = \text{cov}(X_{t+h}, X_t) = \text{cov}(X_h, X_0) = \gamma(h, 0) = \gamma(h) \quad \forall h, t$$

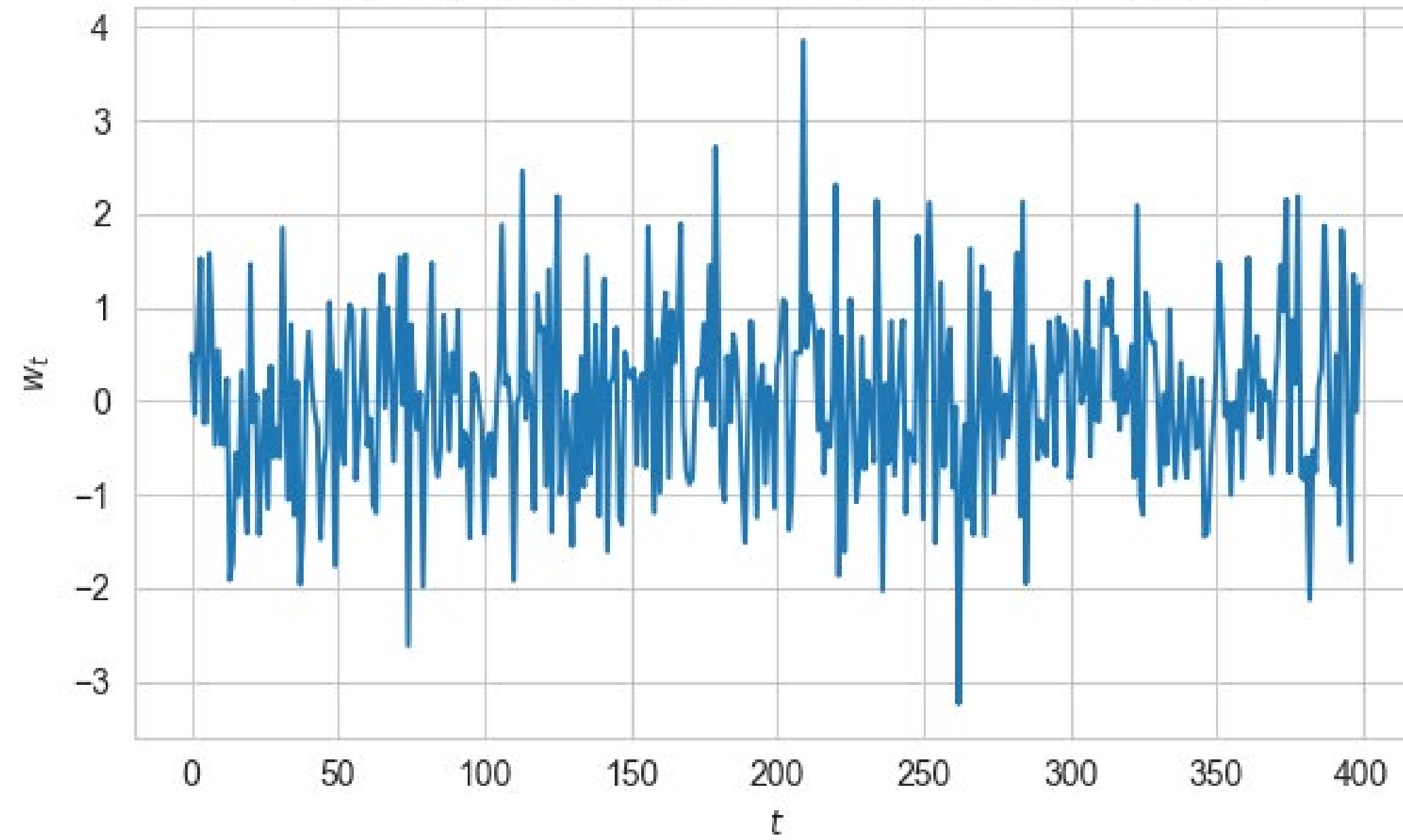
The **auto-correlation** only depends on the **shift** since $\rho(t+h, t) = \frac{\gamma(t+h, t)}{\sqrt{\text{Var}(X_{t+h})\text{Var}(X_t)}} = \frac{\gamma(h, 0)}{\sqrt{\text{Var}(X_h)\text{Var}(X_0)}} = \frac{\gamma(h)}{\gamma(0)} = \rho(h)$

A **white noise process is stationary** since $W_t \sim WN(0, \sigma^2)$ and $\gamma_{W_t}(s, t) = 0 \quad \forall s \neq t$.

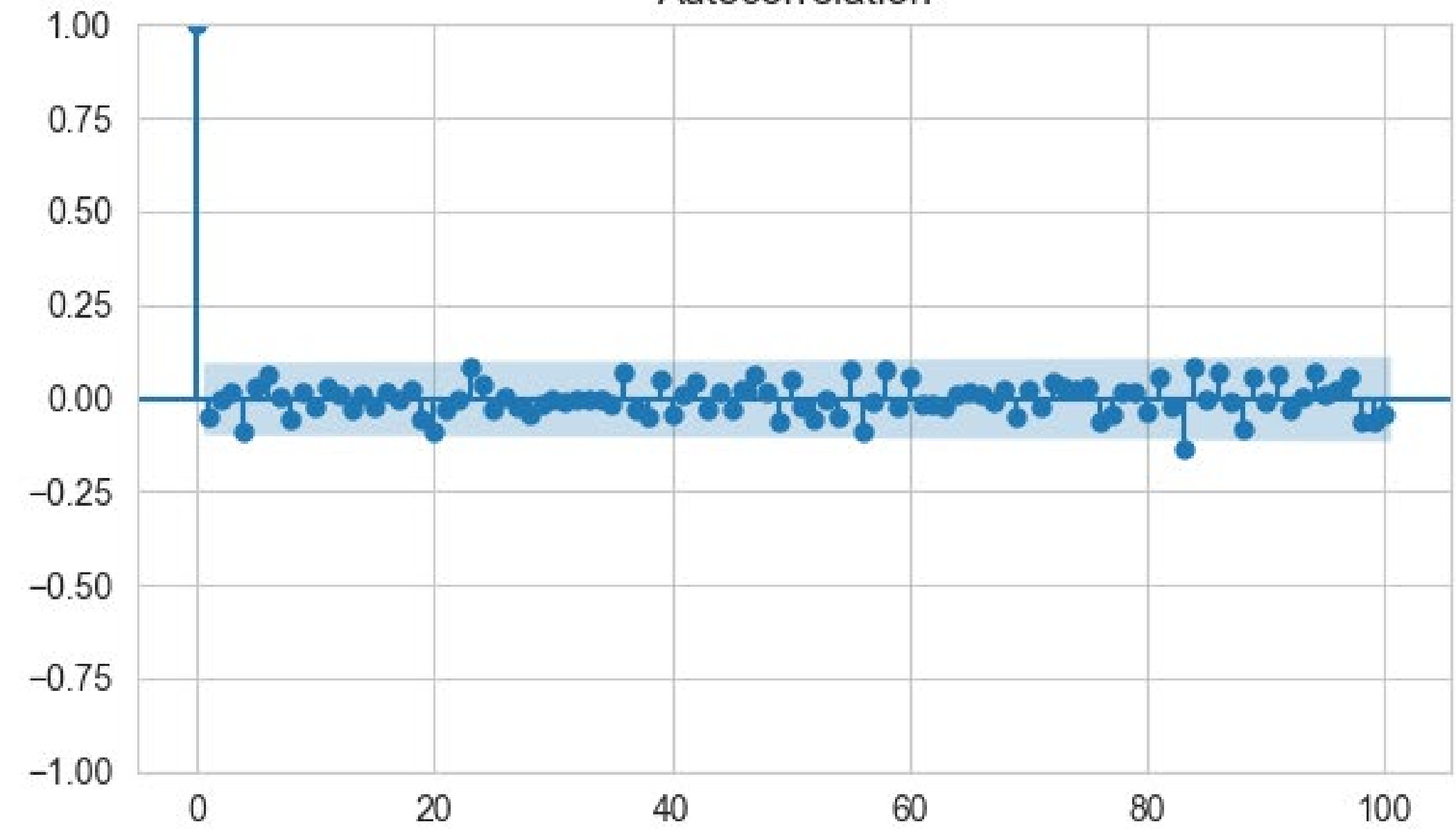
Strict stationarity implies stationarity but the reverse is **not** true in general.

One exception is **Gaussian processes** where **every** set of RVs follows a multivariate Gaussian distribution.

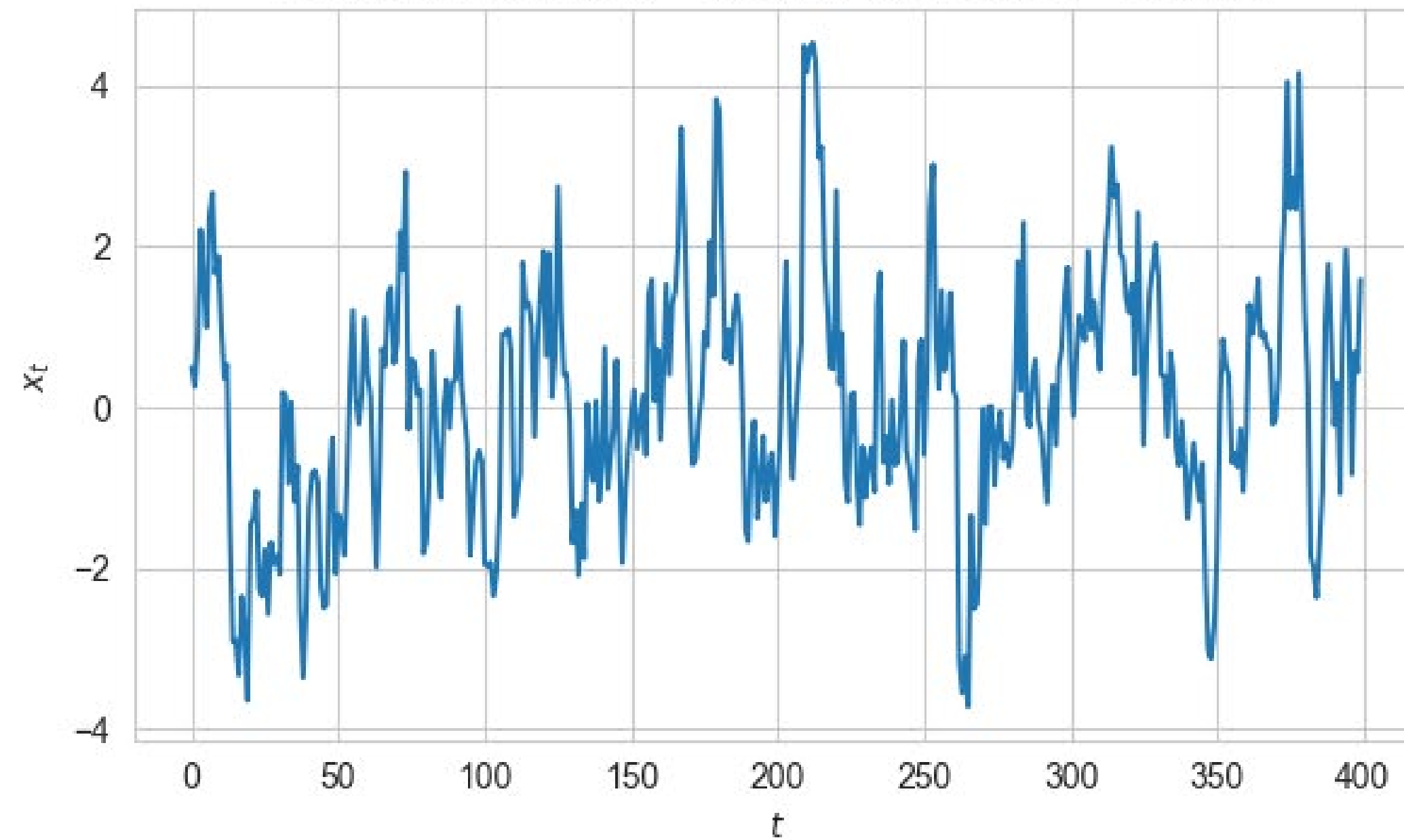
Strict Stationarity: Gaussian White Noise $W_t \sim \mathcal{N}(0, 1)$



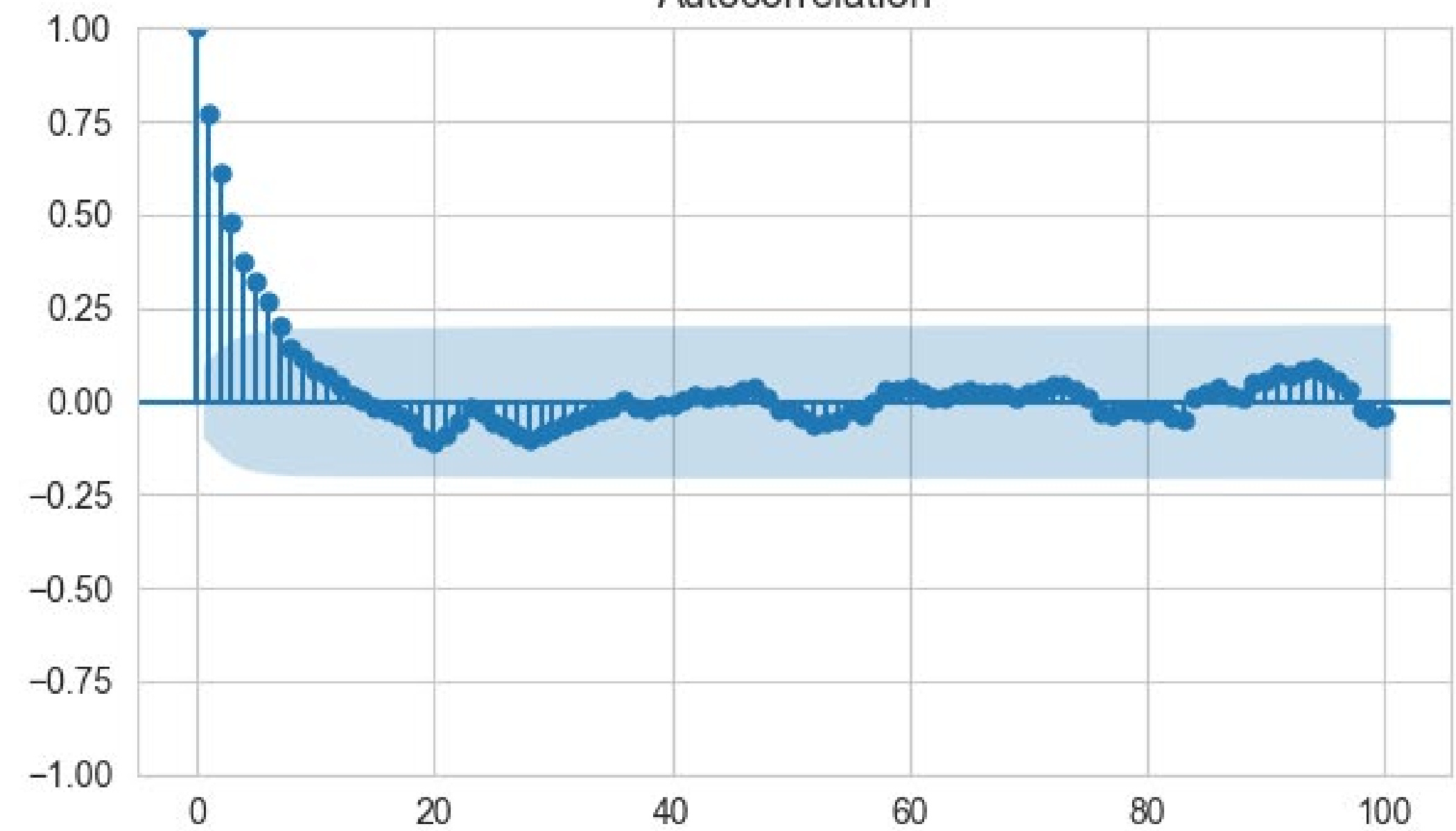
Autocorrelation



Weak Stationarity: $X_t = 0.8X_{t-1} + W_t$ with $W_t \sim \mathcal{N}(0, 1)$



Autocorrelation



Estimating the statistical properties of a time series

Typically, only a **single realization** of a TS is available. With non-stationary TS,

- Actual mean, variance and covariance (statistical properties) may **vary over time**.
- **Cannot** rely on past values to **estimate future** relationships.

With a stationary TS, its **statistical properties are constant over time** and can be estimated from past values.

Sample mean	Sample variance	Sample auto-covariance	Sample auto-correlation
$\hat{\mu} = \frac{1}{n} \sum_{i=0}^n x_i$	$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=0}^n (x_i - \hat{\mu})^2$	$\hat{\gamma}(h) = \frac{1}{n} \sum_{i=0}^{n- h } (x_{i+ h } - \hat{\mu})(x_i - \hat{\mu})$	$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}$

TS analysis typically starts with removing trends, seasonality, and heteroskedasticity to get **stationary residuals**, which can then be modeled.

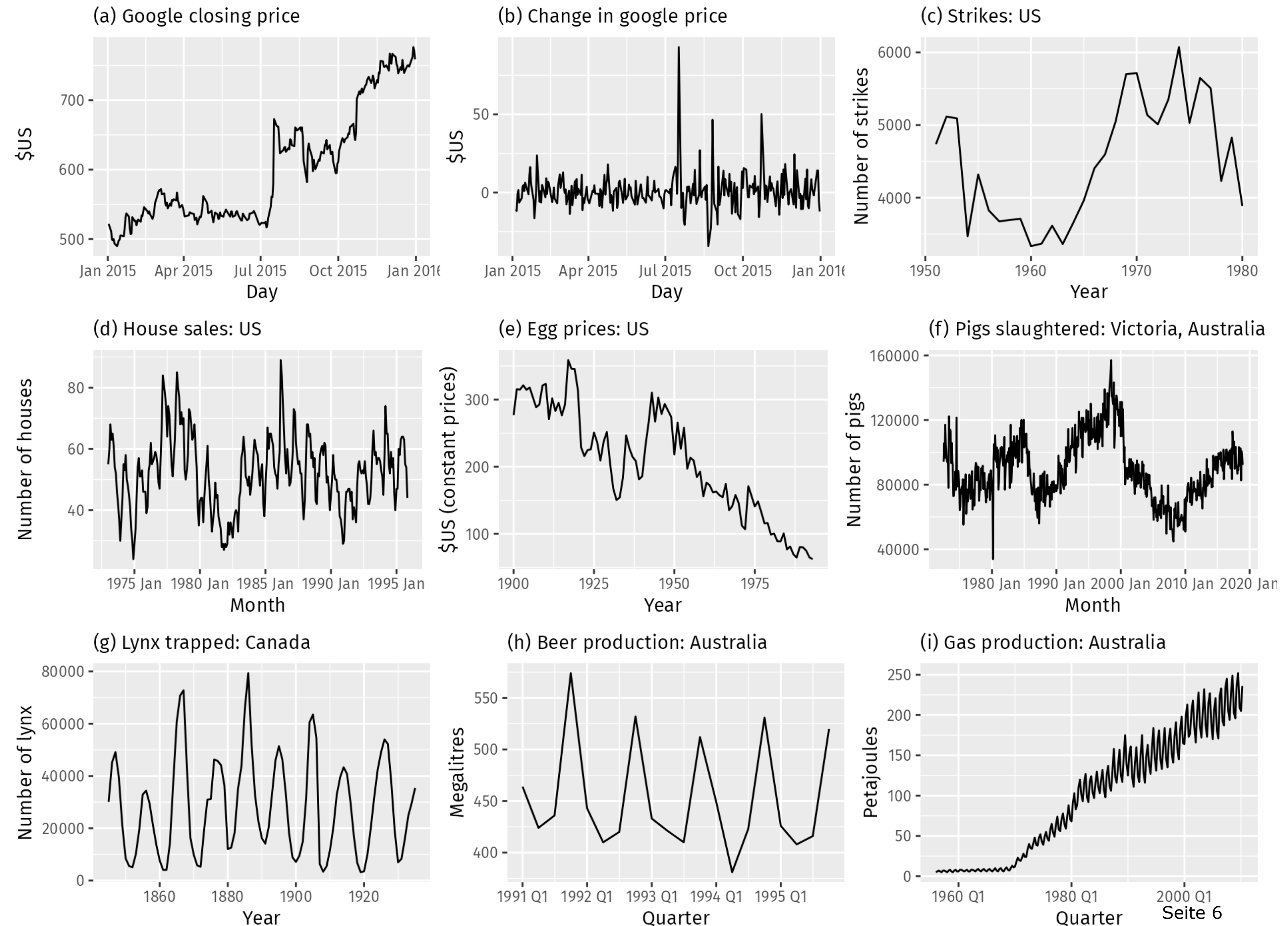
Checking stationarity – Visual inspection of time plots

Which of these TS are stationary?

Non-stationary patterns

- Trends
- Seasonality with trend
- Cycles with trend
- Changing variance
- Apparent breaks in the data

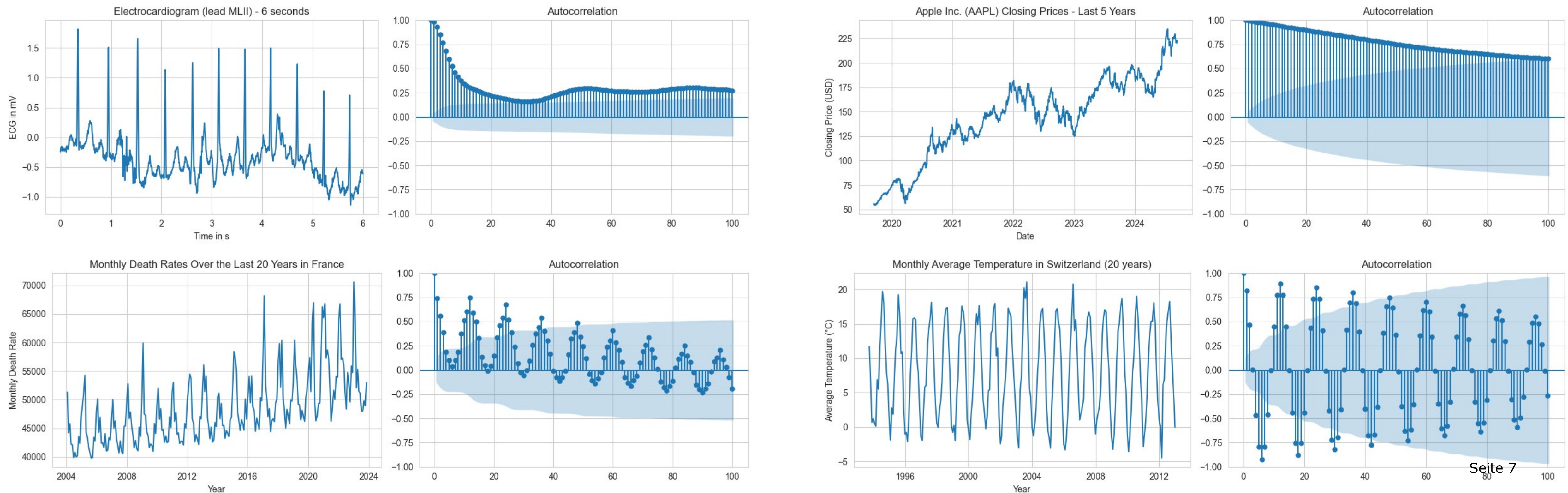
Note that **cyclic patterns may be stationary** if they are **aperiodic**.



Checking stationarity – Visual inspection of correlograms

Correlogram: in stationary TS, the auto-correlations typically **drop off quickly** as the lag increases. If the coefficients drop quickly but then show a pattern, it typically indicates **weak stationarity**.

ACF estimates are biased: shrunken towards zero for higher lags → consider only the **first $\sim n/4$ lags**.



Checking stationarity – Statistical tests

Null hypothesis (H_0): TS has a **unit root** e.g., random walk $x_t = 1 \cdot x_{t-1} + w_t$

- TS value is strongly dependent on its previous value => time-dependent mean/variance, slow-decaying $\rho(h)$.
- Augmented Dickey-Fuller (ADF) test, Phillips-Perron (PP) test

H_0 : TS is **trend stationary** (removing trend leaves a stationary TS)

- Detect whether TS is mean-reverting, has constant variance and auto-correlation over time.
- Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test.

Tests for structural breaks: Zivot-Andrews (ZA) test, Chow test, Bai-Perron test.

Statistical tests focus on specific stationarity aspects, but do **not** test stationarity in a broad sense.

They struggle to detect general forms of non-stationarity → **couple with visual analysis.**

Achieving stationarity

To make a TS stationary, apply either of the following steps.

1. **Non-linear transformations** to stabilize variance (if necessary)
 - Log transform
 - Power transforms
 - Box-Cox transform
2. **Decomposition** to isolate trend and seasonal components (if necessary)
 - STL decomposition
3. **Differencing**
 - On the decomposition remainder if the residuals are non-stationary.
 - On the TS directly when decomposition is either difficult (e.g., weak patterns) or not needed.

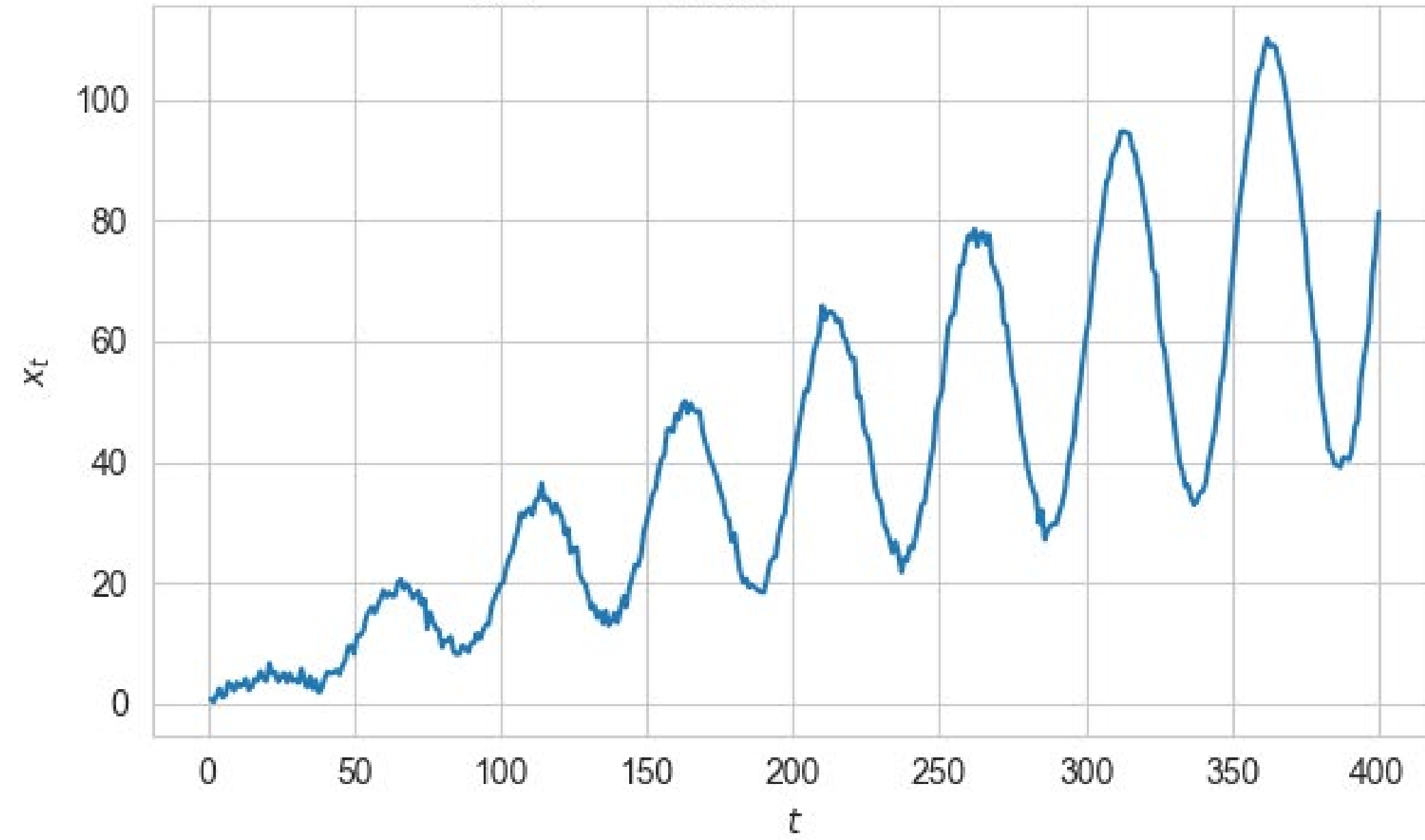
Non-linear transformations

Goal: **normalize** the data and **stabilize** variance.

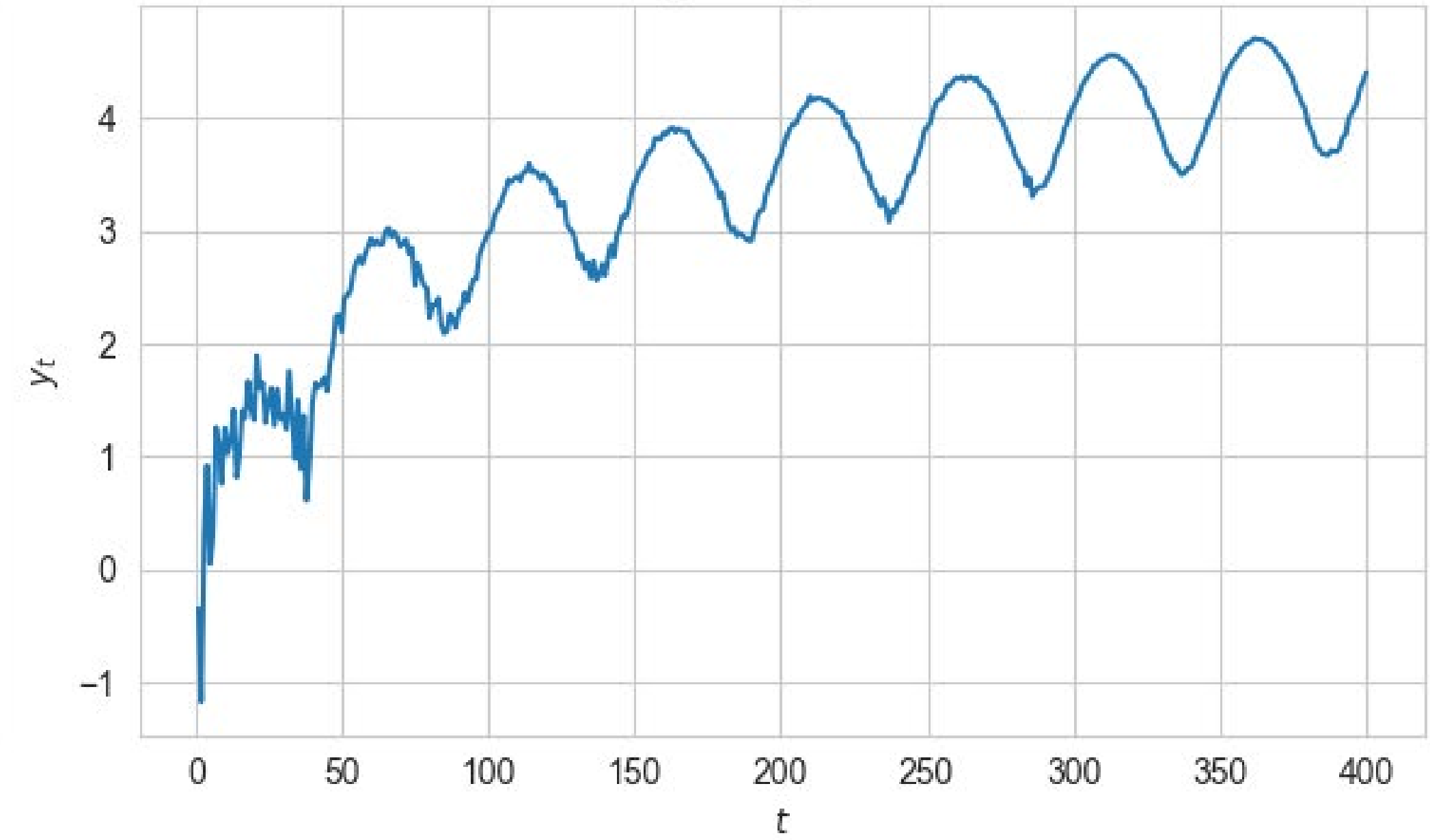
Log transform	Power transform	Box-Cox transform
$y_t = \log(x_t)$	$y_t = x_t^\lambda$	$y_t = \begin{cases} \log(x_t) & \text{if } \lambda = 0 \\ \frac{(\text{sign}(x_t) x_t ^\lambda - 1)}{\lambda} & \text{if } \lambda \neq 0 \end{cases}$
• Interpretable • Use on skewed data with long right tail • Converts multiplicative relationships into additive	• Lack interpretability • Typically, $\lambda = \frac{1}{2}$ or $\frac{1}{3}$ • Use on count data (Poisson distribution) • Reduce impact of large values	• Flexible transformation • Determine optimal λ for each TS

Models and statistical tests typically **assume normally distributed residuals**, which are more likely when the data is normally distributed.

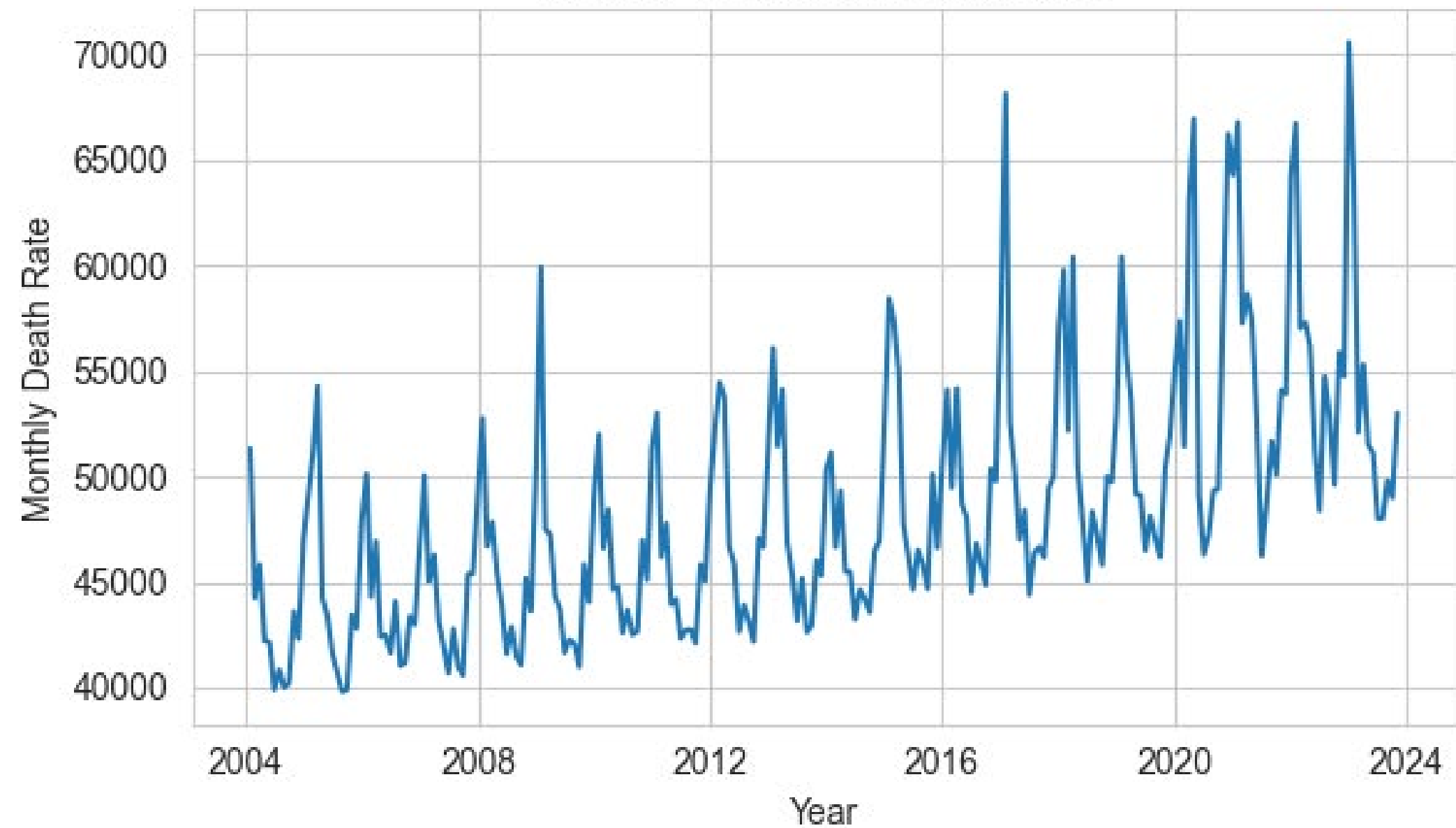
$$x_t = \frac{t}{10} \left(2 + \sin\left(\frac{2\pi t}{50}\right) \right) + w_t \text{ with } w_t \sim \mathcal{N}(0, 1)$$



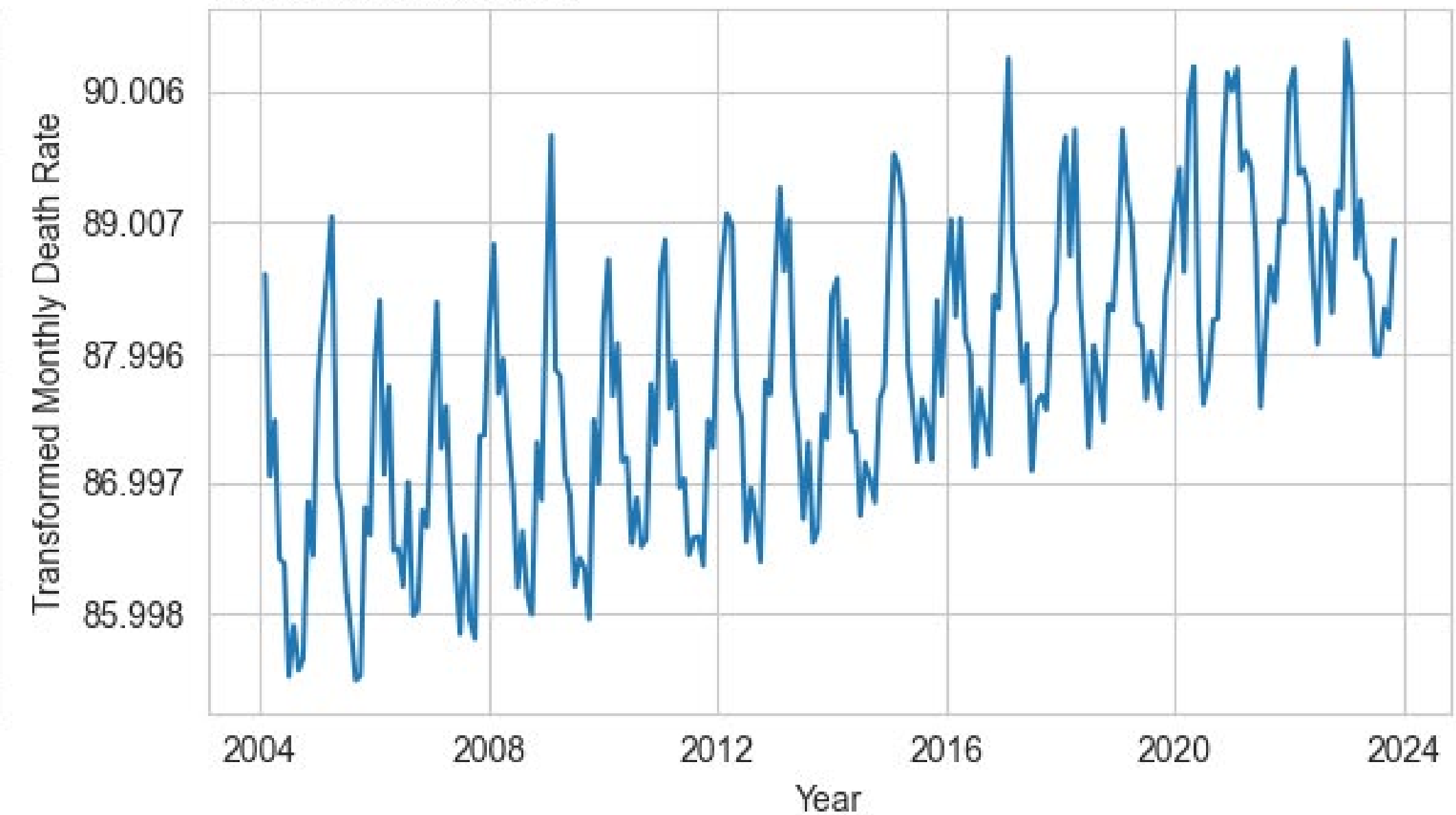
$$y_t = \log(x_t)$$



Monthly Death Rates in France



Box-Cox-transformed ($\lambda = -2.78$) Monthly Death Rates in France
 $1e-14 + 3.603226817e-1$



Recap – Decomposition

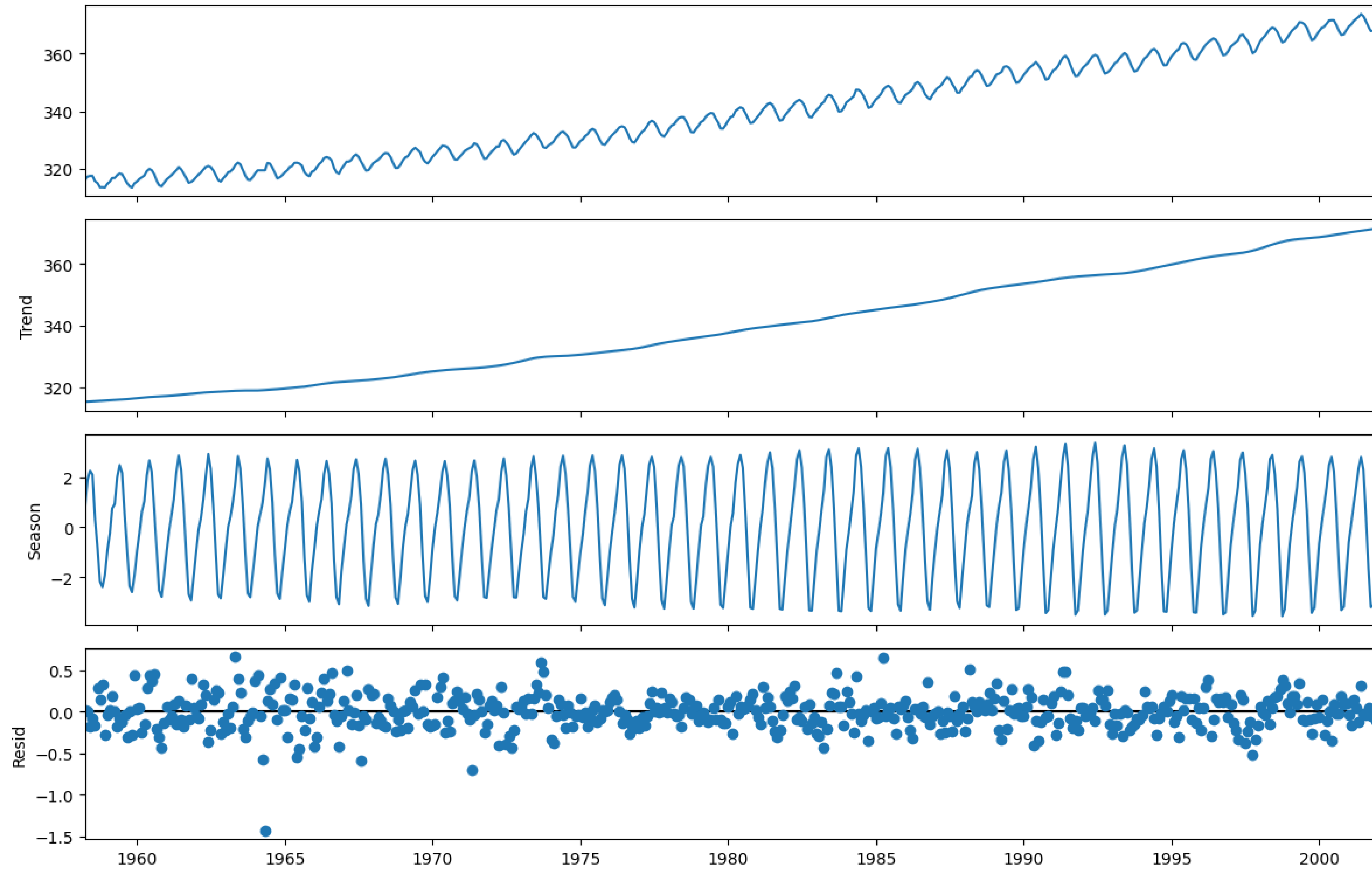
Separate TS into three components: $X_t = U_t + S_t + R_t$

- Trend/trend-cycle U_t : increasing/decreasing pattern in the data.
- Seasonality S_t : repeating pattern with roughly fixed period.
- Remainder R_t : everything remaining (**residuals**).

Given a time series realization $\{x_1, x_2, \dots, x_n\}$:

1. Estimate trend component $\hat{u}_t$
 - LOESS – locally estimated scatterplot smoothing (STL)
2. Detrend time series $\hat{d}_t = x_t - \hat{u}_t$
3. Estimate seasonality component $\hat{s}_t$
 - LOESS – locally estimated scatterplot smoothing (STL)
4. Deseasonalize TS to estimate the remainder $\hat{r}_t = \hat{d}_t - \hat{s}_t$

STL decomposition: Mauna Loa Monthly Atmospheric CO2 Data



Differencing

Studying a TS value **changes** rather than its values: $\nabla X_t = X_t - X_{t-1}$ (**first-order differencing**)

∇_h is called the **lag-h difference operator**: $\nabla_h X_t = X_t - X_{t-h}$ with ∇_1 denoted as ∇ .

Differencing does **not** obtain explicit estimate of trend, season and remainder.

Differencing can **remove the trend** component: consider a TS with linear trend $X_t = \alpha + \beta t + W_t$

Then $\nabla X_t = \alpha + \beta t + W_t - (\alpha + \beta(t-1) + W_{t-1}) = \beta + W_t - W_{t-1}$

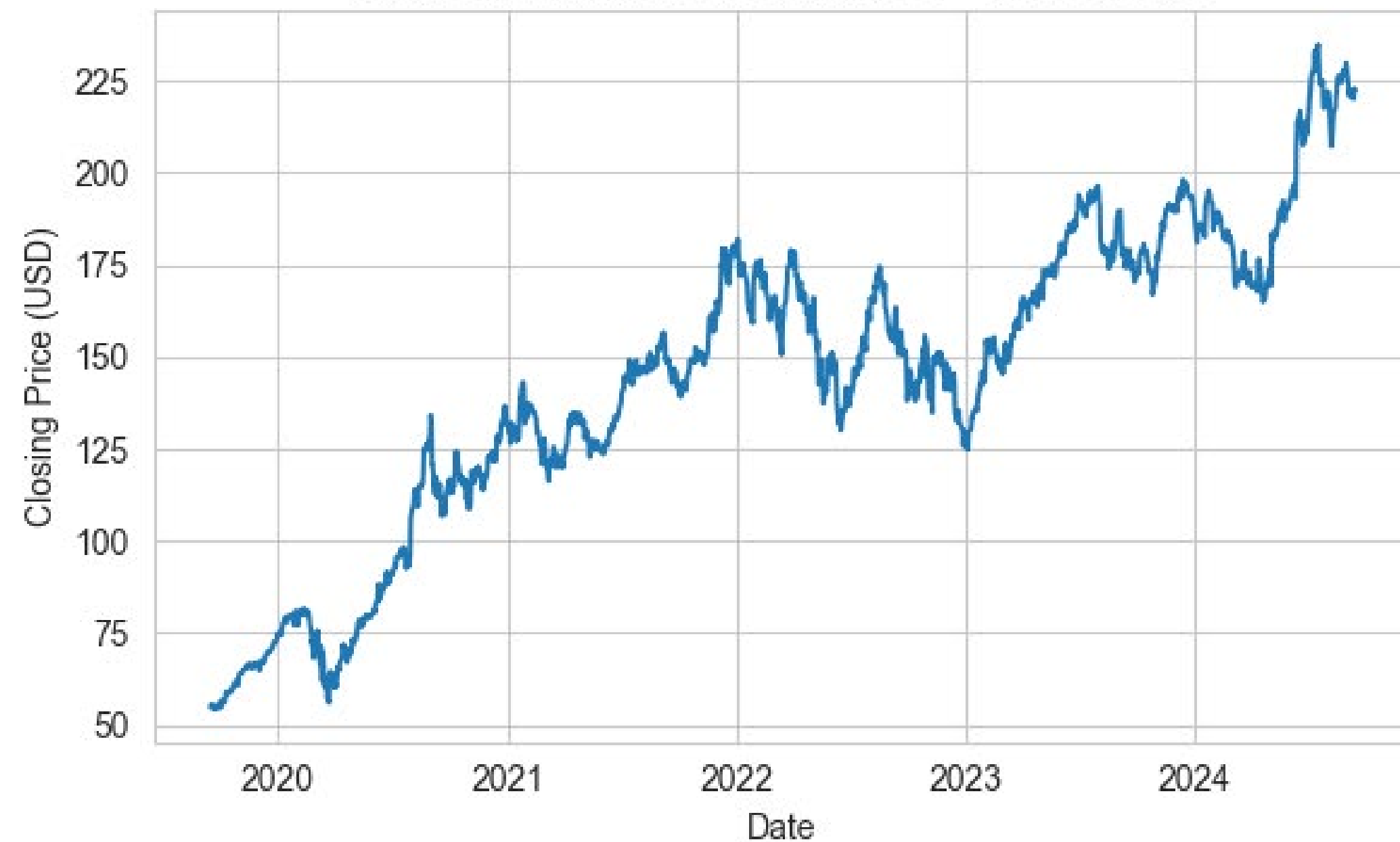
Differencing can remove the seasonal component: consider $X_t = A \sin\left(\frac{2\pi t}{P}\right) + W_t$

Then $\nabla_P X_t = X_t - X_{t-P} = A \sin\left(\frac{2\pi t}{P}\right) + W_t - \left(A \sin\left(\frac{2\pi(t-P)}{P}\right) + W_{t-P}\right) = W_t - W_{t-P}$ (**seasonal differencing**)

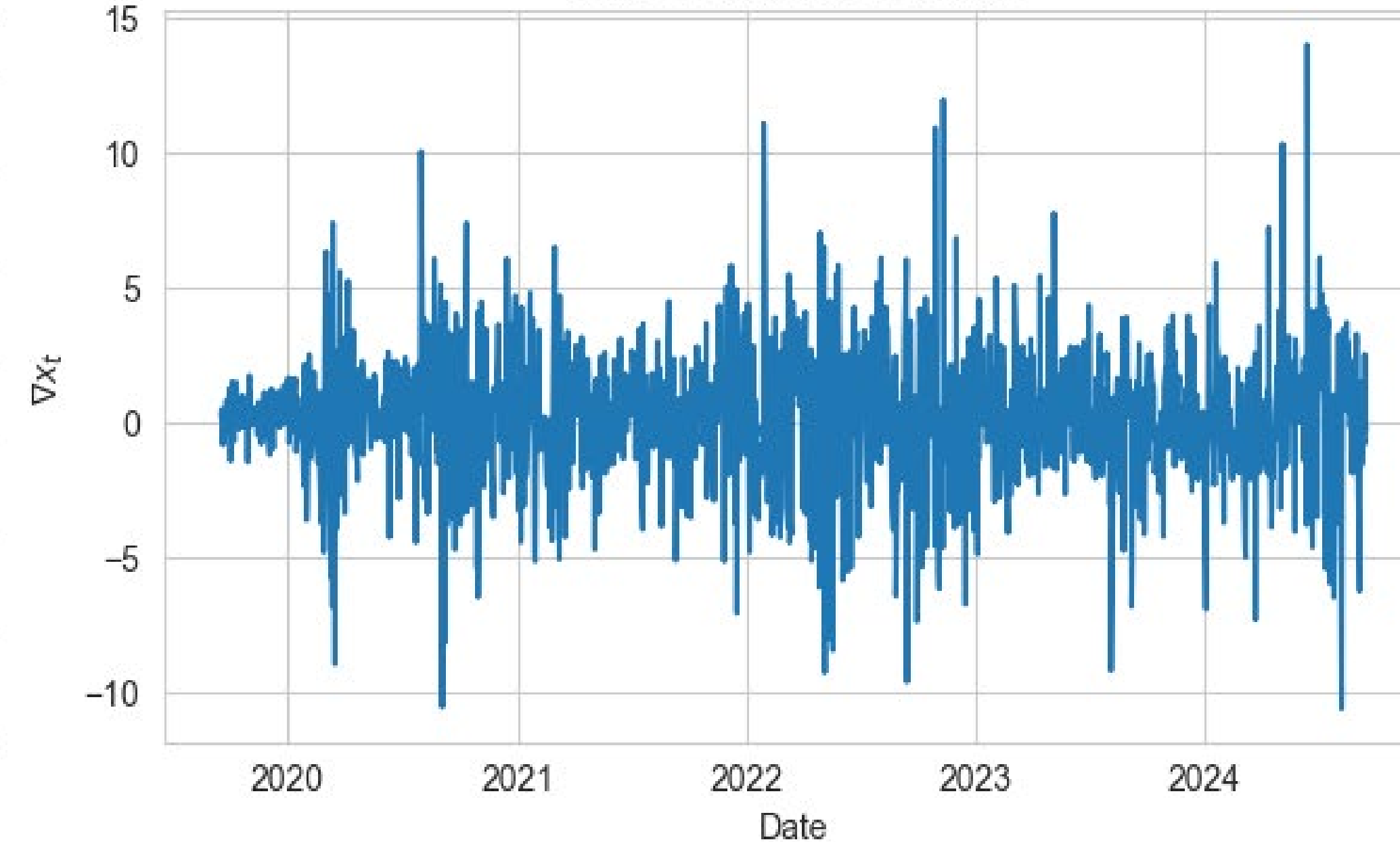
Differencing introduces **artificial dependencies**: $\text{cov}(\nabla X_t, \nabla X_{t-1}) = \text{cov}(W_t - W_{t-1}, W_{t-1} - W_{t-2}) = -\text{cov}(W_{t-1}, W_{t-1}) \neq 0$

→ Keep differences **interpretable**: first-order, seasonal differencing.

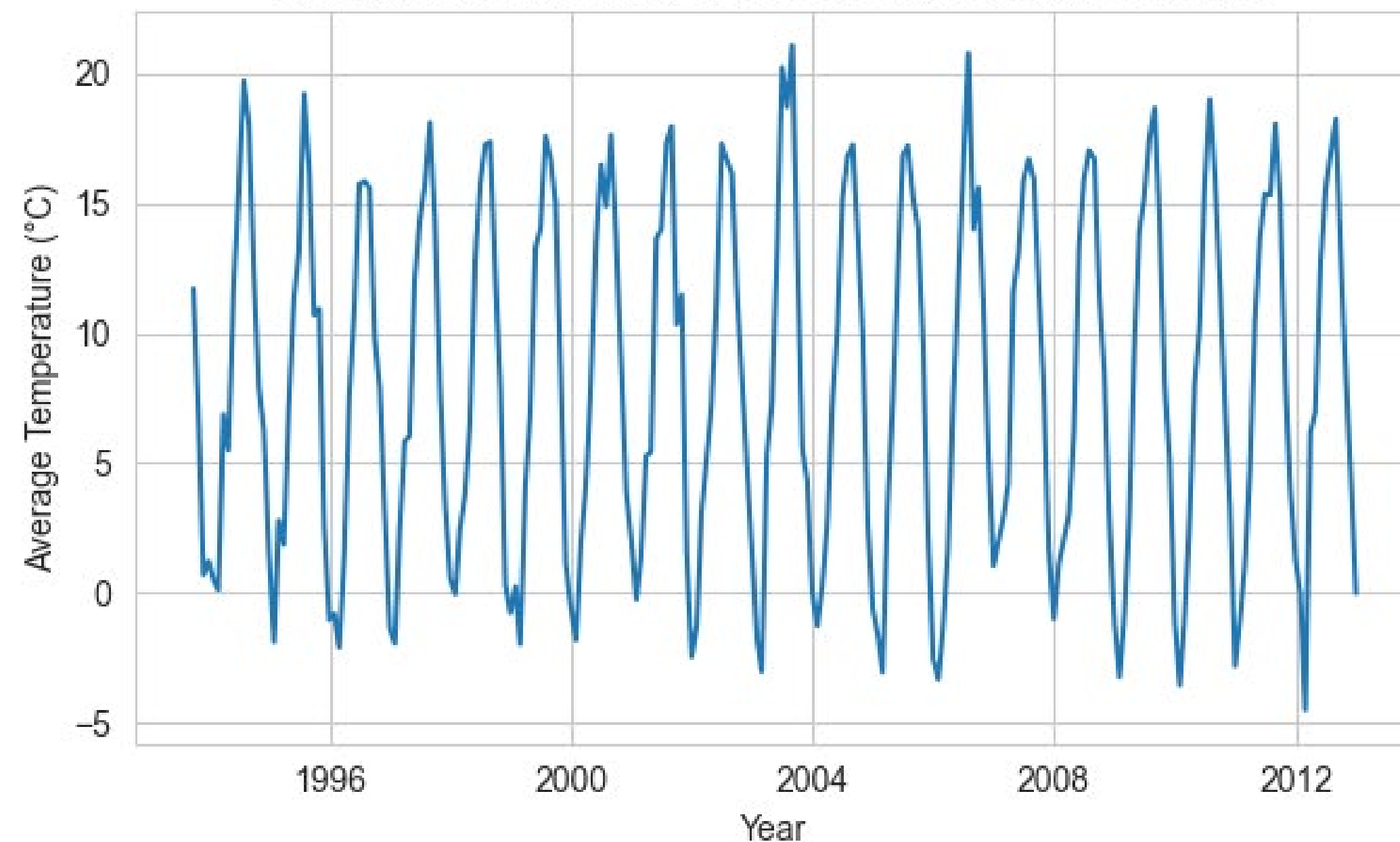
Apple Inc. (AAPL) Closing Prices - Last 5 Years



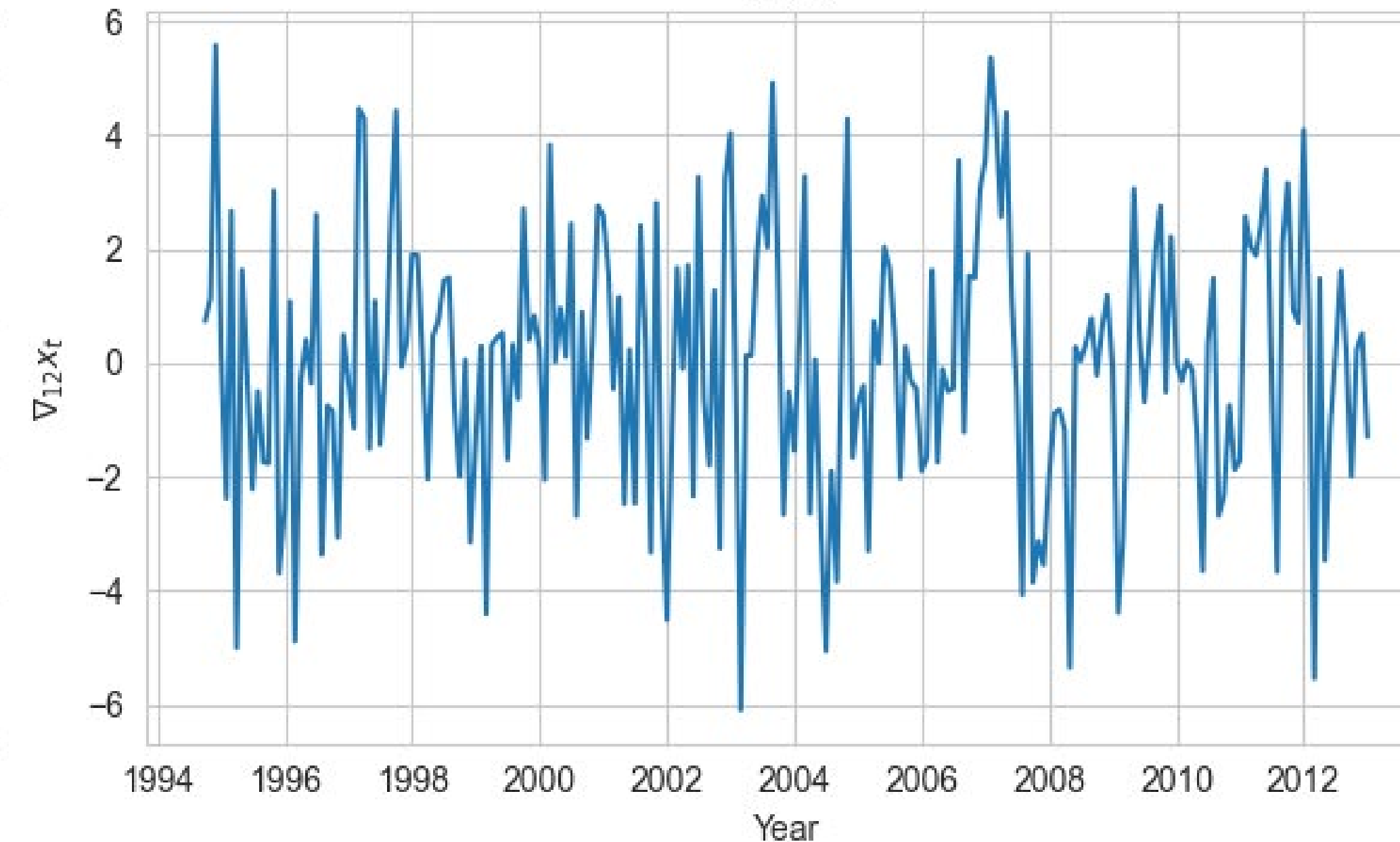
∇X_t : Closing price change



Monthly Average Temperature in Switzerland (20 years)



$\nabla_{12} X_t$



Higher-order differencing

Iterative differencing: $\nabla^2 X_t = \nabla \nabla X_t = \nabla(X_t - X_{t-1}) = X_t - X_{t-1} - (X_{t-1} - X_{t-2}) = X_t - 2X_{t-1} + X_{t-2}$

B is called the **Backshift operator**: $B^h X_t = X_{t-h}$ with B^1 denoted as B and $BX_t = X_{t-1}$

First-order differencing: $\nabla X_t = X_t - X_{t-1} = X_t - BX_t = (1 - B)X_t$

Second-order differencing: $\nabla^2 X_t = \nabla \nabla X_t = \nabla(1 - B)X_t = (1 - B)^2 X_t$

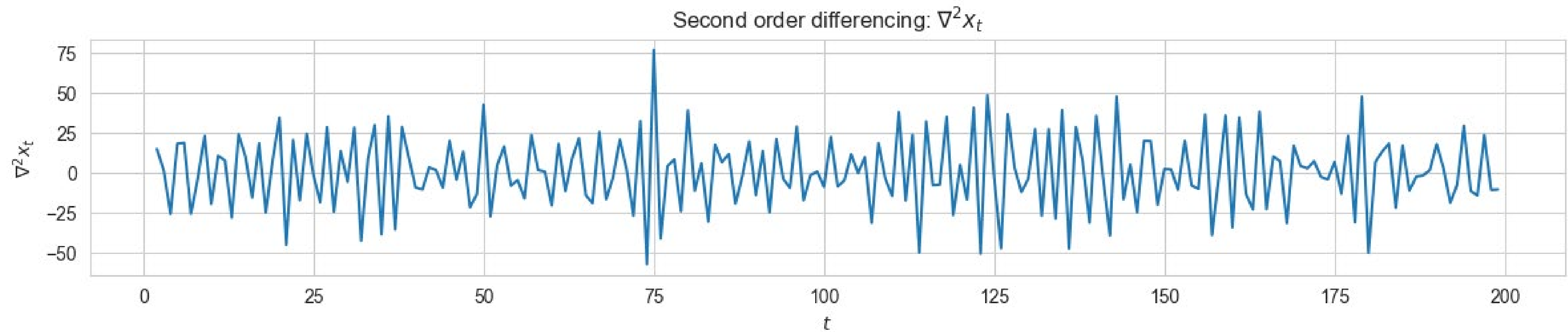
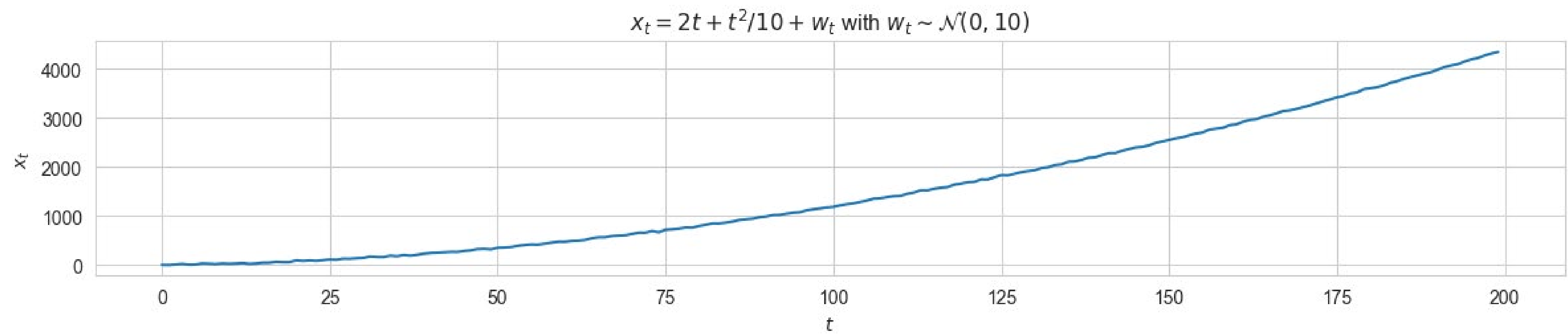
k^{th} -order differencing: $\nabla^k X_t = \nabla^{k-1} \nabla X_t = (1 - B)^k X_t$

Note that $\nabla_h^k = (1 - B^h)^k$

Start with seasonal differencing, continue with differencing if some trend remain e.g., $\nabla^k \nabla_P X_t$

When trend is non-linear, apply non-linear transforms **before** any differencing.

Use higher-order differencing to remove **polynomial trend** e.g., second-order for quadratic trend.



Exercise

Extend previous exercises with stationarity analysis: are the time series **stationary**?

- Visual inspection
- Statistical tests

Make the time series stationary

- Experiment with **non-linear transformations**.
- Experiment with **decomposition**.
- Experiment with **differencing**.

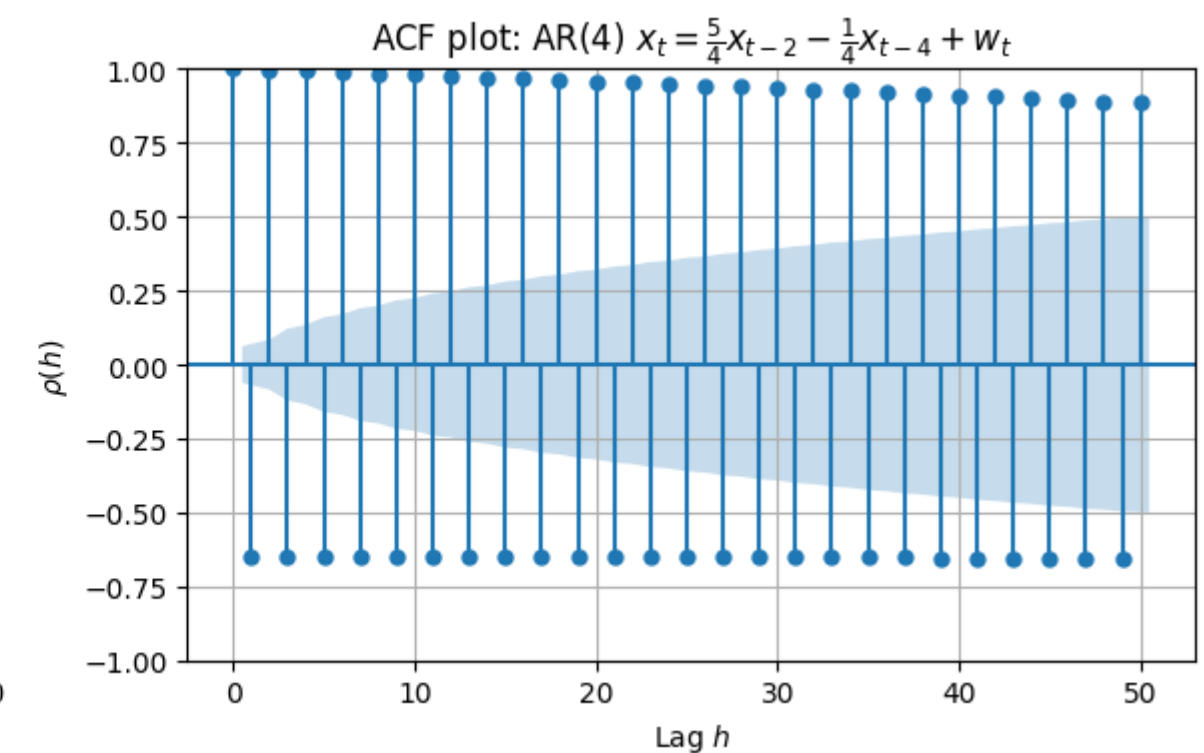
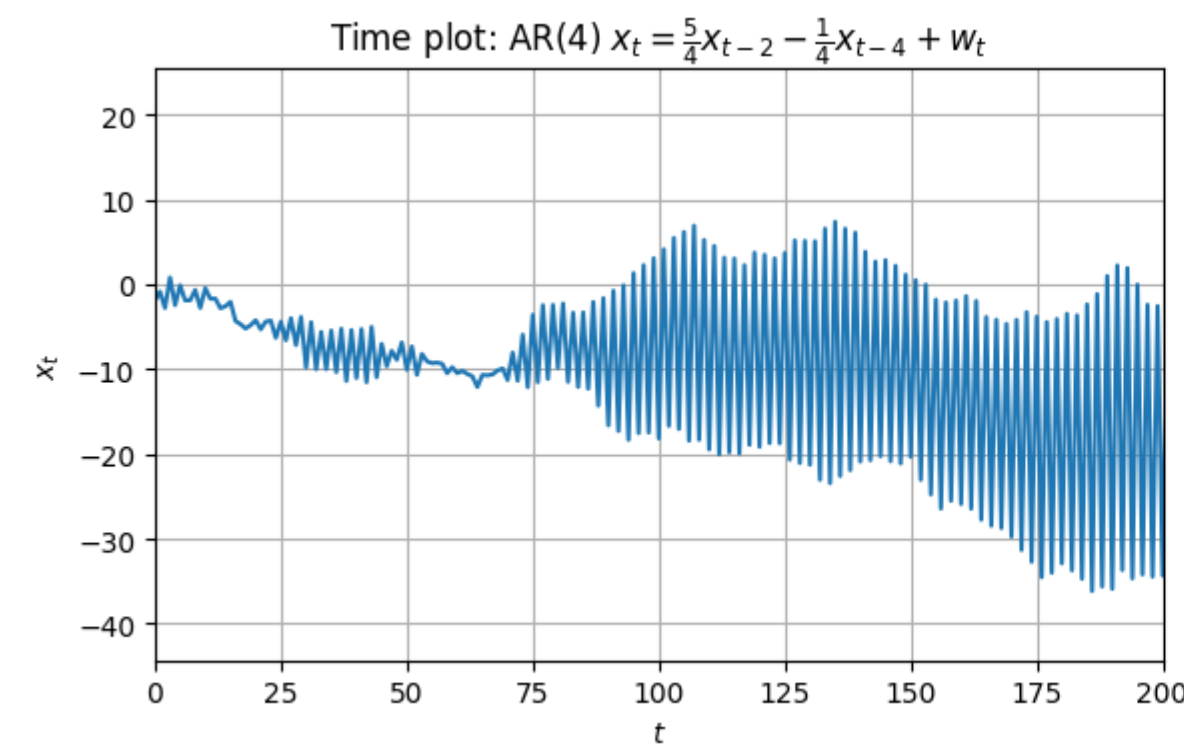
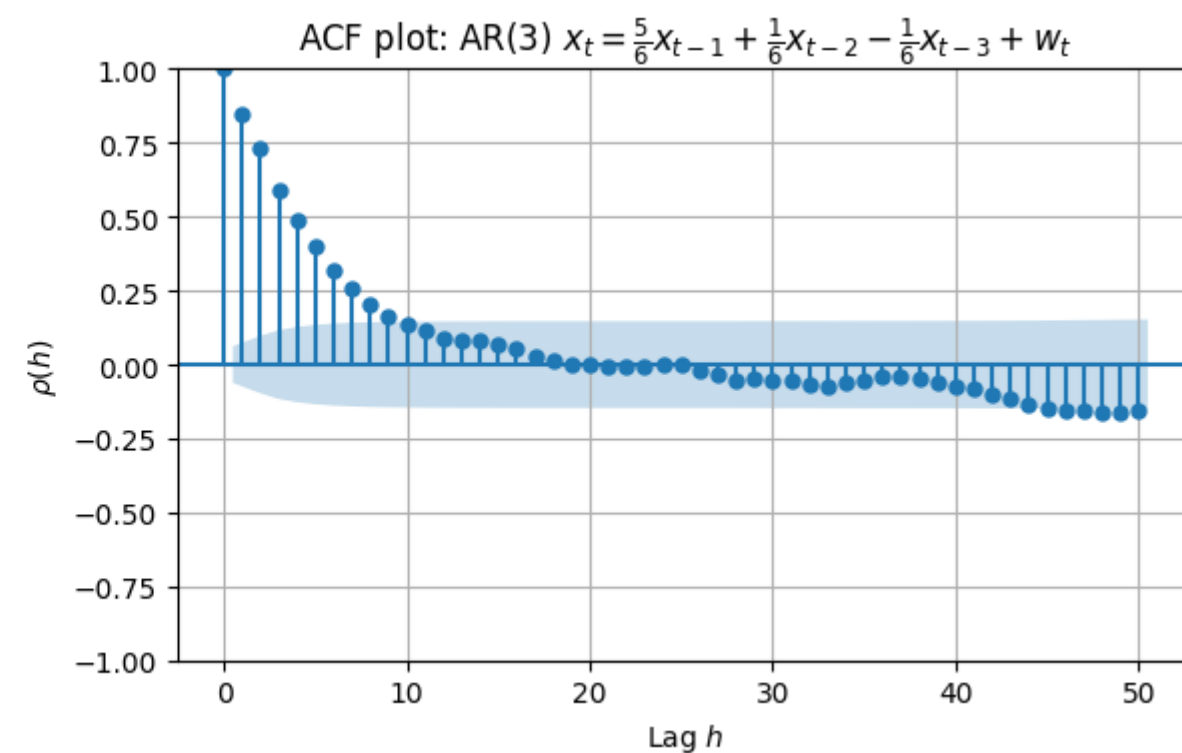
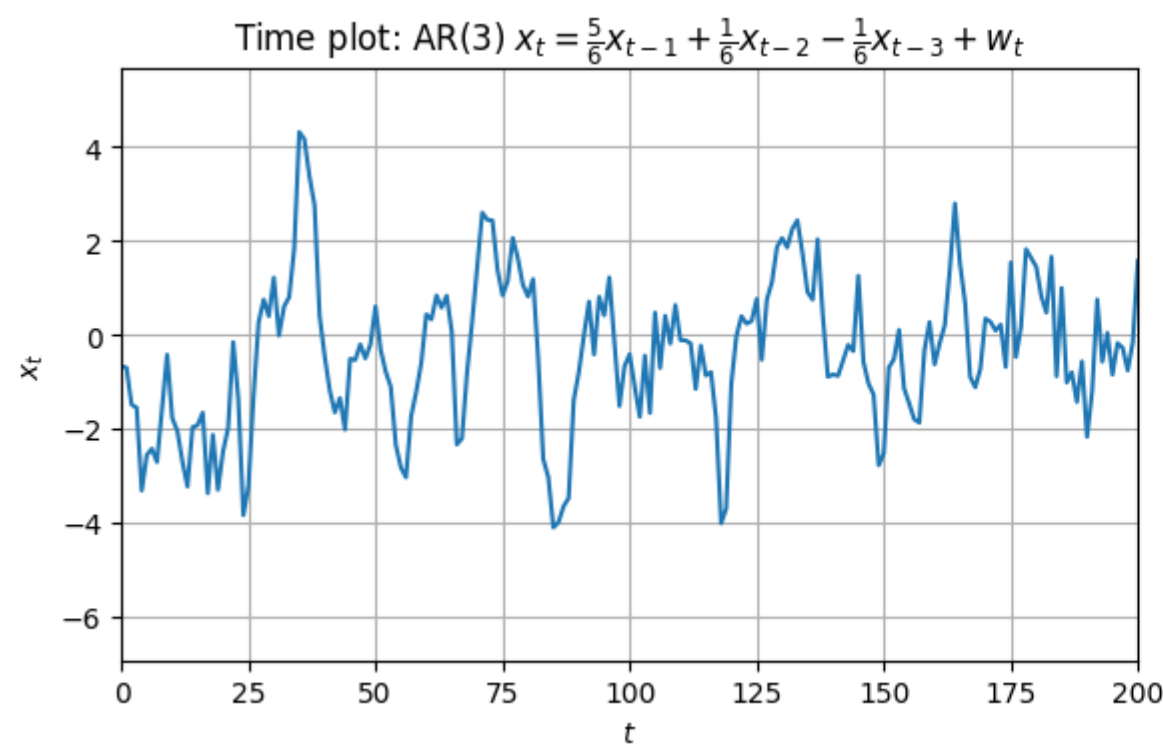
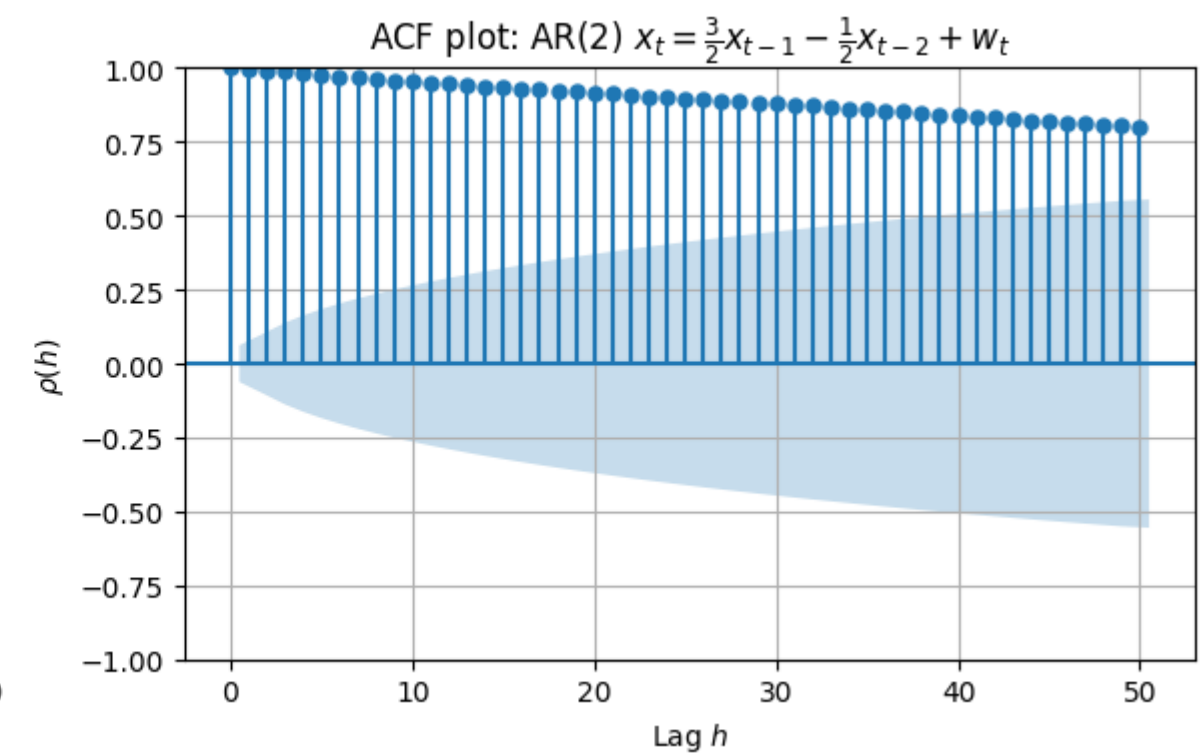
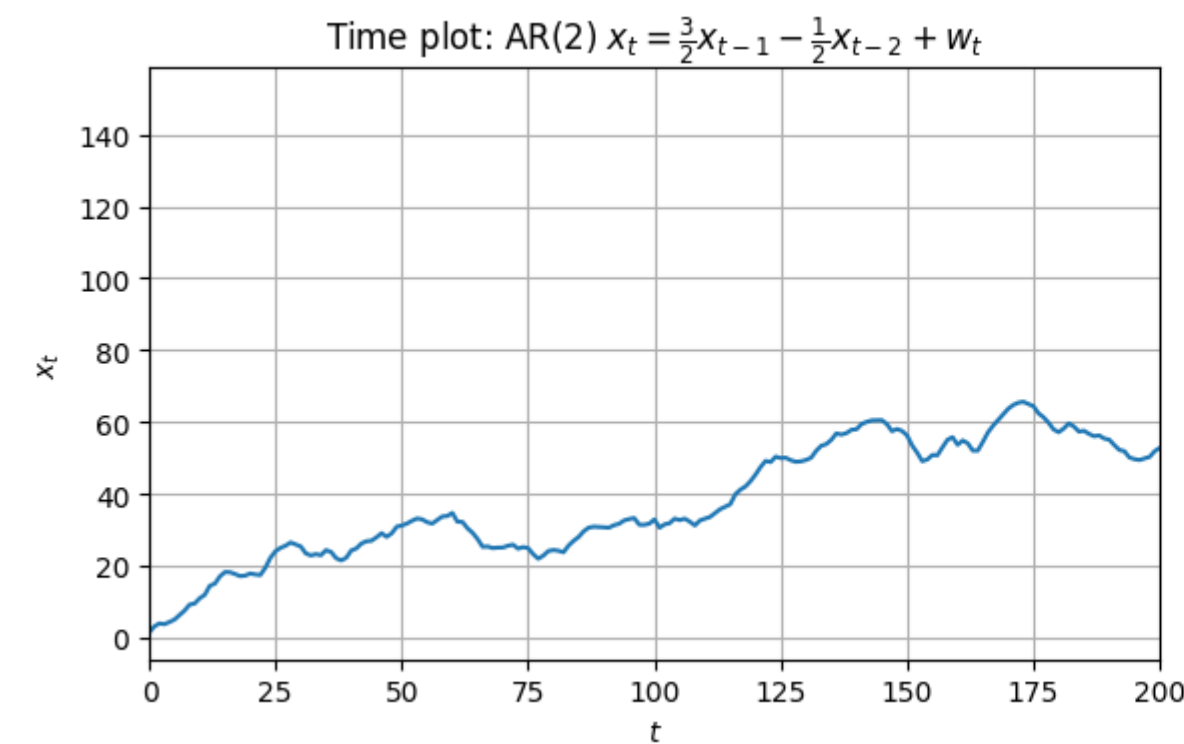
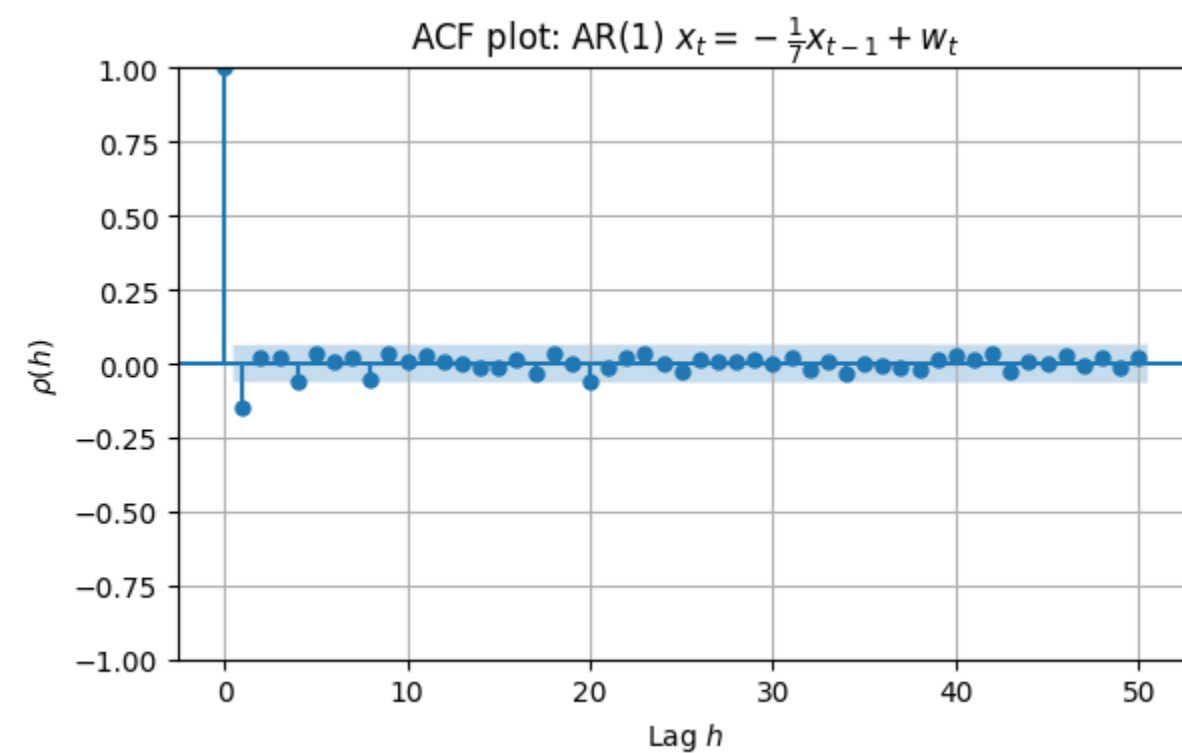
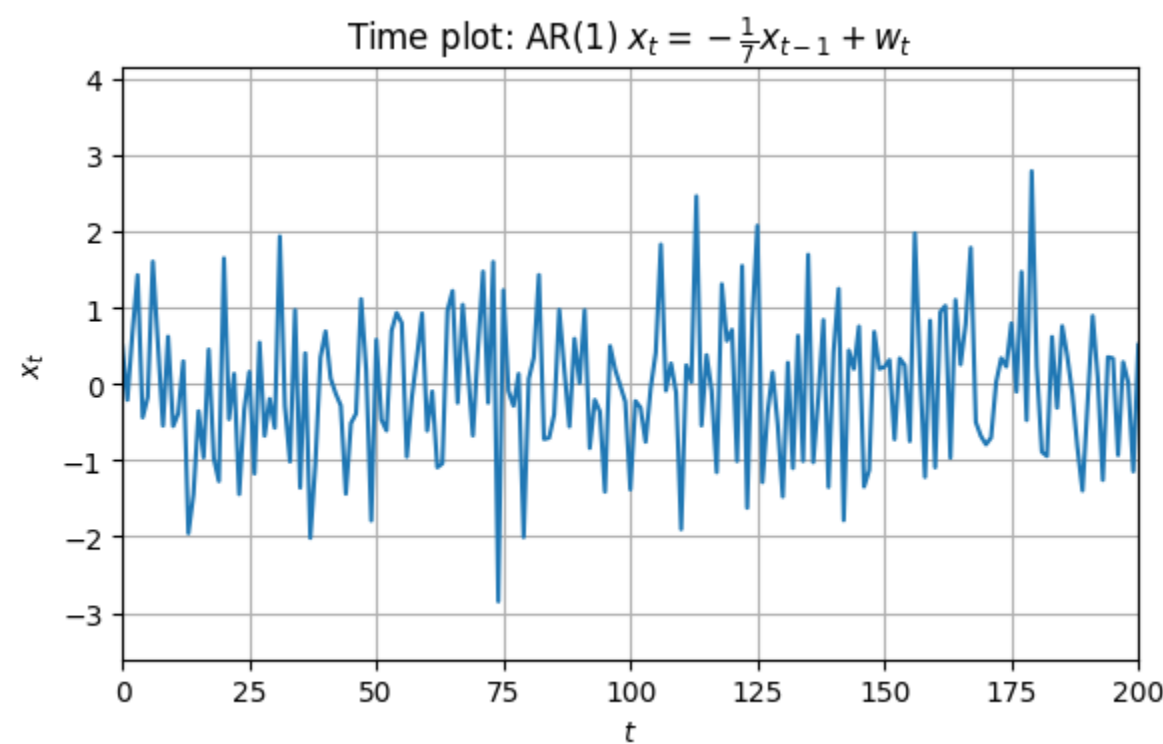
Explore **artificial auto-correlations** induced by differencing

- Start with a stationary time series.
- Apply differencing at various order.
- Compare **ACF**.

Autoregressive (AR) model

Autoregressive model of order p , $AR(p)$: $X_t = c + \sum_{i=1}^p \phi_i X_{t-i} + W_t$ with $W_t \sim WN(0, \sigma^2)$ and c a constant.

We focus on the **zero-mean** AR model ($c = 0$) since we can always choose to model $Y_t = X_t - \mu_X$.



Stationarity of AR(p)

The zero-mean AR(p) equation can be rewritten as $W_t = X_t - \sum_{i=1}^p \phi_i X_{t-i} = (1 - \sum_{i=1}^p \phi_i B^i) X_t = \Phi(B) X_t$ where $\Phi(B) = (1 - \sum_{i=1}^p \phi_i B^i)$ is called the **characteristic polynomial**.

Theorem: An AR(p) process is stationary $\Leftrightarrow$ all roots of $\Phi(B)$ are strictly greater than 1 in absolute value.

Recall that the roots of a polynomial $P(x)$ are the values r that satisfy $P(r) = 0$ e.g., $r_{1,2} = \pm 2$ for $P(x) = x^2 - 4$.

Examples (cf. figure with AR examples):

- AR(1) with $\Phi(B) = 1 + \frac{1}{7}B = \frac{1}{7}(7 + B)$, the root is $r_1 = -7 \Rightarrow$ stationary
- AR(2) with $\Phi(B) = 1 - \frac{3}{2}B + \frac{1}{2}B^2 = \frac{1}{2}(1 - B)(2 - B)$, the roots are $r_1 = 1$ and $r_2 = 2 \Rightarrow$ not stationary
- AR(3) with $\Phi(B) = 1 - \frac{5}{6}B - \frac{1}{6}B^2 + \frac{1}{6}B^3 = \frac{1}{6}(2 - B)(3 - B - B^2)$, then $r_1 = 2$ and $r_{2,3} = \frac{-1 \pm \sqrt{13}}{2} \Rightarrow$ stationary
- AR(4) with $\Phi(B) = 1 - \frac{5}{4}B^2 + \frac{1}{4}B^4 = \frac{1}{4}(1 - B)(1 + B)(2 - B)(2 + B)$, then $r_{1,2} = \pm 1$, $r_{3,4} = \pm 2 \Rightarrow$ not stationary

Partial auto-correlation function (PACF)

The PACF π_h measures the correlation between X_t and X_{t+h} after **removing** the influence of the **intermediate lags**:

$$\pi_h = \text{cor}(X_{t+h}, X_t | X_{t+1}, \dots, X_{t+h-1})$$

This influence can be represented as a linear **regression** of the intermediate lags.

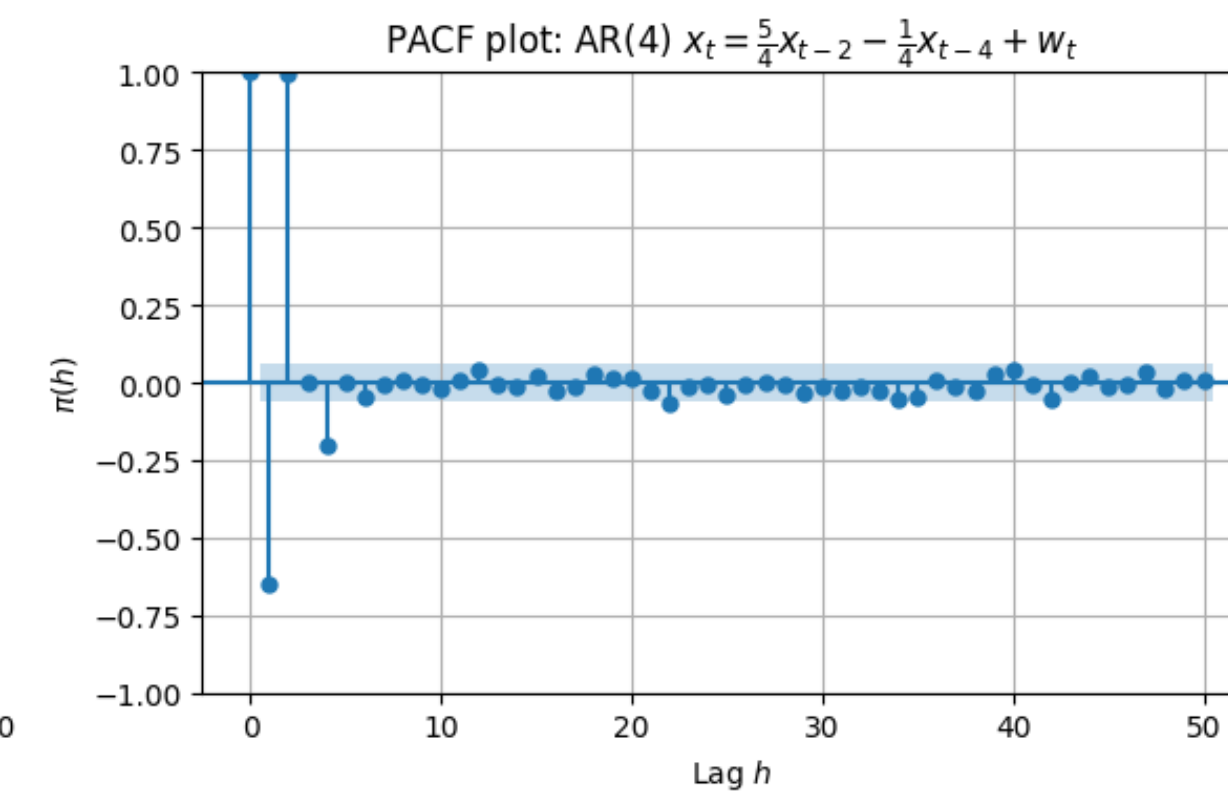
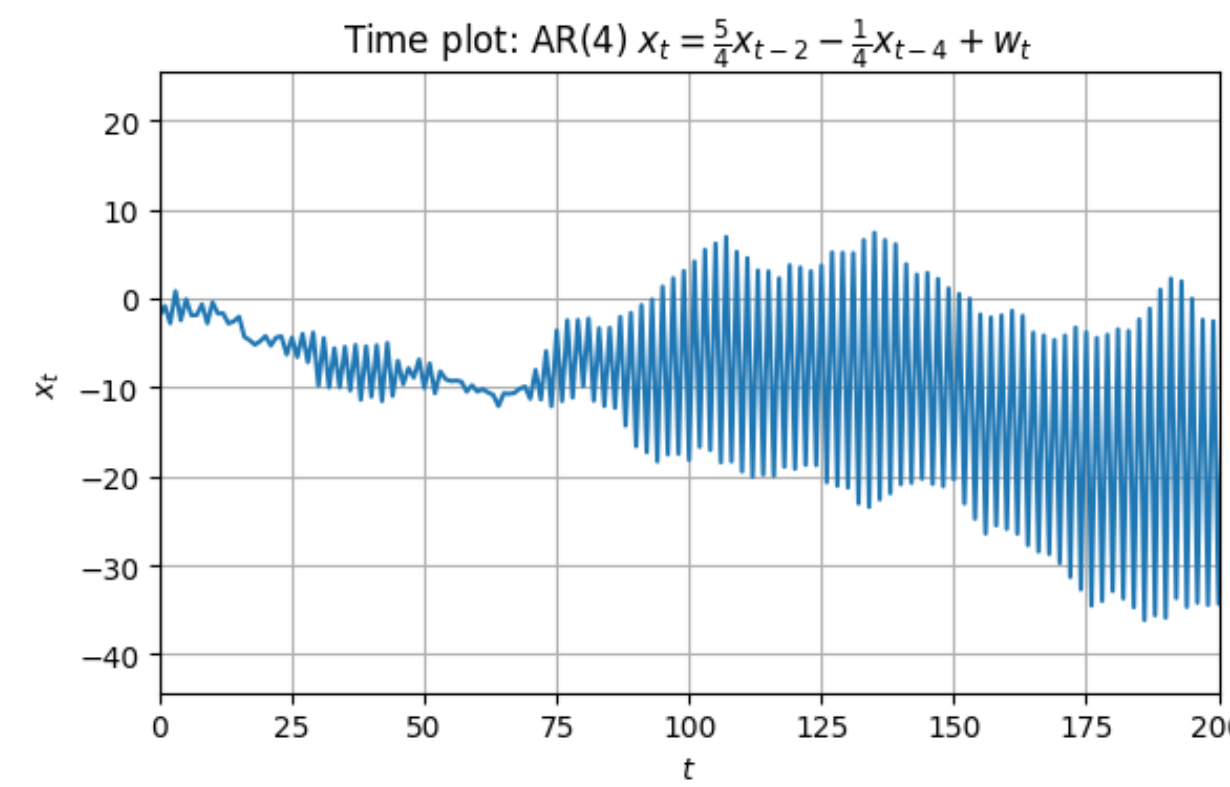
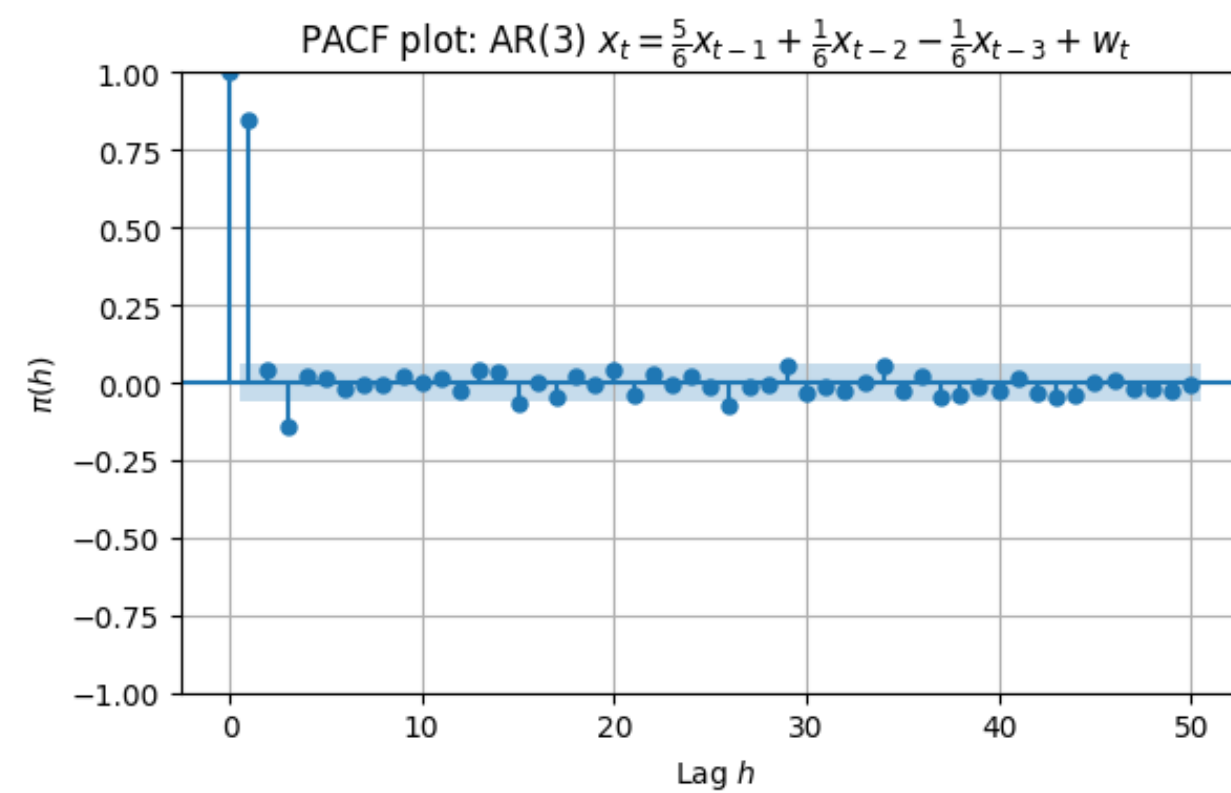
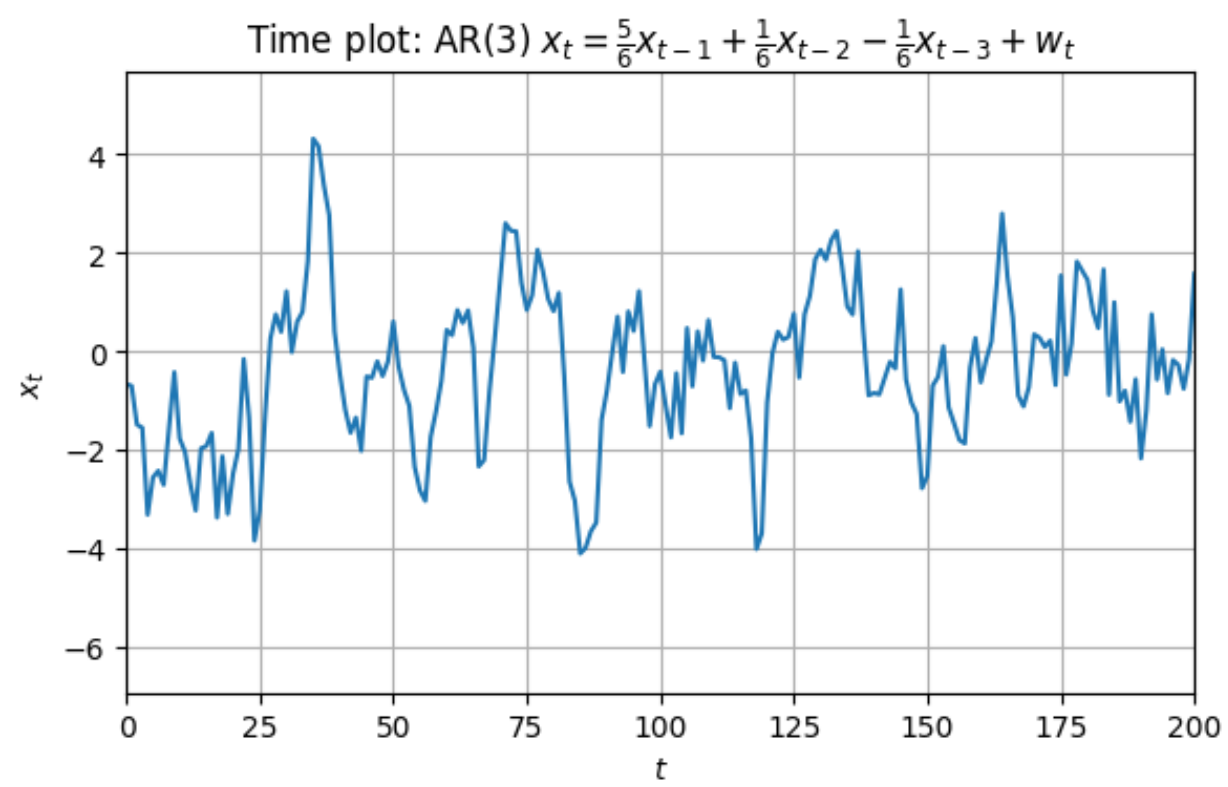
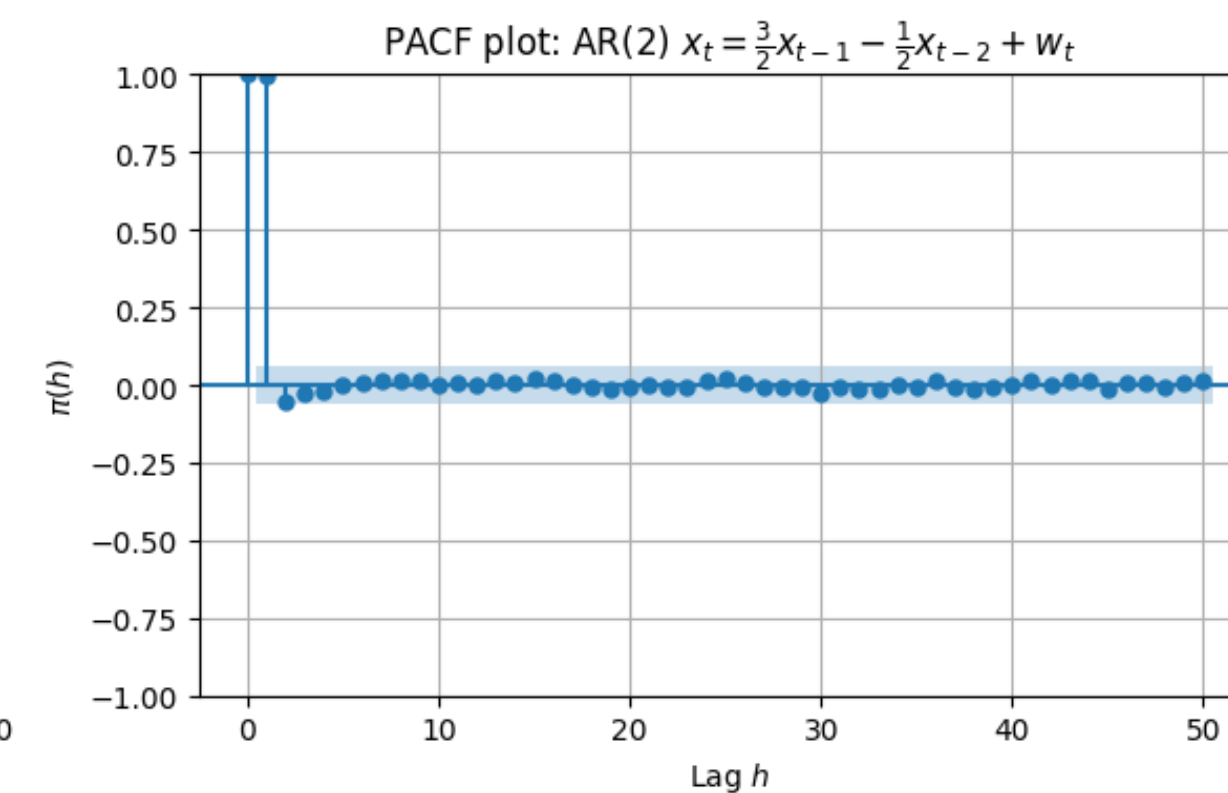
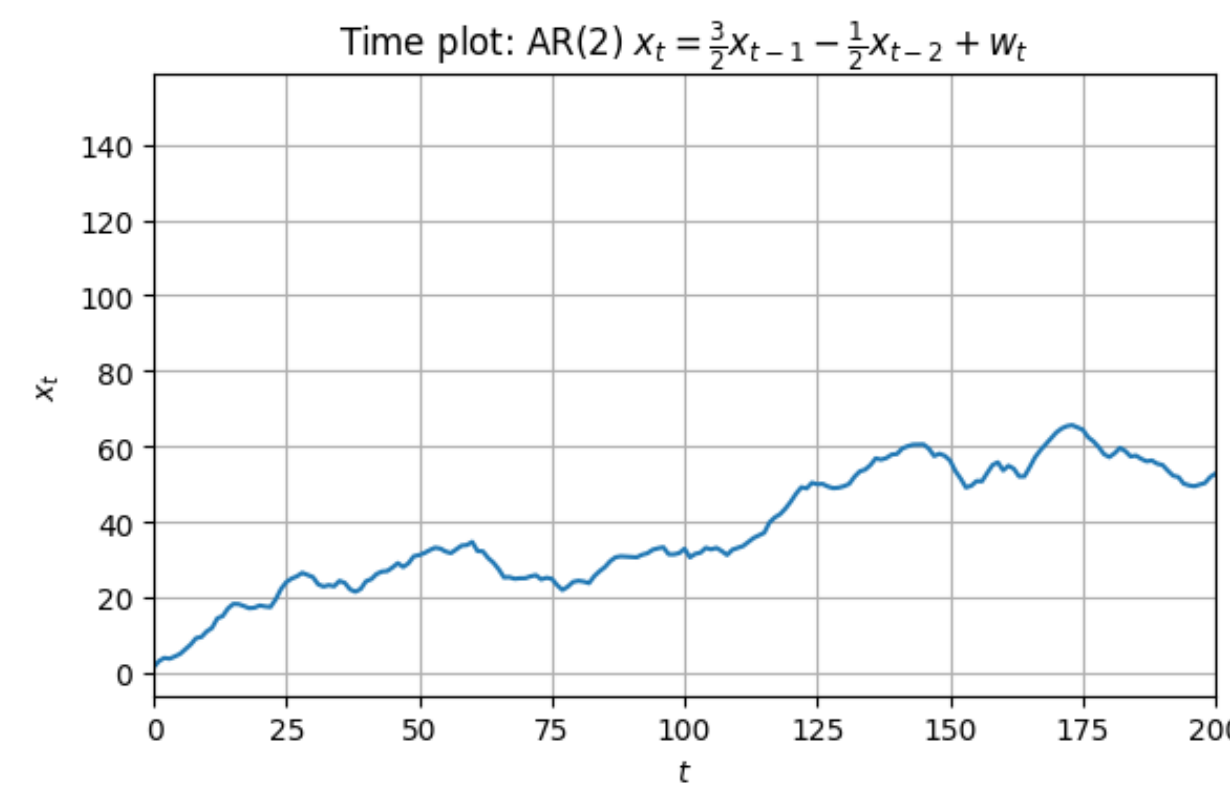
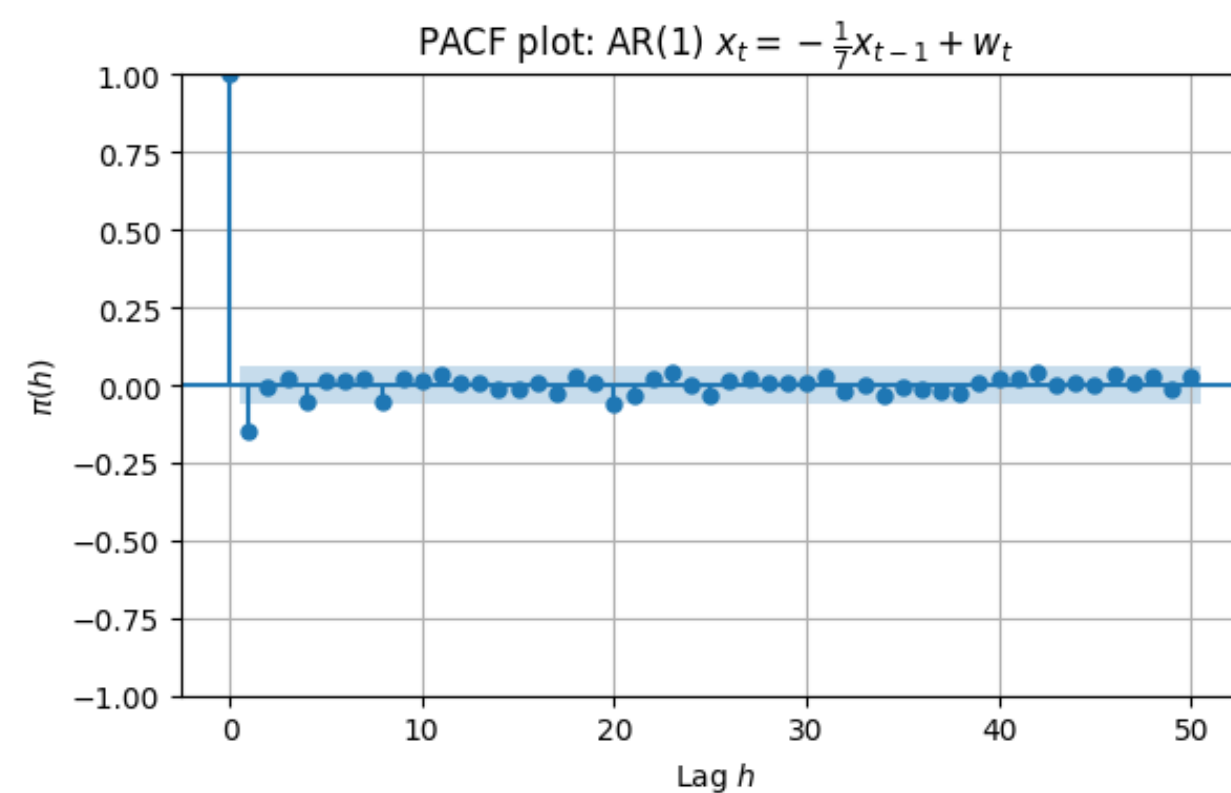
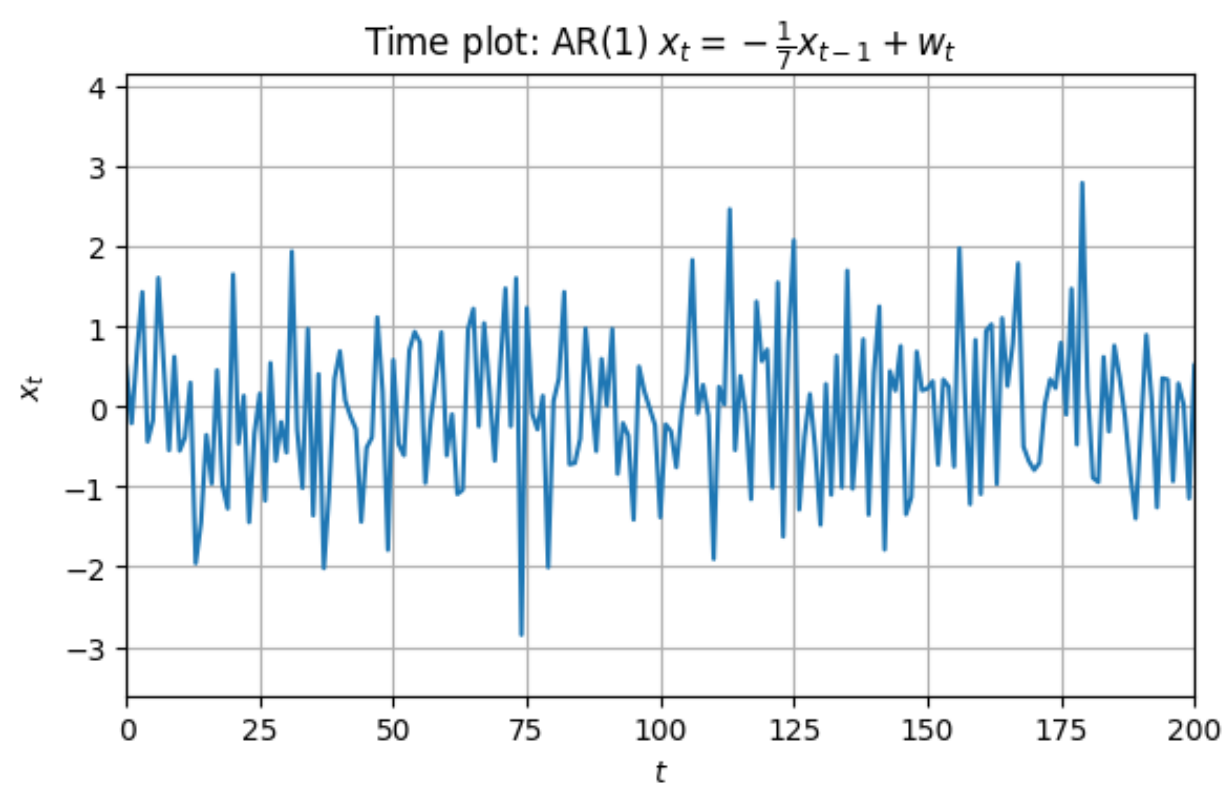
We denote $\hat{X}_{t+h}$ (respectively $\hat{X}_t$) as the linear regression of X_{t+h} (respectively X_t) on $X_{t+1}, \dots, X_{t+h-1}$.

$$\pi_h = \begin{cases} 1 & h = 0 \\ \rho_1, & h = 1 \\ \text{cor}(X_{t+h} - \hat{X}_{t+h}, X_t - \hat{X}_t) & h > 1 \end{cases}$$

For $h = p$, we have $\hat{X}_{t+p} \approx X_{t+p} - \phi_p X_t - W_{t+p} \Rightarrow \pi_p \approx \phi_p$

For $h > p$, the regression results in $\hat{X}_{t+h} \approx X_{t+h} - W_{t+h} \Rightarrow$ the **PACF is zero for all lags greater than the AR order p**.

Partial auto-correlation function (PACF)



Fitting an AR model

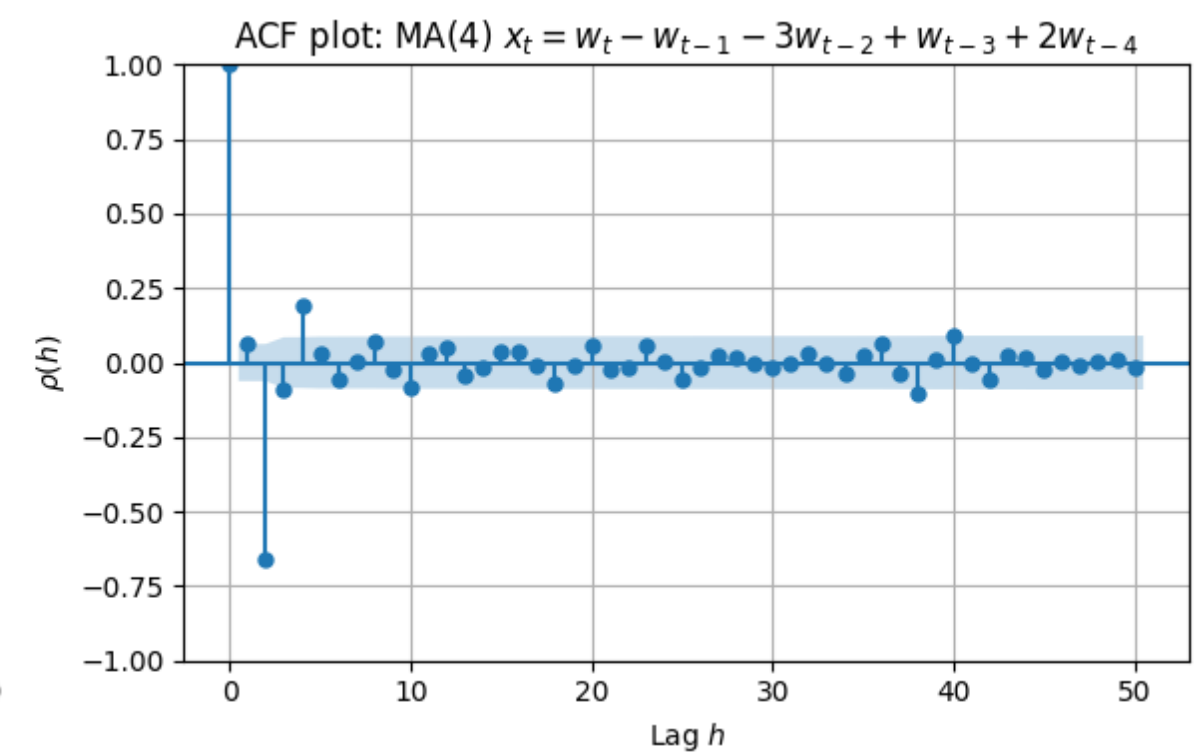
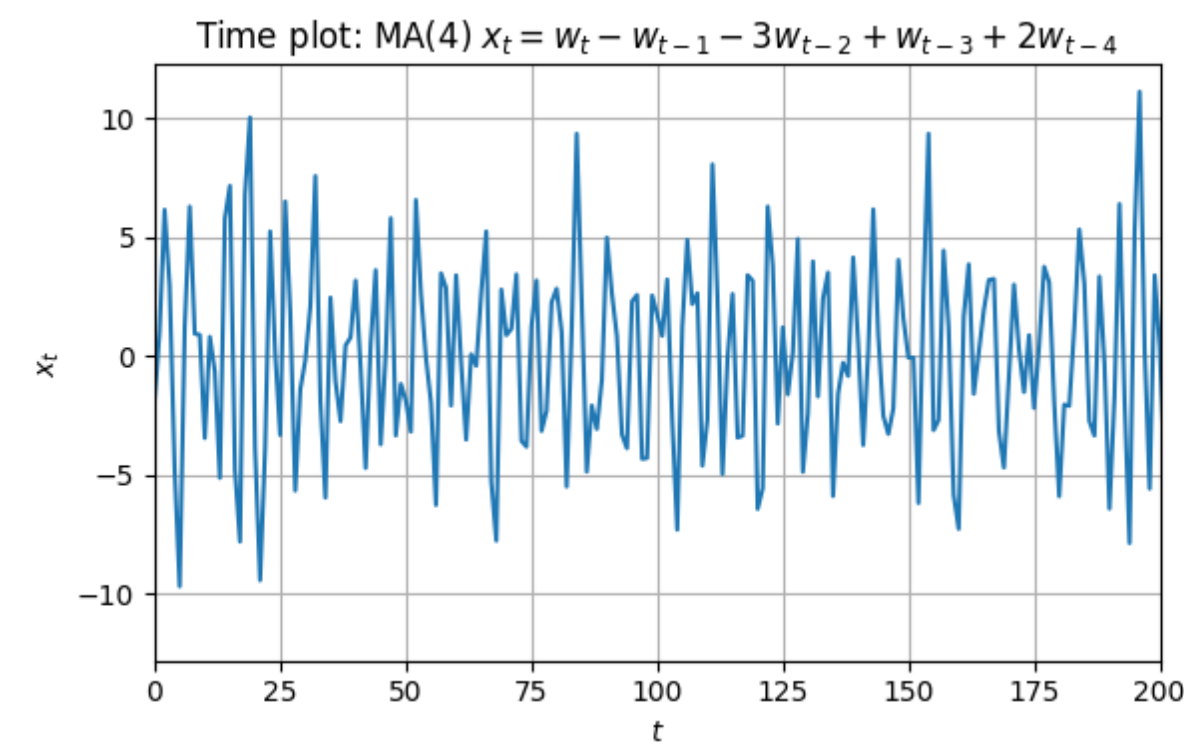
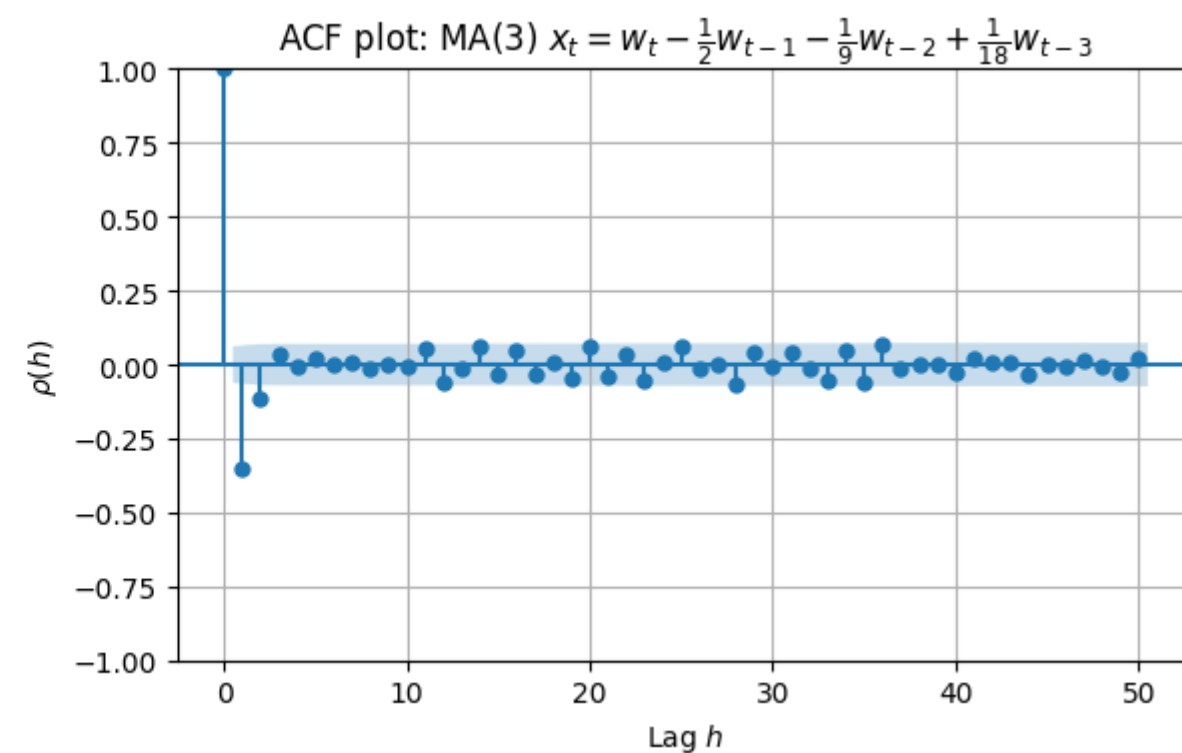
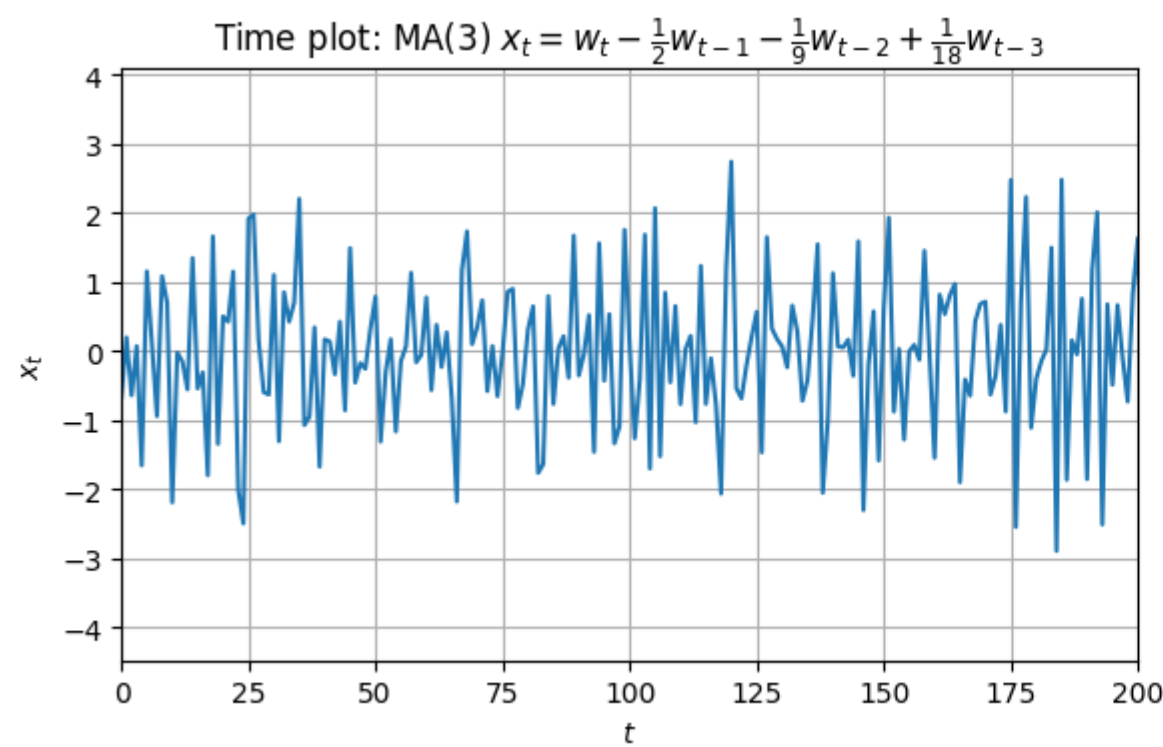
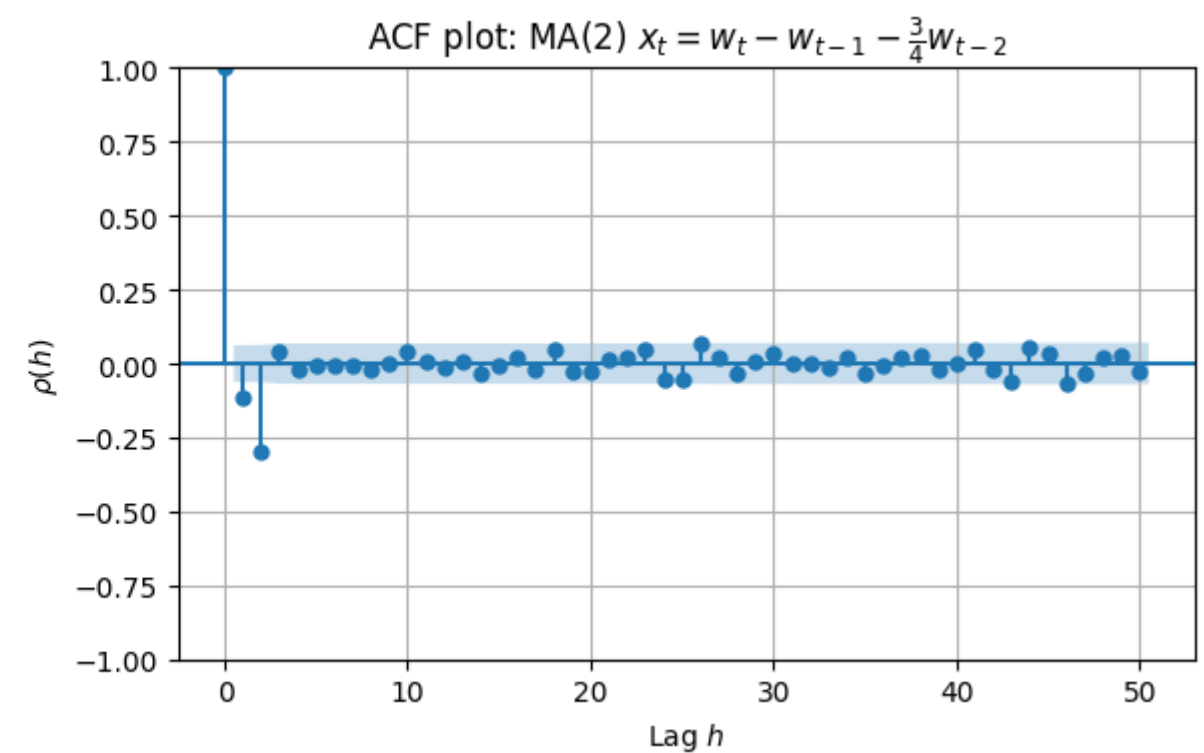
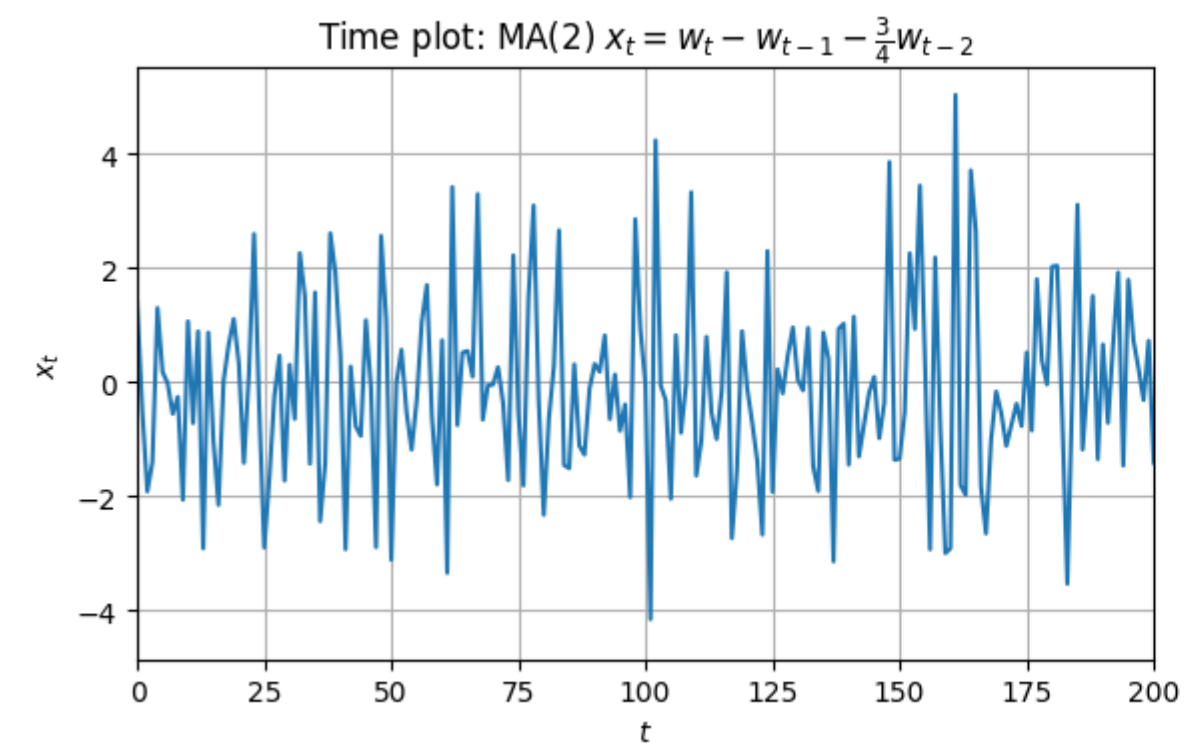
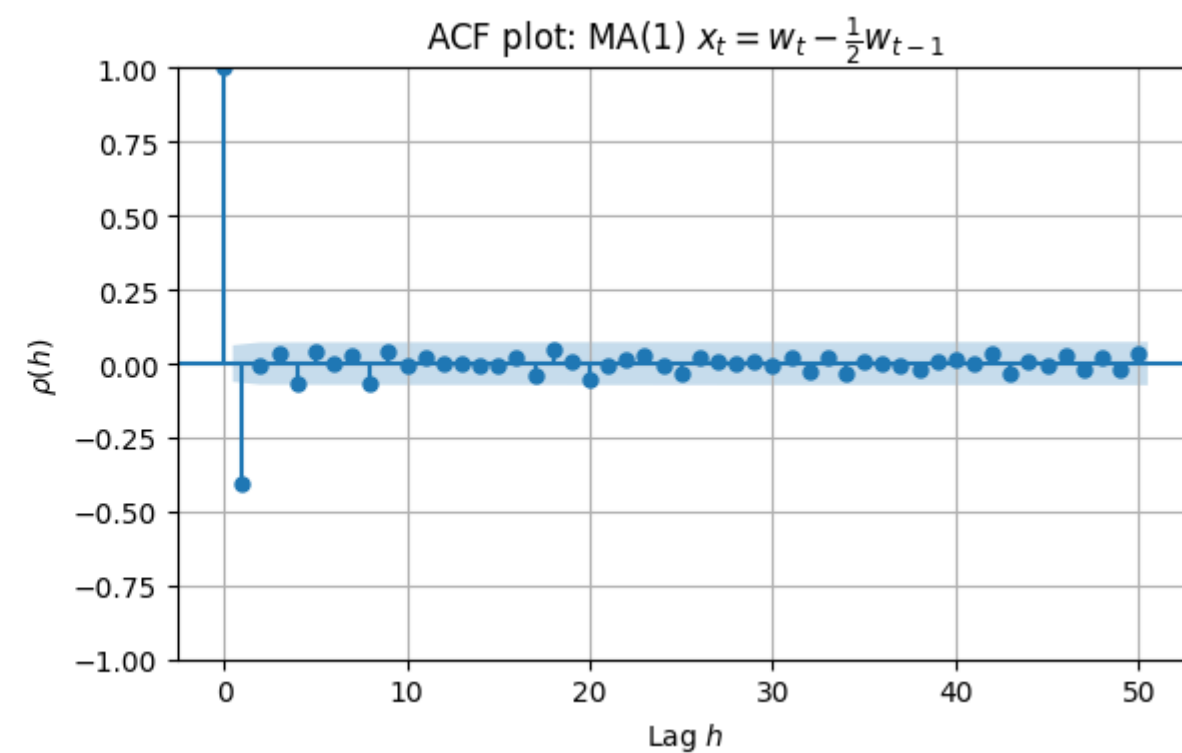
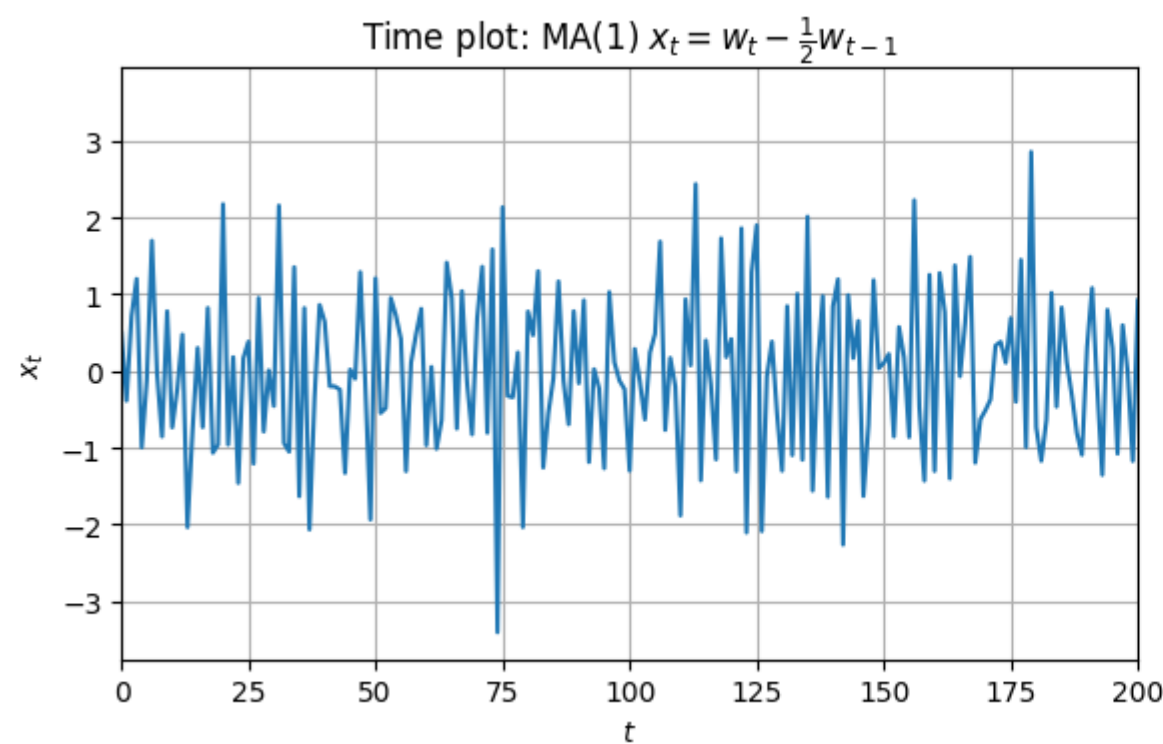
Given a **stationary** time series realization $\{x_1, x_2, \dots, x_n\}$ with **zero-mean**,

1. Is the autoregressive model **appropriate**?
 - ACF shows an exponential (sinusoidal) decay pattern.
 - PACF shows a cut-off at some lag p after which the PACF is close to zero.
2. Select model order based on the PACF, favoring smaller p values.
3. Estimate the AR(p) model **coefficients**.
 - Burg's algorithm
 - Maximum-likelihood estimation
4. Analyze **residuals** to evaluate model goodness of fit.

Moving average (MA) model ($\neq$ moving average filter)

Moving average model of order q , $\text{MA}(q)$: $X_t = c + \sum_{i=1}^q \theta_i W_{t-i} + W_t$ with $W_t \sim WN(0, \sigma^2)$ and c a constant.

We focus on the **zero-mean** MA model ($c = 0$) since we can always choose to model $Y_t = X_t - \mu_X$.



Duality between the MA(q) and AR(p) models

MA(q) processes can be represented as **AR(∞) processes**. Taking a MA(1) as example,

$$X_t = \theta W_{t-1} + W_t = \theta(X_{t-1} - \theta W_{t-2}) + W_t = \sum_{i=1}^{\infty} (-1)^{i+1} \theta^i X_{t-i} + W_t = \sum_{i=1}^{\infty} \phi_i X_{t-i} + W_t$$

with $\phi_i = (-1)^{i+1} \theta^i$

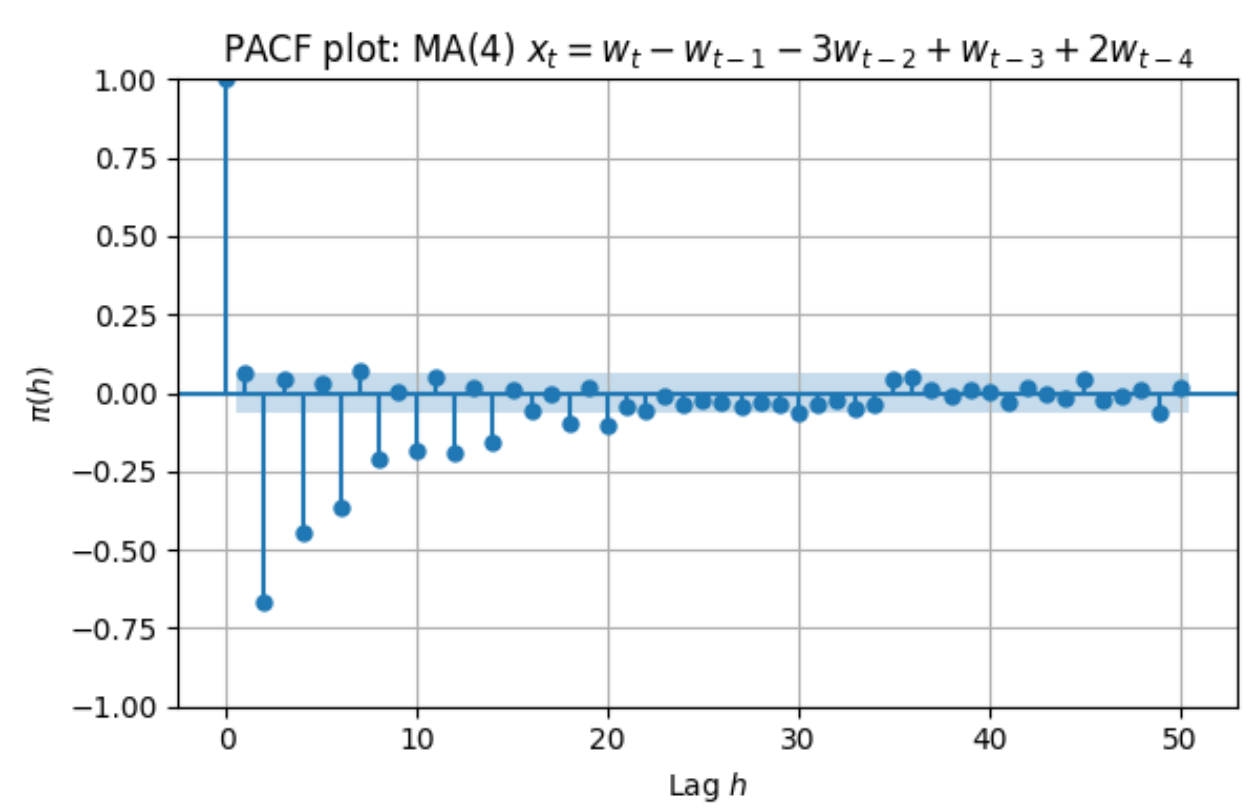
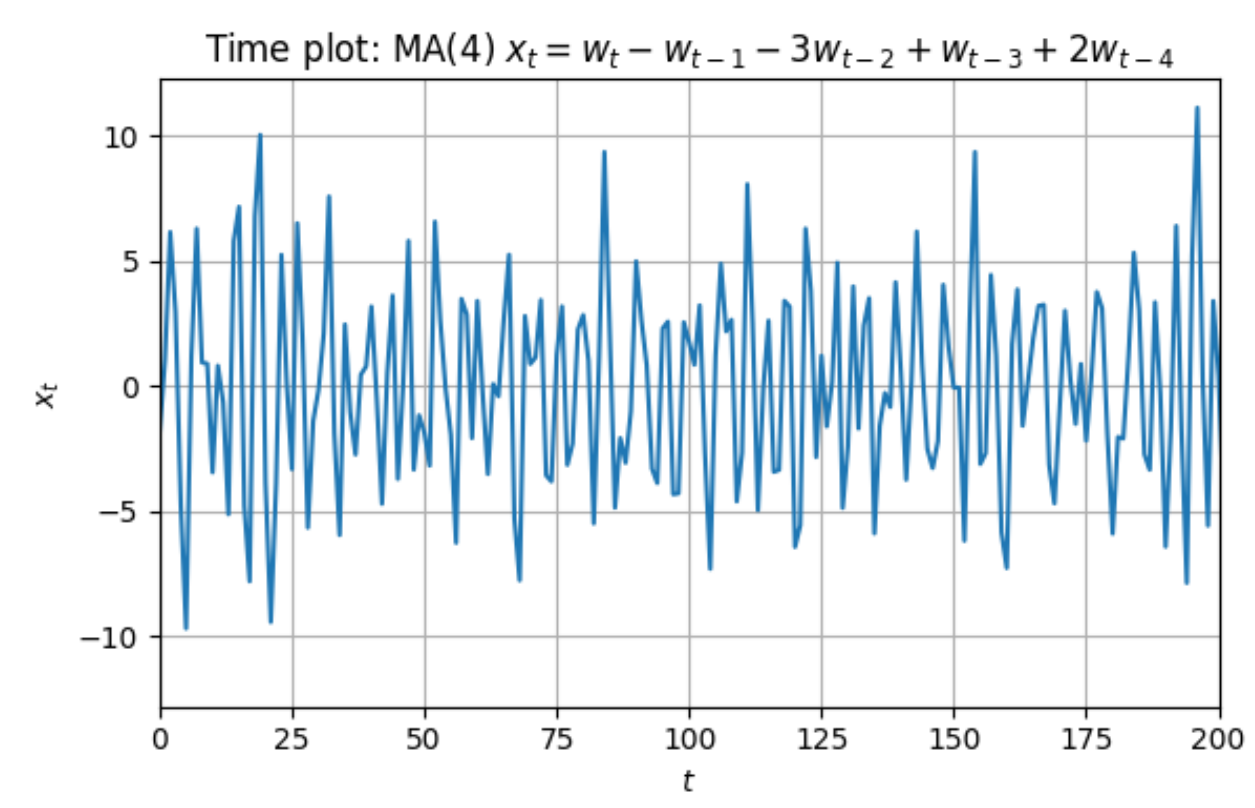
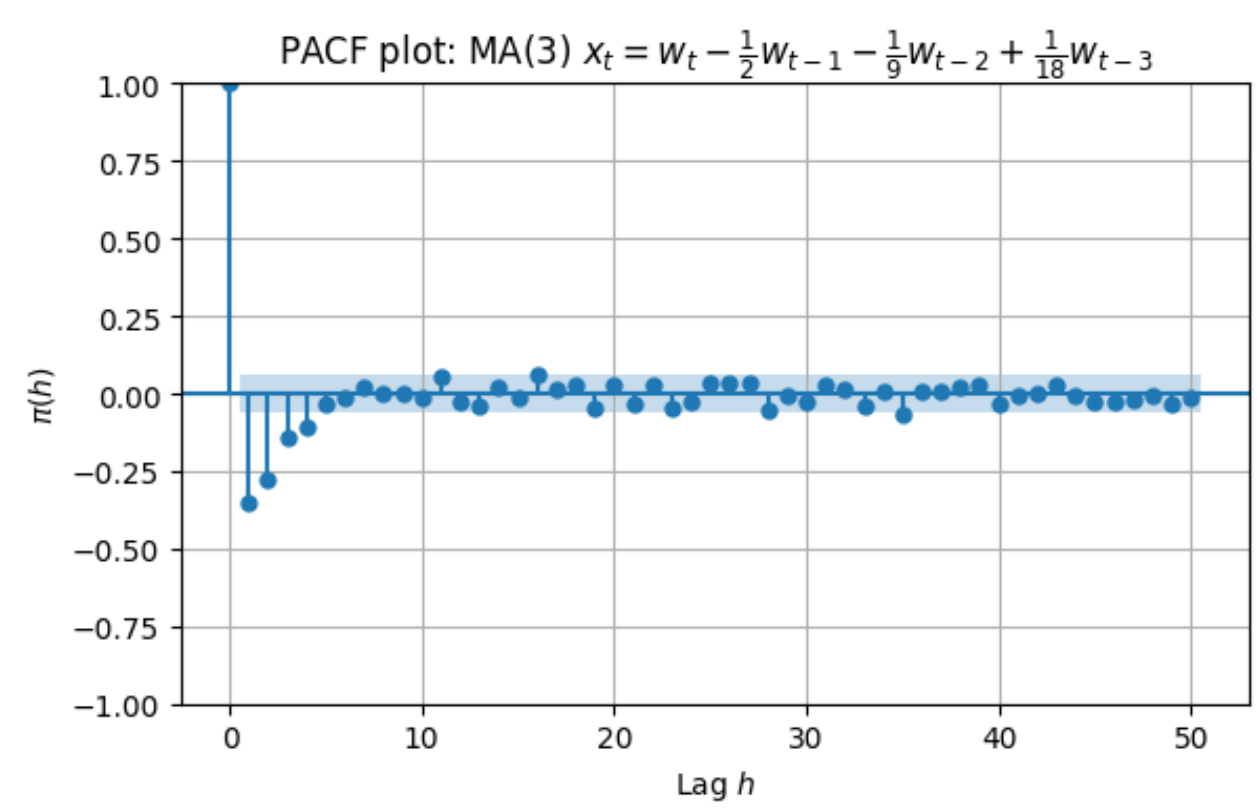
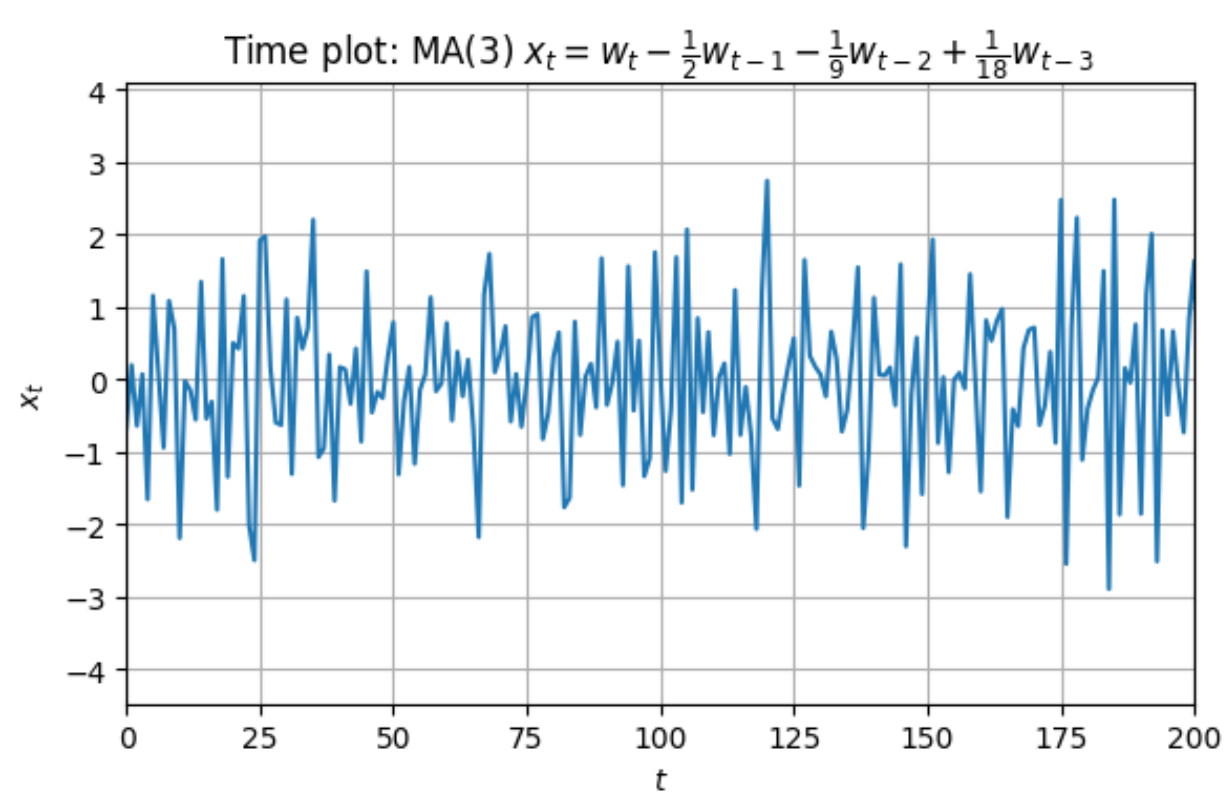
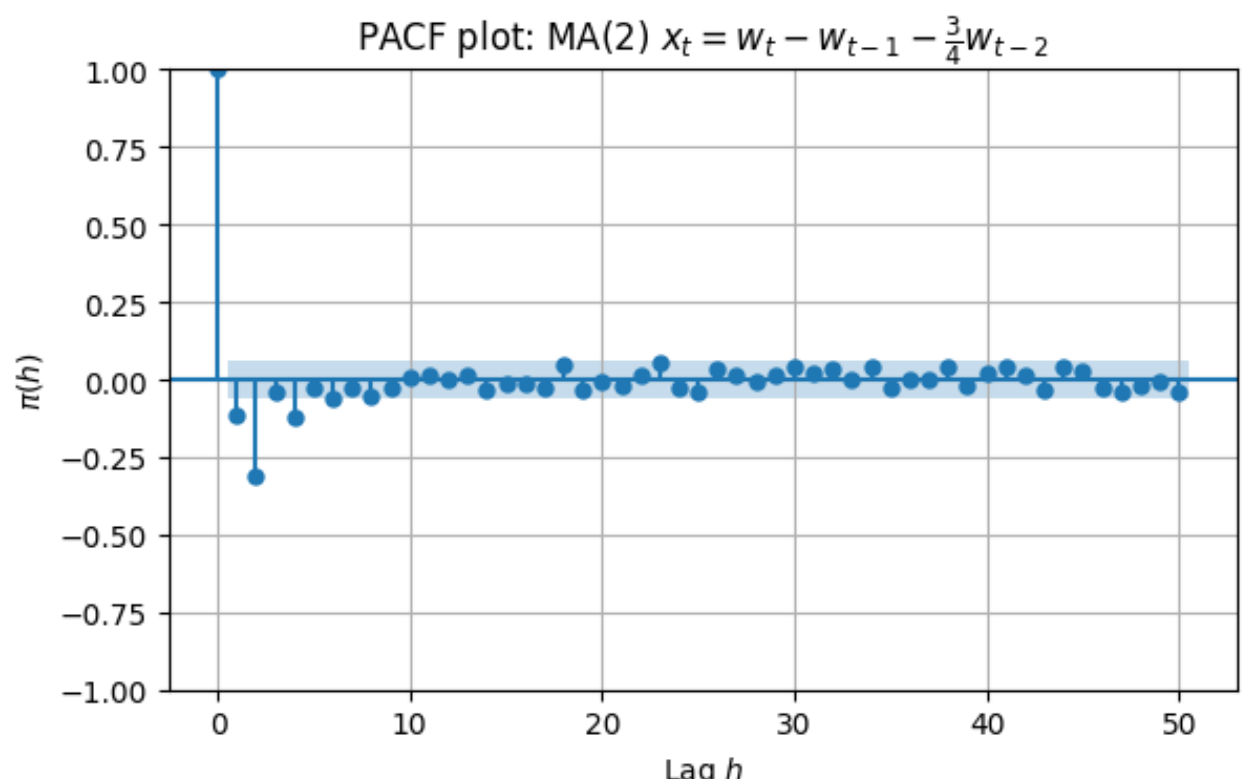
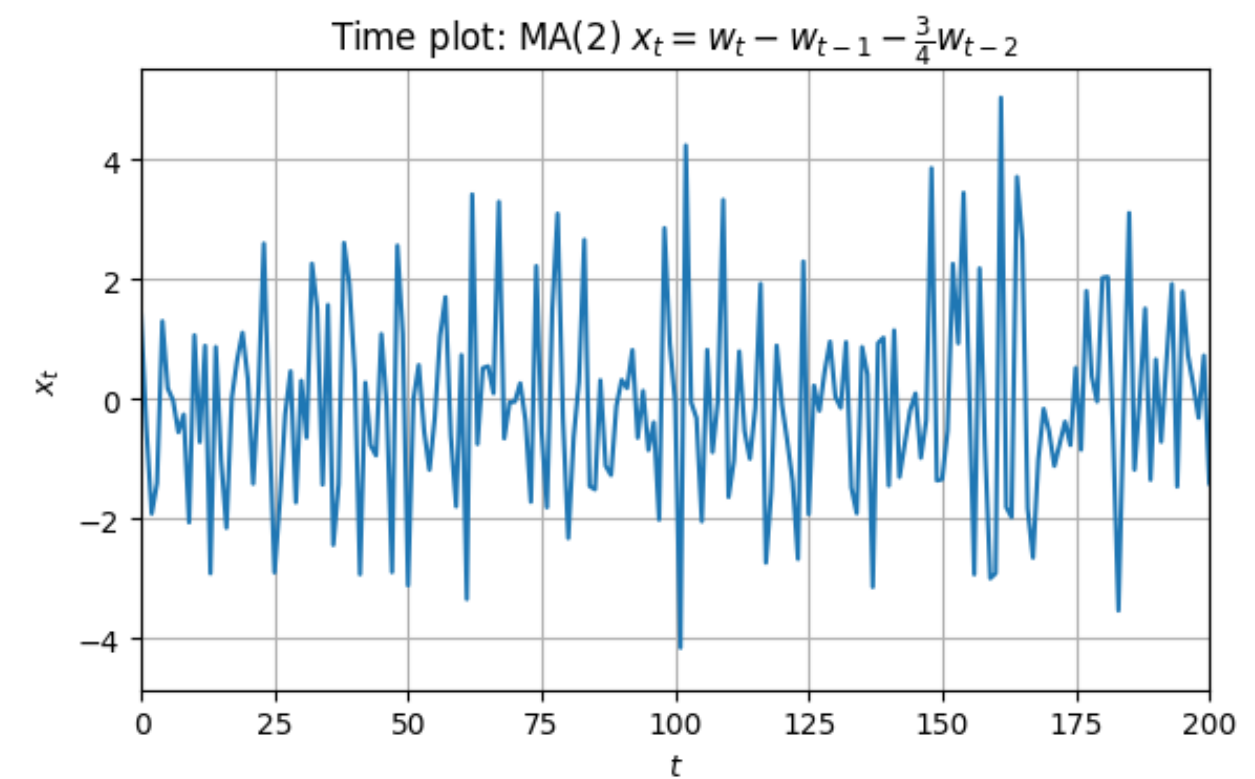
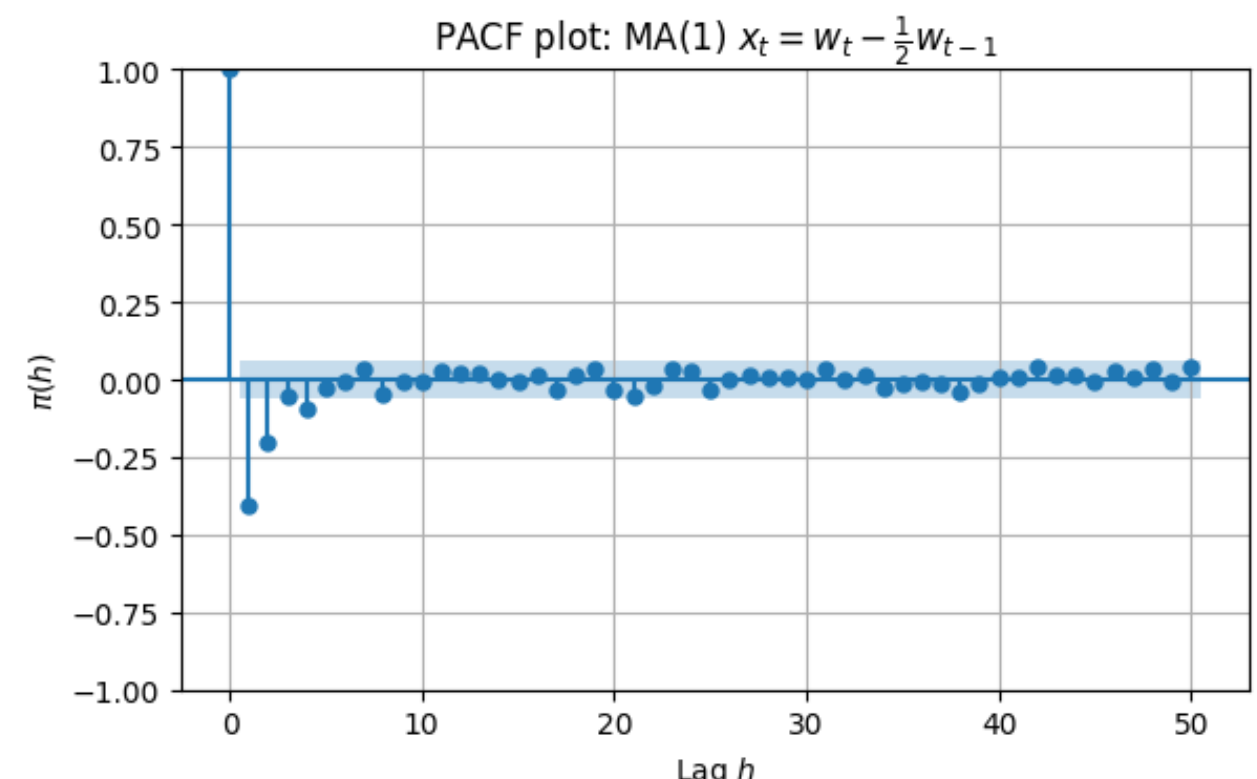
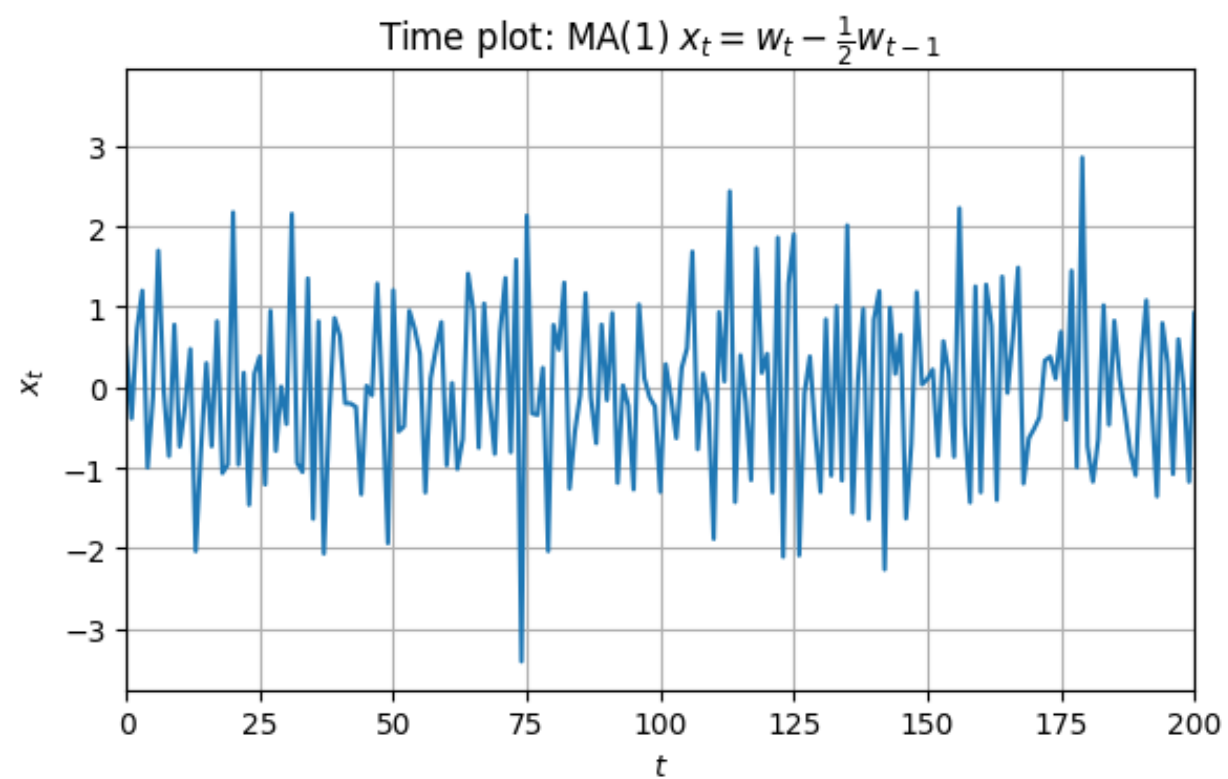
AR(p) processes can be represented as an **MA(∞) processes**. Taking an AR(1) as example,

$$X_t = \phi X_{t-1} + W_t = \phi(\phi X_{t-2} + W_{t-1}) + W_t = \sum_{i=0}^{\infty} \phi^i W_{t-i} = \sum_{i=1}^{\infty} \theta_i W_{t-i} + W_t$$

with $\theta_i = \phi^i$

Model	ACF ρ_h	PACF π_h
AR(p)	Infinite exponential decay	Cut-off at $h = p$
MA(q)	Cut-off at $h = q$	Infinite exponential decay

Partial auto-correlation function (PACF)



Invertibility of MA(q)

Different MA(q) parameters can lead to the same ACF, **preventing their unique identification.**

The zero-mean MA(q) equation can be rewritten as $X_t = \sum_{i=1}^q \theta_i W_{t-i} + W_t = (1 + \sum_{i=1}^q \theta_i B^i) W_t = \Theta(B) W_t$ where $\Theta(B) = (1 + \sum_{i=1}^q \theta_i B^i)$ is called the **characteristic polynomial**.

Theorem: A MA(q) process is invertible $\Leftrightarrow$ all roots of $\Theta(B)$ are strictly greater than 1 in absolute value.

By imposing the **invertibility condition**, we ensure that there is a unique MA(q) process for a given ACF.

Examples (cf. figure with MA examples):

- MA(1) with $\Theta(B) = 1 - \frac{1}{2}B$, the root is $r_1 = 2 \Rightarrow$ invertible
- MA(2) with $\Theta(B) = 1 - B - \frac{3}{4}B^2 = \frac{3}{4}(B + 2)(B - \frac{2}{3})$, the roots are $r_1 = -2$ and $r_2 = \frac{2}{3} \Rightarrow$ not invertible
- MA(3) with $\Theta(B) = 1 - \frac{1}{2}B - \frac{1}{9}B^2 + \frac{1}{18}B^3 = \frac{1}{18}(B - 3)(B + 3)(B - 2)$, then $r_{1,2} = \pm 3$ and $r_3 = 2 \Rightarrow$ invertible
- MA(4) with $\Theta(B) = 1 - B - 3B^2 + B^3 + 2B^4 = (2B - 1)(B + 1)^2(B - 1)$, then $r_{1,2} = \pm 1$, and $r_3 = \frac{1}{2} \Rightarrow$ not invertible

Fitting a MA model

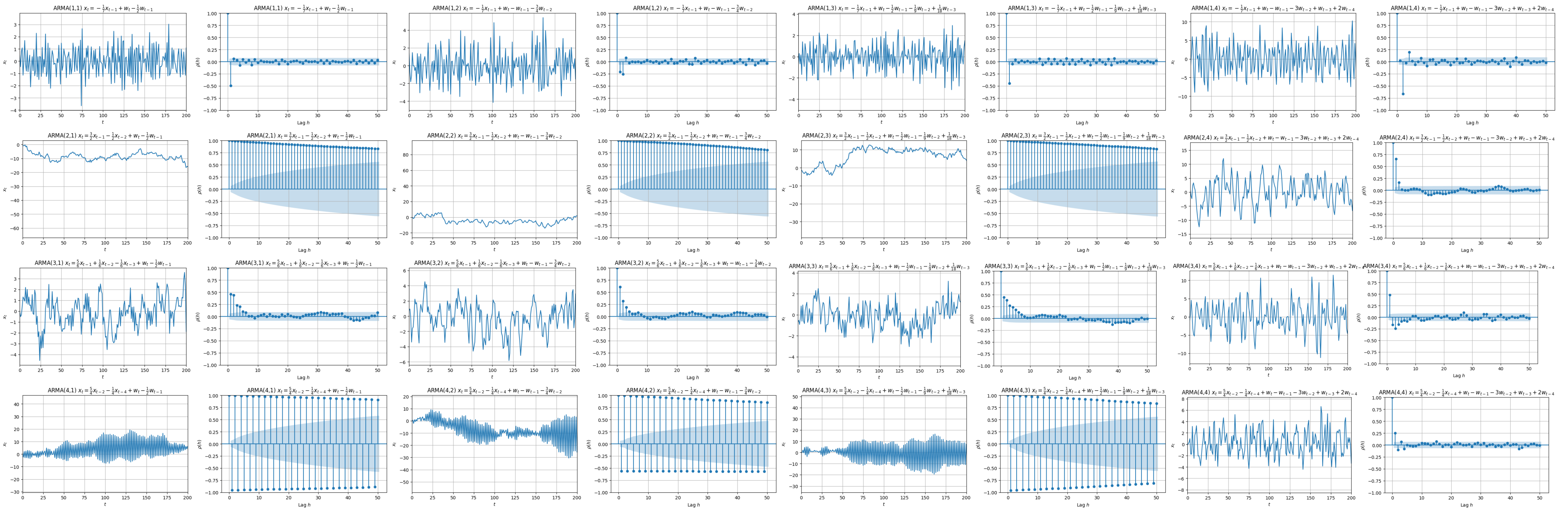
Given a **stationary** time series realization $\{x_1, x_2, \dots, x_n\}$ with **zero-mean**,

1. Is the moving average model **appropriate**?
 - ACF shows a cut-off at some lag q after which the ACF is close to zero.
 - PACF shows an exponential (sinusoidal) decay pattern.
2. Select model order based on the ACF, favoring smaller q values.
3. Estimate the MA(q) model **coefficients**.
 - Conditional sum of squares
 - Maximum-likelihood estimation
4. Analyze **residuals** to evaluate model goodness of fit (discussed in following lectures).

Autoregressive moving average (ARMA) model

ARMA(p,q): $X_t = c + \sum_{i=1}^p \phi_i X_{t-i} + \sum_{i=0}^q \theta_i W_{t-i}$ with $\theta_0 = 1$ and $W_t \sim WN(0, \sigma^2)$ and c a constant.

We focus on the **zero-mean** ARMA model ($c = 0$) since we can always choose to model $Y_t = X_t - \mu_X$.



Properties of ARMA(p,q)

ARMA processes can capture both **long-term dependencies** (AR part) and **short-term shocks** (MA part).

The zero-mean ARMA(p,q) can be rewritten with the **characteristic polynomials** of the AR and MA parts:

$$\Phi(B)X_t = \left(1 - \sum_{i=1}^p \phi_i B^i\right) X_t = \left(1 + \sum_{i=1}^q \theta_i B^i\right) W_t = \Theta(B)W_t$$

We **only** consider ARMA models **without common roots** between $\Phi(B)$ and $\Theta(B)$ (**no parameter redundancy**).

Theorem: An ARMA(p,q) process is stationary $\Leftrightarrow$ all roots r of $\Phi(B)$ lie outside the unit circle ($|r| > 1$).

- Stationarity depends on the AR part since any MA(q) is stationary.

Theorem: An ARMA(p,q) process is invertible $\Leftrightarrow$ all roots r of $\Theta(B)$ lie outside the unit circle ($|r| > 1$).

- Invertibility depends on the MA part since any AR(p) is uniquely represented.
- The invertibility condition ensures that an ARMA(p, q) process can be uniquely represented.

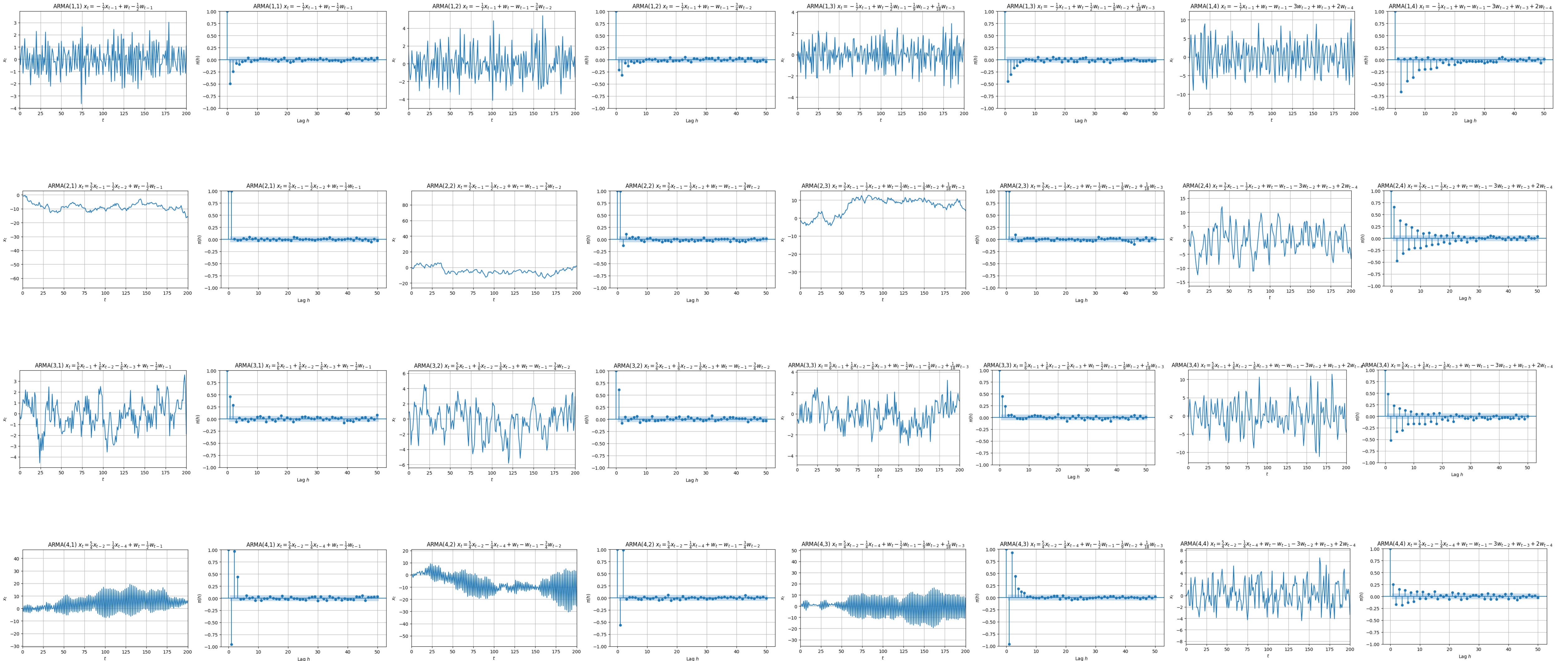
ACF and PACF of ARMA(p,q)

Theorem: an ARMA(p, q) process can be expressed as either an AR(∞) process or an MA(∞) process.

- Expand either the MA(q) part as an AR(∞) process or the AR(p) part as an MA(∞) process.
- Typically, ARMA(p,q) requires fewer parameters than pure AR or MA processes would.

Model	ACF ρ_h	PACF π_h
AR(p)	Infinite exponential decay	Cut-off at $h = p$
MA(q)	Cut-off at $h = q$	Infinite exponential decay
ARMA(p,q)	Infinite exponential decay with drop-off for $h > q$	Infinite exponential decay with drop-off for $h > p$

Partial auto-correlation function (PACF)



Fitting an ARMA model

Given a **stationary** time series realization $\{x_1, x_2, \dots, x_n\}$ with **zero-mean**,

1. Is the ARMA model **appropriate**?
 - Neither the ACF nor the PACF show a pronounced cut-off.
 - ACF shows a drop-off after some lag q and an exponential decay pattern.
 - PACF shows a drop-off after some lag p and an exponential decay pattern.
2. Select p and q based on the ACF and PACF, favoring smaller values.
 - With unclear patterns, test multiple pairs and calculate information criteria (discussed in following lectures)
3. Estimate the ARMA(p, q) model **coefficients**.
 - Conditional sum of squares
 - Maximum-likelihood estimation
4. Analyze **residuals** to evaluate model goodness of fit (discussed in following lectures).

Autoregressive integrated moving average (ARIMA) model

X_t follows an ARIMA(p,d,q) model if its d^{th} order **differencing** $\nabla^d X_t$ is an ARMA(p,q) process:

$$\Phi(B)\nabla^d X_t = \Phi(B)(1 - B)^d X_t = c + \Theta(B)W_t$$

We say that X_t is given by the d^{th} order **integration** of an ARMA(p,q).

Differencing enables to **handle non-stationarity**. Setting $d = 1$ or 2 is typically sufficient to achieve stationarity.

Model	ARIMA equivalent
White noise	ARIMA(0,0,0) with $c = 0$
Random walk	ARIMA(0,1,0) with $c = 0$
Random walk with drift	ARIMA(0,1,0) with $c \neq 0$
AR(p)	ARIMA(p,0,0)
MA(q)	ARIMA(0,0,q)
ARMA(p,q)	ARIMA(p,0,q)

ARIMA	Long-term forecast
$c = 0$ and $d = 0$	Converge to 0
$c = 0$ and $d = 1$	Converge to a non-zero constant
$c = 0$ and $d = 2$	Follow a linear trend
$c \neq 0$ and $d = 0$	Converge to the historical mean
$c \neq 0$ and $d = 1$	Follow a linear trend
$c \neq 0$ and $d = 2$	Follow a quadratic trend

Seasonal autoregressive integrated moving average (SARIMA) model

Seasonal ARMA with **seasonal orders** P, Q and **period** s, ARMA(p,q)(P,Q)_s:

$$\Phi(B)\acute{\Phi}(B^s)X_t = c + \Theta(B)\acute{\Theta}(B^s)W_t$$

$$\text{with } \acute{\Phi}(B^s) = (1 - \sum_{i=1}^p \acute{\phi}_i B^{i \cdot s}) \text{ and } \acute{\Theta}(B^s) = (1 + \sum_{i=1}^q \acute{\theta}_i B^{i \cdot s})$$

Seasonal orders can be inferred from ACF and PACF but for **lags on the seasonal scale** i.e., in multiples of s.

To **handle non-stationarity**, we add integration on both the original and seasonal scales, ARIMA(p,d,q)(P,D,Q)_s:

$$\Phi(B)\acute{\Phi}(B^s)\nabla^d \nabla_s^D X_t = \Phi(B)\acute{\Phi}(B^s)(1 - B)^d (1 - B^s)^D X_t = c + \Theta(B)\acute{\Theta}(B^s)W_t$$

Comments

- Setting $d = D = 1$ is typically sufficient to achieve stationarity.
- As with other models, the ACF and PACF help guide initial estimates for the orders p,q,P,Q.
- In practice different combinations of these values are tested.
- Best model is selected based on information criteria (discussed in following lectures).

Exercise

Simulate and analyze AR, MA, ARMA, ARIMA, SARIMA models

- Generate time series data then compute ACF and PACF.
- Determine model orders.
- Compare empiric ACF and PACF with theoretical counterpart.
- Estimate the model parameters from the generated data.

Model real-world time series

- Is differencing required?
- Determine model order from ACF and PACF (review this [summary](#)).
- Estimate the model parameters.

Forecasting

Extrapolating past observations to predict future data.

- Works well provided future data follows past patterns.
- Strong signals (low noise) can lead to accurate forecasts.
- Noise increases **uncertainty**, making predictions reliable only for the **short term**.

Sources of uncertainty

- Data: unexpected **disruption** from past patterns.
- Model: chosen model may not represent the **true data-generating process**.
- Parameters: even with the correct model, estimated **parameters may be inaccurate**.
- Forecasts: model typically yield an estimate of the **conditional mean** of future instances, which may be strongly influenced by future **unpredictable innovations**.

Forecasts must be complemented with a measure of the model uncertainty, typically **prediction intervals**.

Forecasting

Given a time series realization $\{x_1, x_2, \dots, x_T, \dots, x_n\}$,

The h -step forecast of x_{T+h} based on the data $\{x_1, x_2, \dots, x_T\}$ is represented as $\hat{x}_{T+h|T}$.

- T is the **forecast time**.
- h is the **forecast horizon** i.e., how far into the future the forecast is made.
- $T + h$ is the **target time** i.e., the time point of the forecast.

Considering a monthly time series and a 1-year forecast horizon,

- **Point** forecast is $\hat{x}_{T+12|T}$.
- **Multi-step** forecast is $\{\hat{x}_{T+1|T}, \dots, \hat{x}_{T+12|T}\}$.

Forecasting baselines

Mean: forecasts are equal to the average of the observed data, $\hat{x}_{T+h|T} = \frac{1}{T} \sum_{i=1}^T x_i$

Naïve: forecasts are equal to the last observed value of the series, $\hat{x}_{T+h|T} = x_T$

- Naïve is optimal for random walk process.

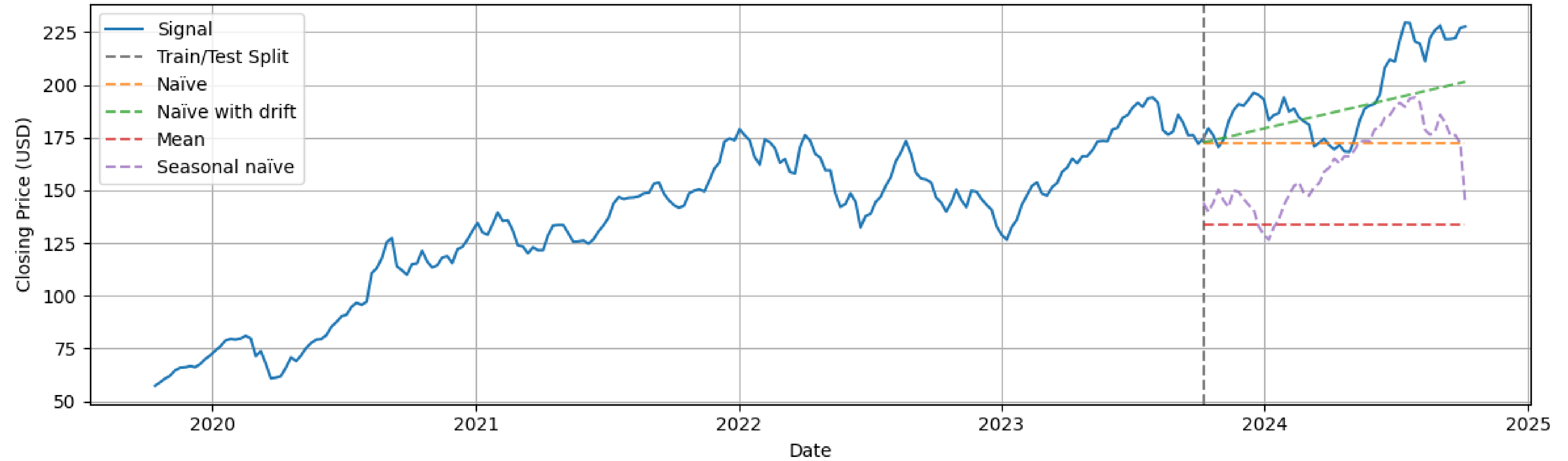
Seasonal naïve: forecasts are equal to the last observed value from the same season, $\hat{x}_{T+h|T} = x_{T+h-[h/P]P}$

- Setting $P = 1$ (non-seasonal data) results in the naïve forecast

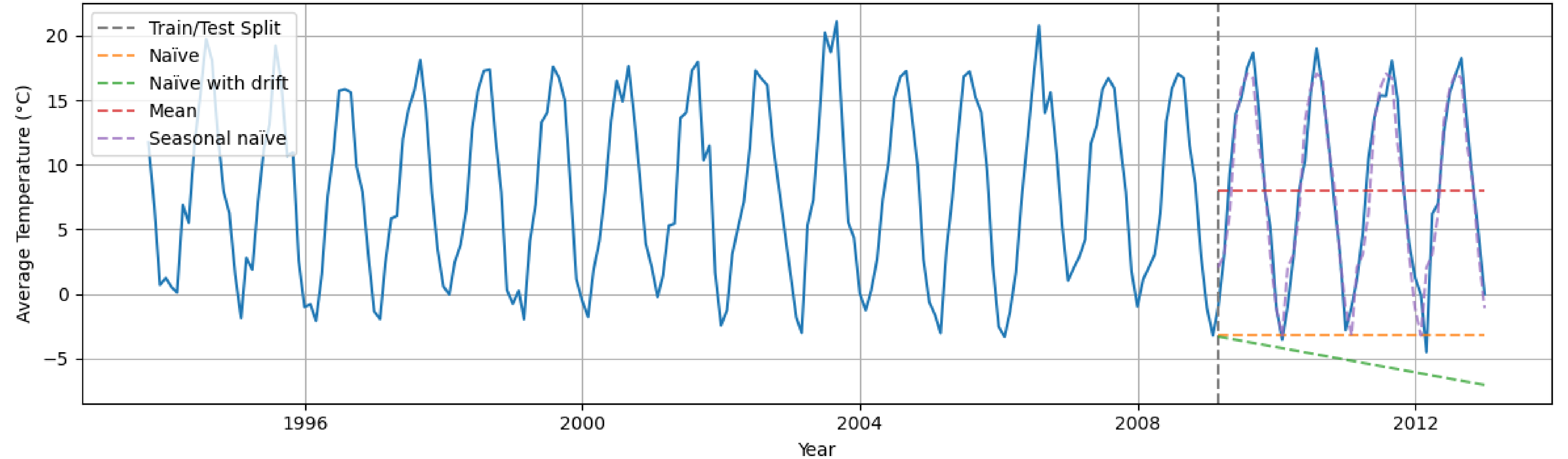
Naïve with drift: naïve forecast with linear drift, $\hat{x}_{T+h|T} = x_T + h \frac{x_T - x_1}{T-1}$

Baselines serve as **benchmark** to evaluate the added value of more complex methods.

Time plot: Weekly Apple Inc. (AAPL) Closing Prices



Time plot: Monthly Average Temperature in Switzerland



Forecasting with ARIMA

The forecast $\hat{x}_{T+h|T}$ from an ARIMA model can be computed as follows:

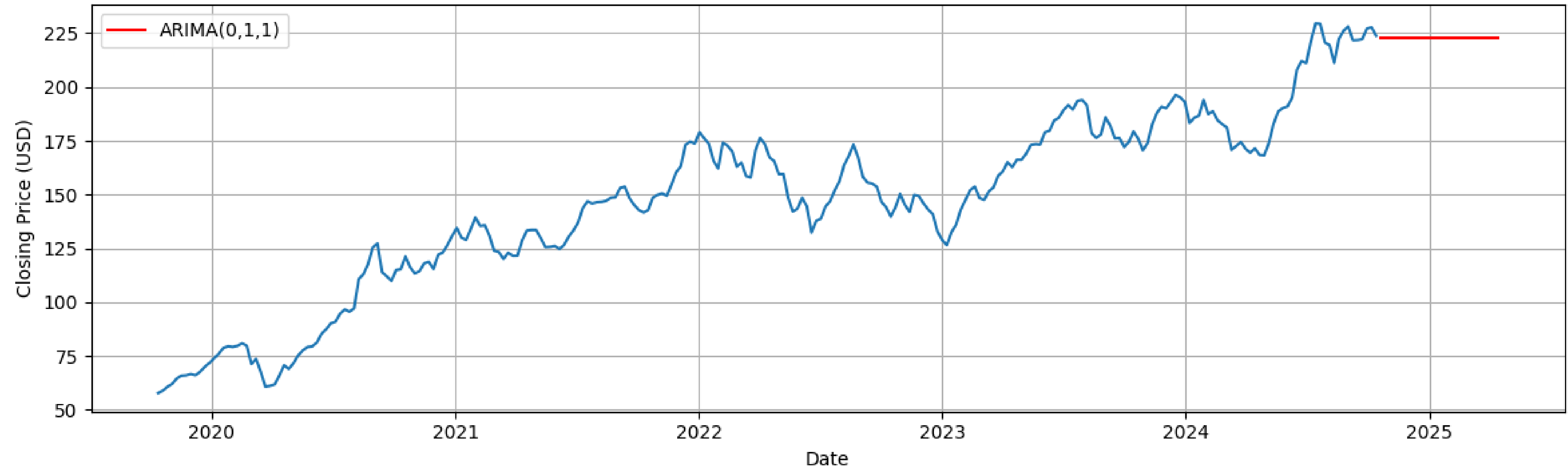
1. Rewrite the ARIMA equation $\Phi(B)\nabla^d X_t = \Phi(B)(1-B)^d X_t = c + \Theta(B)W_t$ with x_t on the left-hand side.

- For an ARIMA(1,1,1): $(1 - \hat{\phi}B)(x_T - x_{T-1}) = \hat{c} + (1 + \hat{\theta}B)w_T \Leftrightarrow x_T = \hat{c} + w_T + \hat{\theta}w_{T-1} + (\hat{\phi} + 1)x_{T-1} - \hat{\phi}x_{T-2}$

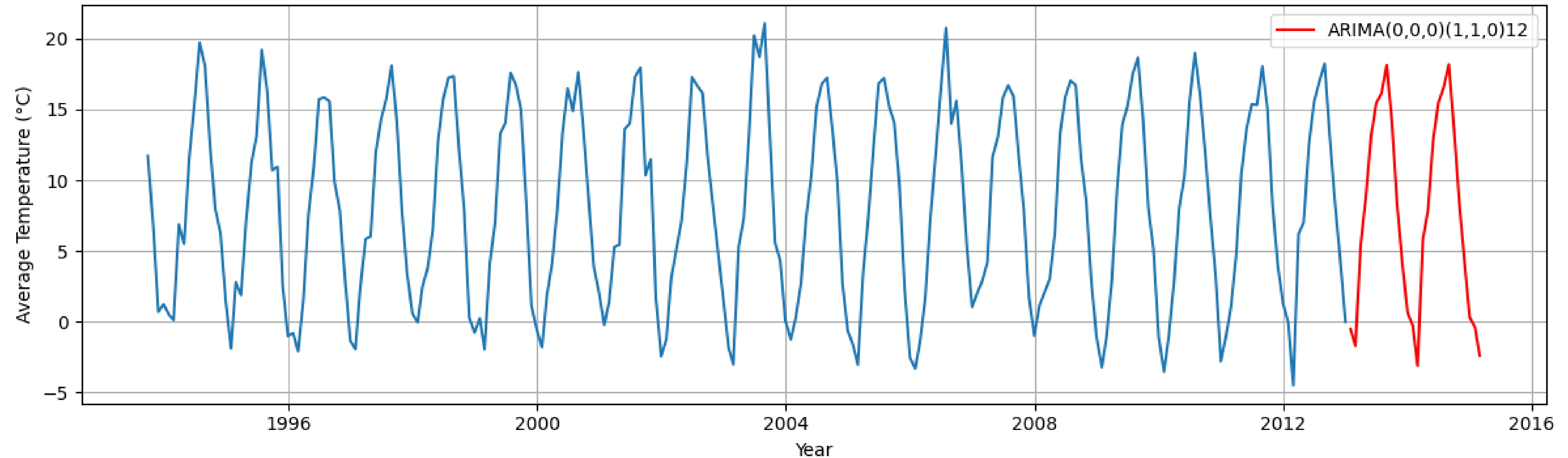
2. Replace future observations with their forecast, future errors with zero, and past errors with the ARIMA residuals.

- $T + h \rightarrow x_{T+h} = \hat{c} + w_{T+h} + \hat{\theta}w_{T+h-1} + (\hat{\phi} + 1)x_{T+h-1} - \hat{\phi}x_{T+h-2}$
- $h = 1 \rightarrow \hat{x}_{T+1|T} = \hat{c} + 0 + \hat{\theta}\hat{w}_T + (\hat{\phi} + 1)x_T - \hat{\phi}x_{T-1}$
- $h = 2 \rightarrow \hat{x}_{T+2|T} = \hat{c} + 0 + 0 + (\hat{\phi} + 1)\hat{x}_{T+1|T} - \hat{\phi}x_T$
- $h = 3 \rightarrow \hat{x}_{T+3|T} = \hat{c} + 0 + 0 + (\hat{\phi} + 1)\hat{x}_{T+2|T} - \hat{\phi}\hat{x}_{T+1|T}$
- ...

Time plot: Forecast of Monthly Average Apple Closing Prices



Time plot: Forecast of Monthly Average Temperature in Switzerland



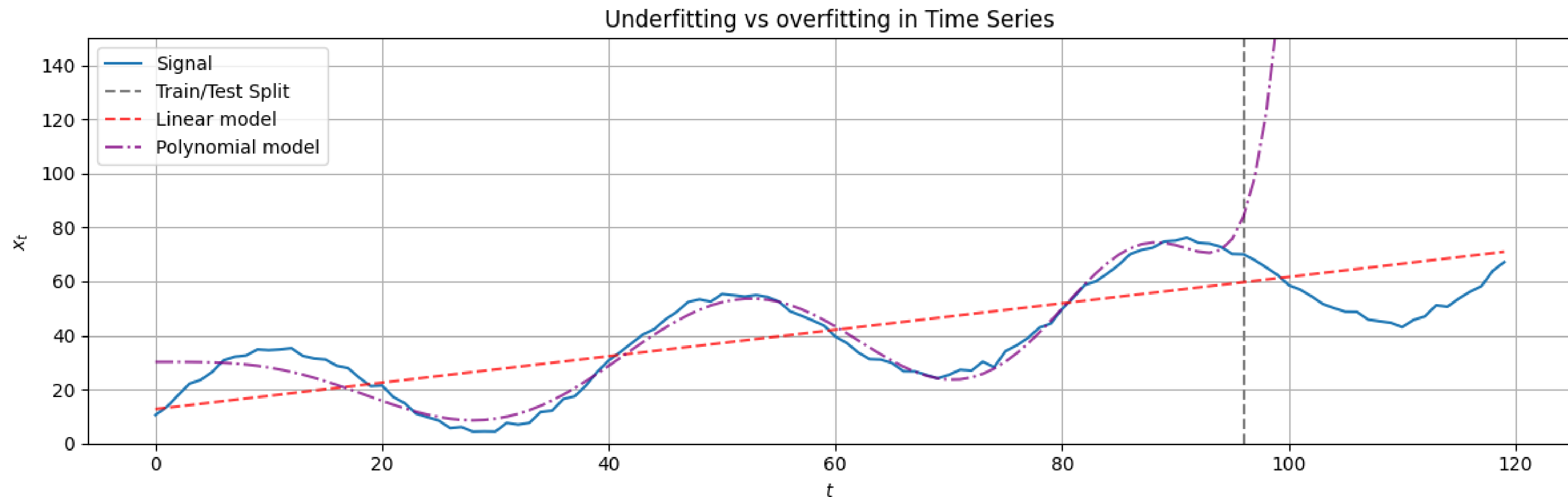
Underfitting and overfitting

Underfitting occurs when a model fits training data poorly and fails to generalize to new data.

- Model is too simple to capture the underlying patterns in the data.

Overfitting occurs when a model fits training data very well but fails to generalize to new data.

- Model is too complex and learns to reproduce the training data exactly.
- Training data is too small.
- Training procedure does not involve regularization (discussed in following lectures).



Evaluation workflow

Given a time series realization $\{x_1, x_2, \dots, x_T, \dots, x_V, \dots, x_n\}$,

1. **Split dataset** into **training** $\{x_1, \dots, x_T\}$, **validation** $\{x_{T+1}, \dots, x_V\}$, and **test** $\{x_{V+1}, \dots, x_n\}$ sets.
 - Validation set is required either when model fitting involves hyperparameters tuning or when model selection is based on performance metrics.
 - Multiple versions of training and validation sets can be considered with cross-validation.
2. **Train** candidate models on training set.
 - Tune hyperparameters using the validation set.
3. **Select model** based on model fit, complexity and performance on the validation set.
 - Information criteria, residual analysis, uncertainty, performance metrics
4. **Train selected model** on training + validation sets then **evaluate performance** on test set.
 - Metrics provide an indication of how well the model will forecast new data.

Splitting time series data & rolling cross-validation

Split dataset into **training** $\{x_1, \dots, x_T\}$, **validation** $\{x_{T+1}, \dots, x_V\}$, and **test** $\{x_{V+1}, \dots, x_n\}$ sets

- Time series must be split **chronologically** → **no random splits**.
- **Seasonality and trends**: ensure the splits account for any patterns in the data.
- Validation and test set should be at least as large as the **forecast horizon** i.e., $h \leq V - T$ and $h \leq n - V$.

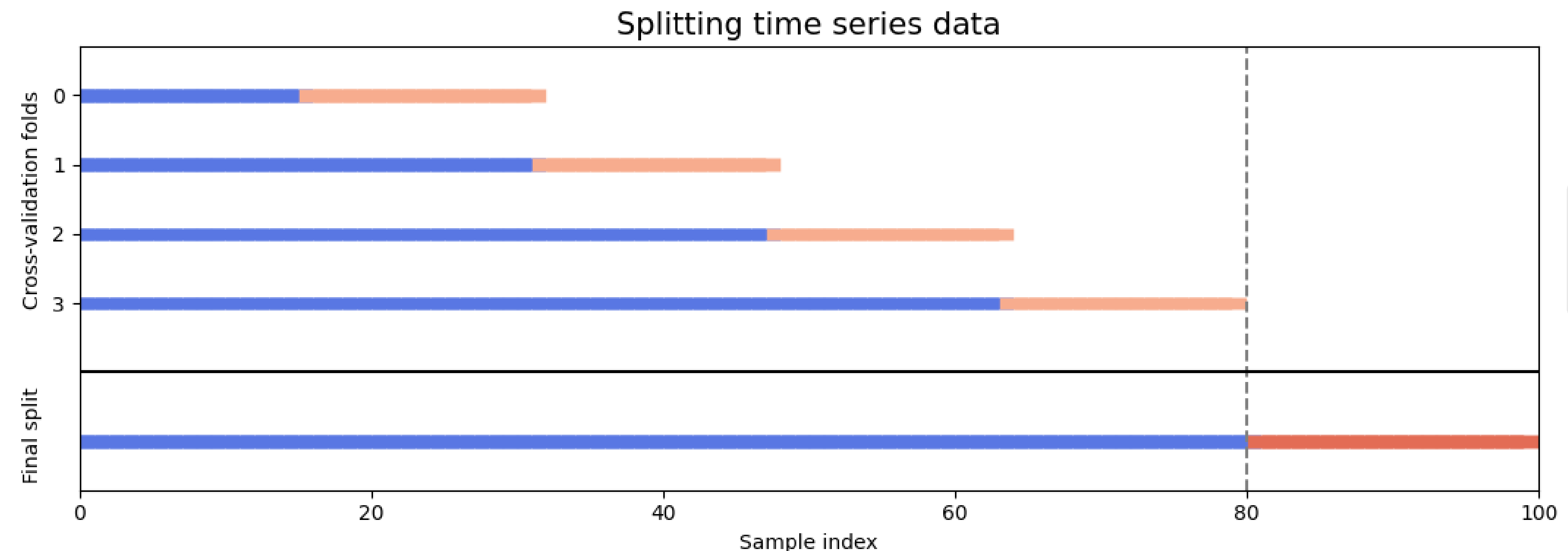
Use test data once for final evaluation, otherwise risk of **over-estimating** performance on new data

→ compare candidate models performance on the validation set.

Rolling Cross-Validation:

sequentially increase the training set, while moving the validation set forward.

HSLU



Residual analysis

Residuals are the difference between observed values and predicted values: $e_i = x_i - \hat{x}_{i|T}$ for $i = 1, \dots, T$.

- Also called **training set errors**, it is an estimate of the noise/innovation component of the data.
- With most methods, residuals are expected to be normal, uncorrelated, zero-mean with constant variance.
- In practice, it is preferred to perform residual analysis on **standardized residuals** $\tilde{e}_i = e_i / \hat{\sigma}_e$

Identify patterns or autocorrelations that the model did not capture.

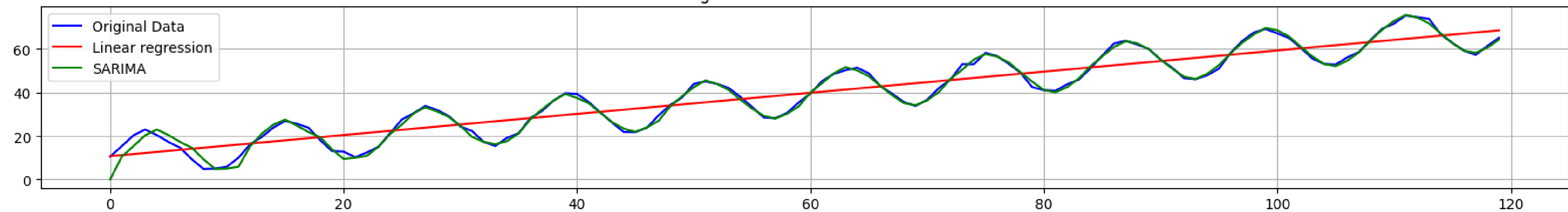
- Time plot, correlogram
- Ljung-Box test: null hypothesis (H_0) states that residuals are **uncorrelated** up to a certain lag.

Validate model assumptions, typically $E_t \sim \mathcal{N}(0, \sigma^2)$.

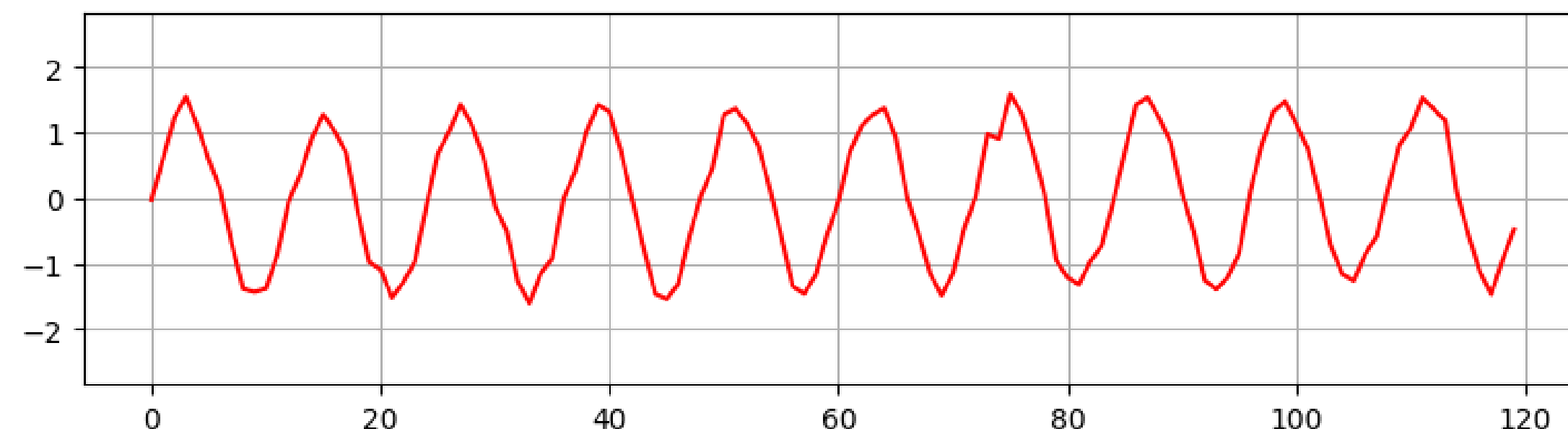
- Q-Q plot: compare the residuals quantiles with normal quantiles.
- Histogram: visual representation of the residuals distribution.

Residual analysis evaluates whether a forecasting method **utilizes all available information** in the data, however, it **does not help with model selection**.

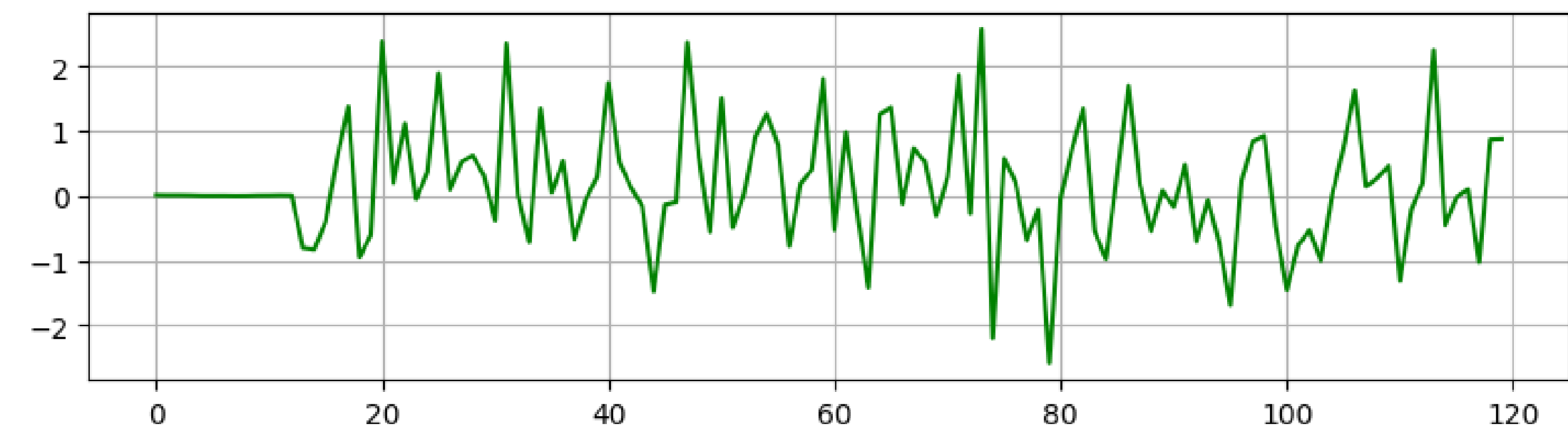
Original Data and Fitted Values



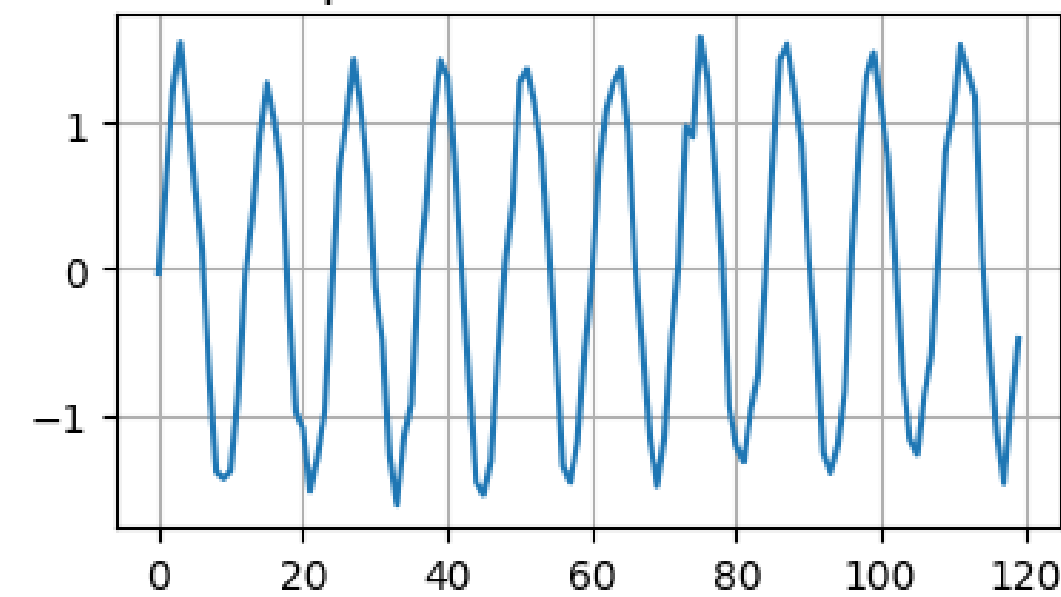
Standardized residuals of the linear model



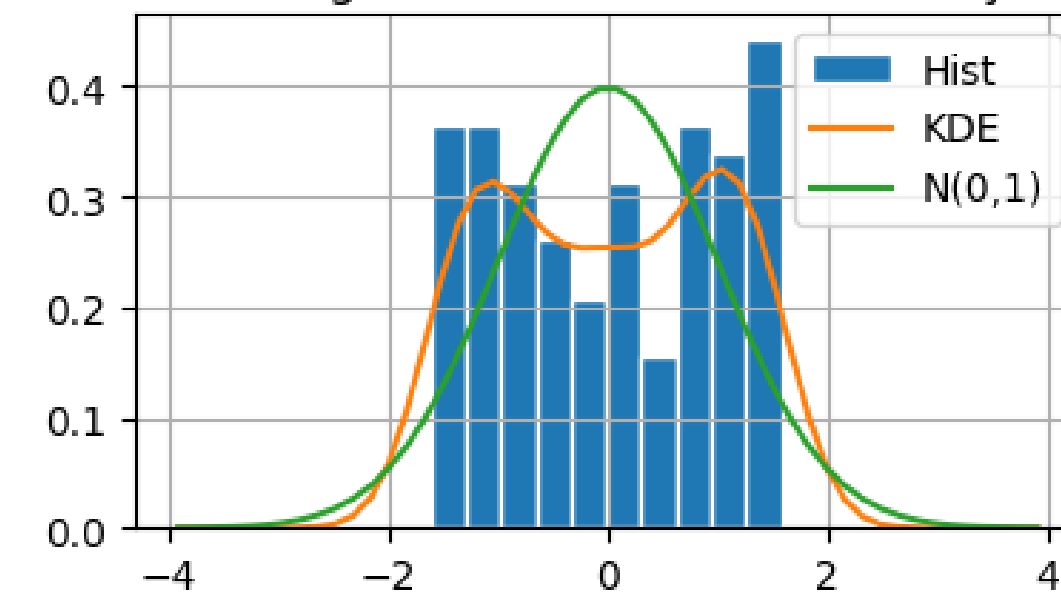
Standardized residuals of the SARIMA model



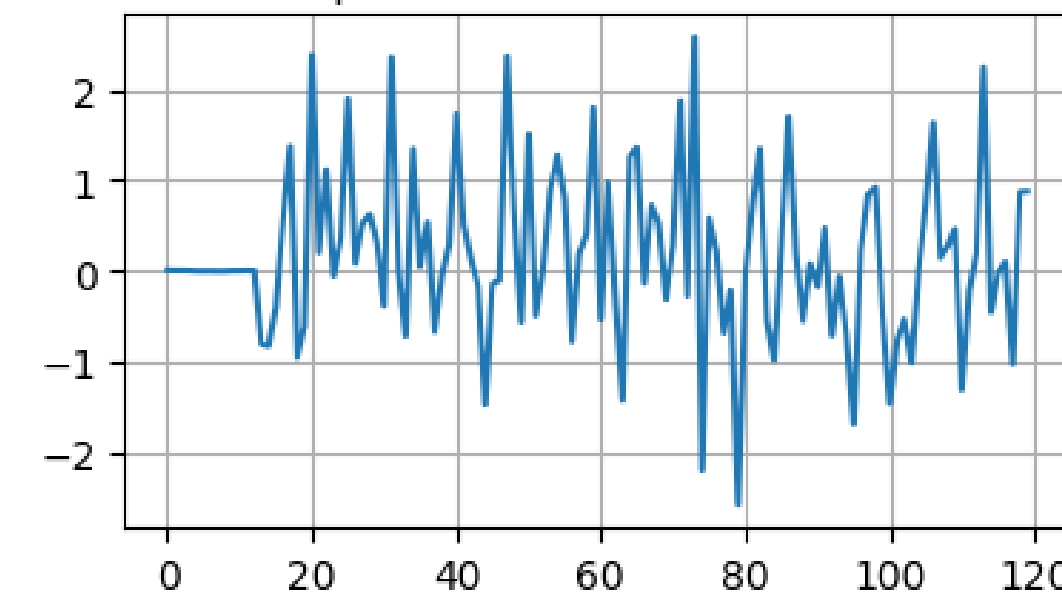
Time plot: Standardized residuals



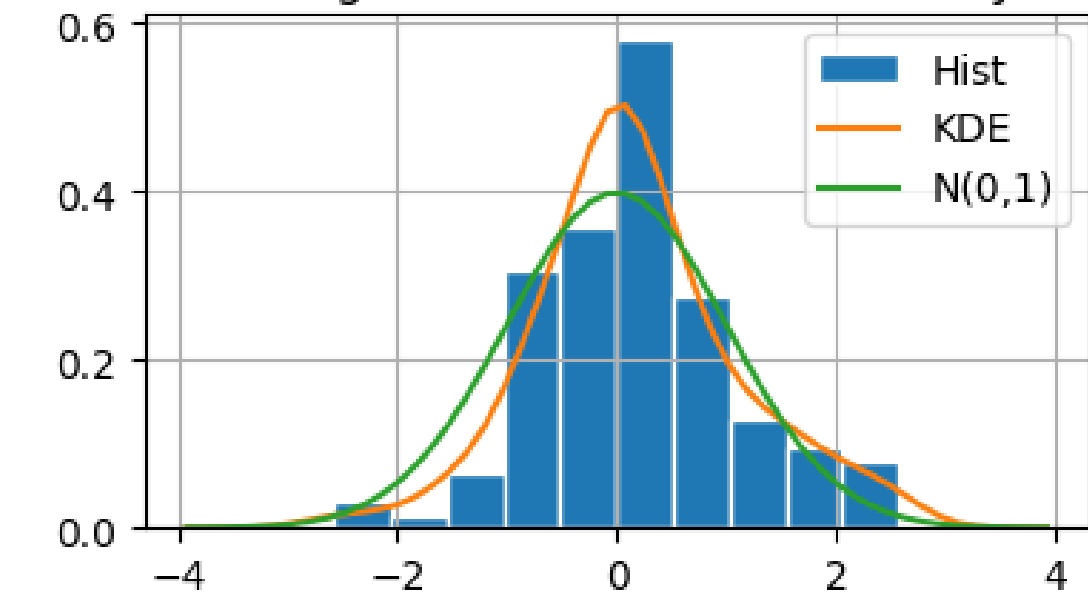
Histogram with estimated density



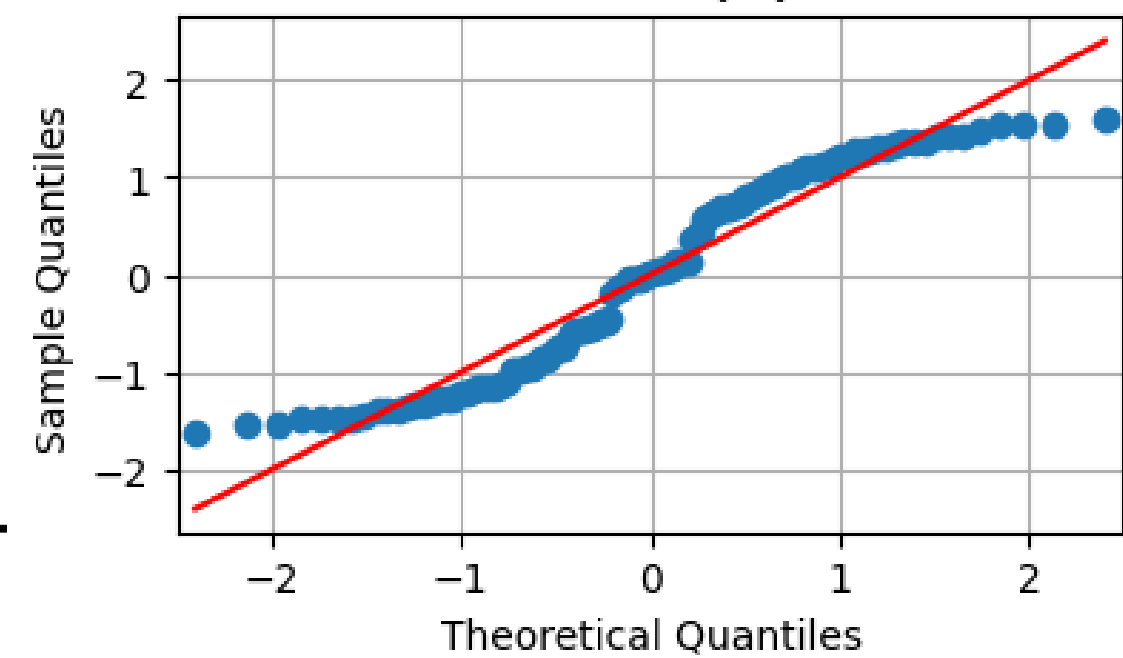
Time plot: Standardized residuals



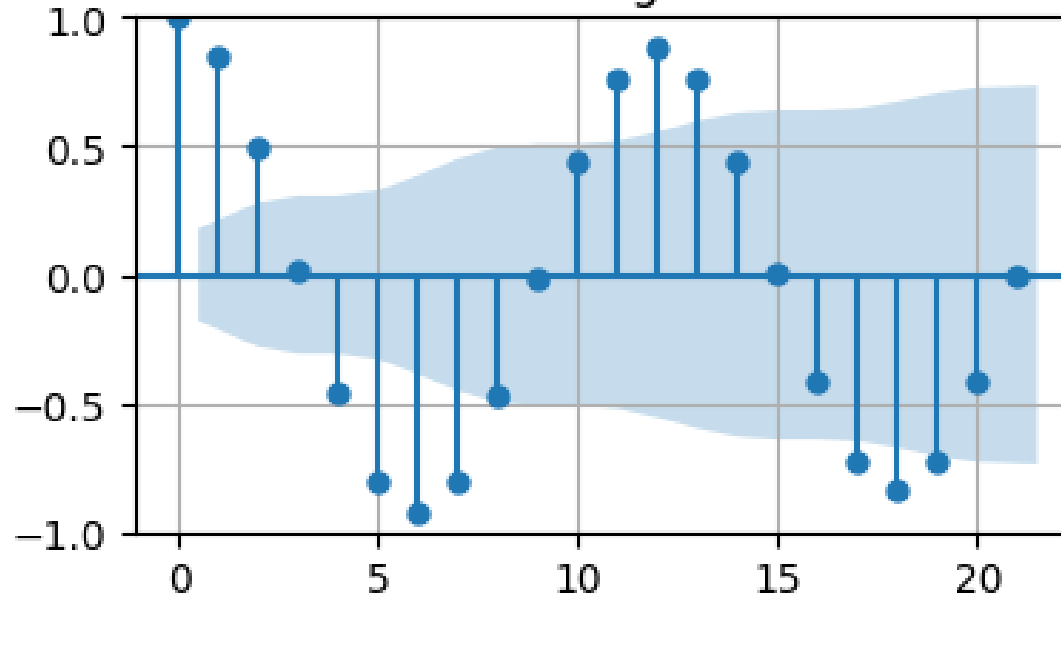
Histogram with estimated density



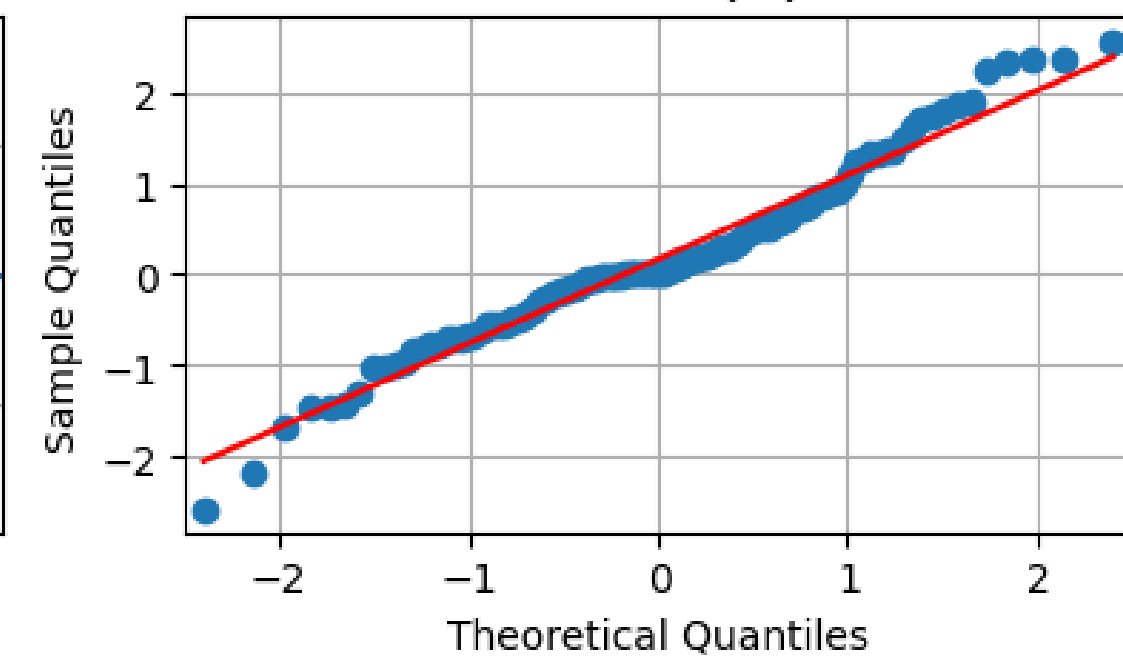
Normal Q-Q



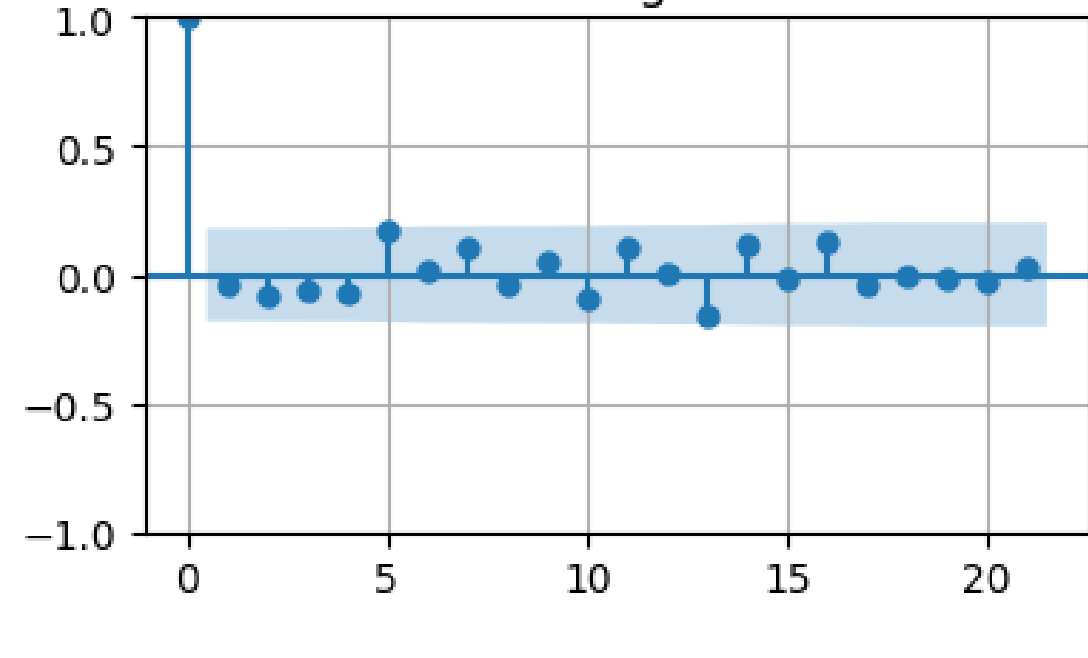
Correlogram



Normal Q-Q



Correlogram



Confidence intervals (CI) for normal residuals

Confidence intervals provide a measure of **forecast uncertainty**.

When the residuals are **normally distributed**, the CI of $\hat{x}_{T+h|T}$ is $\hat{x}_{T+h|T} \pm z_{\alpha/2} \hat{\sigma}_h$

- $z_{\alpha/2}$ is the critical value of the normal distribution e.g., for 95% CI $z_{\alpha/2} = 1.96$
- $\hat{\sigma}_h$ is an **estimate of the standard deviation** of the h -step forecast with $\hat{\sigma}_1 = \hat{\sigma}_e$ (residuals std)

$$\hat{\sigma}_e = \sqrt{\frac{1}{T-k-m} \sum_{i=1}^T e_i^2}$$

with k the number of parameters of the forecasting method and m the number of missing values in the residuals.

For $h > 1$, $\hat{\sigma}_h$ depends on the forecasting method and how the **innovations accumulate**:

Mean	Naïve	Seasonal naïve	Naïve with drift
$\hat{\sigma}_h = \hat{\sigma}_e \sqrt{1 + \frac{1}{T}}$	$\hat{\sigma}_h = \hat{\sigma}_e \sqrt{h}$	$\hat{\sigma}_h = \hat{\sigma}_e \sqrt{\lceil h/p \rceil}$	$\hat{\sigma}_h = \hat{\sigma}_e \sqrt{h \left(1 + \frac{h}{T-1} \right)}$

Confidence intervals (CI) for non-normal residuals

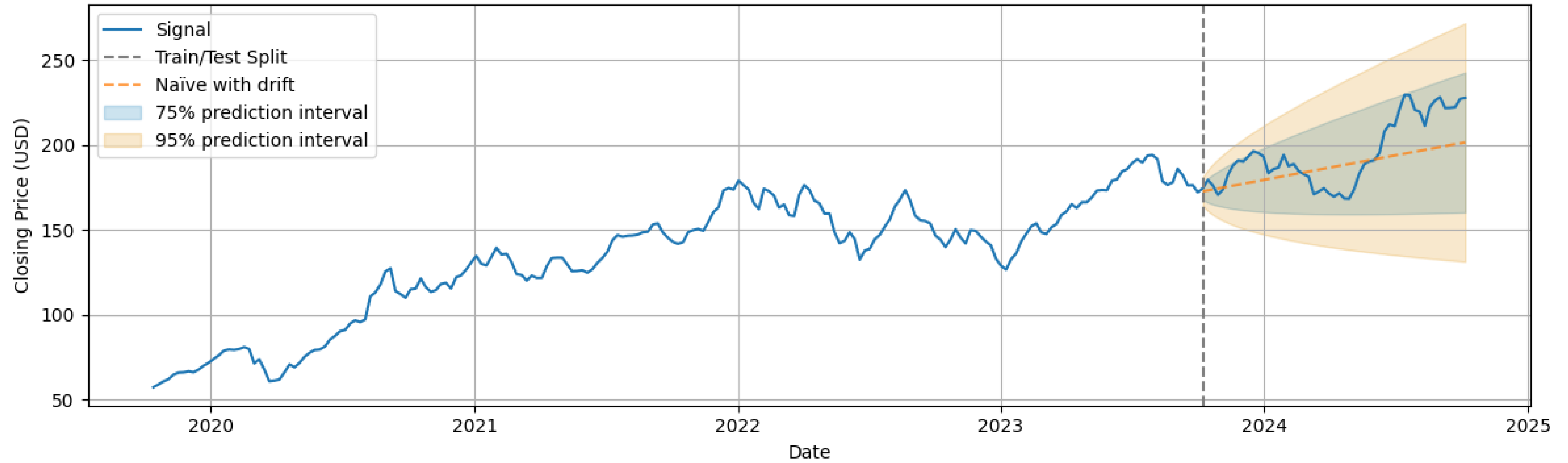
Use **bootstrapping** when the residuals are **uncorrelated** and have **constant variance**.

Assuming future and past errors will be similar, generate possible futures:

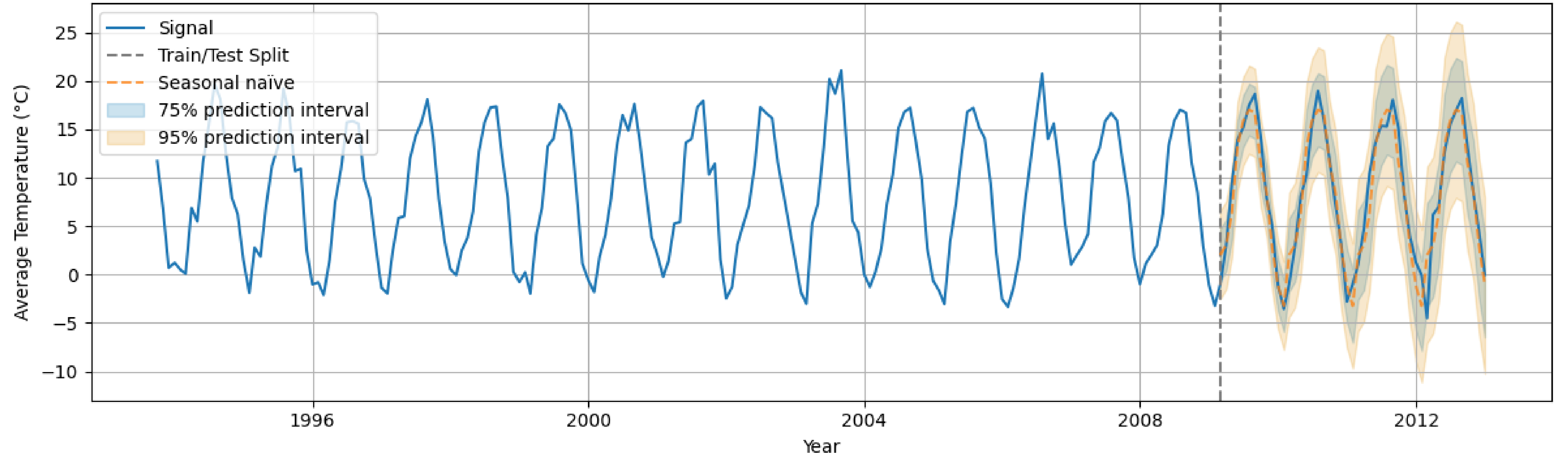
1. Fit forecasting model on $\{x_1, x_2, \dots, x_T\}$ and compute residuals $\{e_1, e_2, \dots, e_T\}$.
2. **Resample residuals with replacement** to simulate future forecast errors $\{e_{k_1}, e_{k_2}, \dots, e_{k_h}\}$.
3. Generate futures by adding resampled residuals to model forecasts $\{\hat{x}_{T+1|T} + e_{k_1}, \hat{x}_{T+2|T} + e_{k_2}, \dots, \hat{x}_{T+h|T} + e_{k_h}\}$.
4. Repeat step 2, 3 multiple times e.g., 1000 iterations.
5. Derive confidence intervals by computing the **percentiles**.
 - e.g., for 95% CI use the 2.5th and 97.5th percentiles.

Note that the bootstrapped confidence intervals are **not symmetric**.

Time plot: Weekly Apple Inc. (AAPL) Closing Prices



Time plot: Monthly Average Temperature in Switzerland



Information criteria

Measure of the **goodness of fit** of a model while penalizing for **model complexity**.

Goodness of fit is measured by the **likelihood** of the data under the model:

$$L = \prod_{i=1}^T \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{e_i^2}{2\sigma^2}\right) \text{ with } e_i = x_i - \hat{x}_{i|T} \text{ and assuming } e_i \sim \mathcal{N}(0, \sigma^2)$$

Akaike's Information Criterion: $AIC = 2k - 2\log(L)$ with k the number of model parameters.

Bayesian Information Criterion: $BIC = \log(T) k - 2\log(L)$

For ARIMA $e_i = \hat{w}_i$ and $k = \begin{cases} p + q & \text{if } c = 0 \\ p + q + 1 & \text{if } c \neq 0 \end{cases}$

Select model **minimizing either AIC or BIC** (for models in the same class)

- BIC tends to favor simpler models than AIC due to a larger penalty term.
- AIC prioritizes **model fit** (potentially better performance), while BIC emphasizes **model simplicity** (faster inference, simpler model interpretation).

Performance metrics

Given a validation set $\{x_{T+1}, \dots, x_V\}$, the forecast performance can be evaluated based on

- **Scale-dependent errors:** $e_i = x_i - \hat{x}_{i|T}$ (same unit as the data $\rightarrow$ not comparable for TS with different units)
- **Scaled errors:** $e_i = (x_i - \hat{x}_{i|T}) / \left(\frac{1}{T} \sum_{j=1}^T |x_j - \hat{x}'_{j|T}| \right)$ with $\hat{x}'_{j|T}$ a baseline **training** forecasts.

Considering the **multi-step** forecasts $\{\hat{x}_{T+1|T}, \dots, \hat{x}_{V|T}\}$, the errors can be aggregated as follows:

- Mean absolute (scaled) error MAE/MASE: $\frac{1}{T-V} \sum_{i=T+1}^V |e_i|$ $\rightarrow$ robust to outliers
- Root mean squared (scaled) error RMSE/RMSSE: $\sqrt{\frac{1}{T-V} \sum_{i=T+1}^V e_i^2}$ $\rightarrow$ sensitive to outliers

When units has a **meaningful zero**, consider the mean absolute **percentage errors** MAPE $\frac{1}{T-V} \sum_{i=T+1}^V \left| \frac{100(x_i - \hat{x}_{i|T})}{x_i} \right|$

Rolling forecast performance

When the validation set encompasses multiple forecast horizons $\{x_{T+1}, \dots, x_h, \dots, x_V\}$,

1. With $i = 0$ for the first iteration, forecast $\{\hat{x}_{T+i+1|T}, \dots, \hat{x}_{T+i+h|T}\}$.
2. Compute performance metrics of the forecast.
3. Increment $i = i + k$ with k the chosen step-size, typically $k = h$.
4. Refit model with the newly available values $\{x_{T+i+1}, \dots, x_{T+i+h}\}$.
 - On the test set choosing between **refit** vs **update** strategy depends on the training objective.
 - Update: recalculate model internal state given new data points **without refitting its parameters**.
5. Repeat until the end of the validation set and then aggregate performance.

Back-transforms

To obtain forecasts on the **original scale**, we need to **reverse transformations** applied to the data.

The back-transform for differencing is

- First-order: $\hat{x}_{T+h|T} = x_T + \sum_{i=1}^h \hat{y}_{T+i|T}$ with $y_t = \nabla x_t$
- Seasonal differencing: $\hat{x}_{T+(kP+n)|T} = x_{T+n} + \sum_{i=1}^k \hat{y}_{T+iP+n|T}$ with $y_t = \nabla_P x_t$

Reversing non-linear transforms does **not** preserve the **mean** from the transformed scale but the **median** (assuming the distribution on the transformed scale is symmetric)

- Considering a log-normal distribution $y_t = \log(x_t)$, the mean of y_t corresponds to the median of x_t .
- **Bias correction** is needed to account for the variance in the transformed space.

Reverse Box-Cox transform	Bias adjusted reverse Box-Cox transform
$\hat{x}_{T+h T} = \begin{cases} \exp(\hat{y}_{T+h T}) & \text{if } \lambda = 0 \\ \text{sign}(\lambda \hat{y}_{T+h T} + 1) \lambda \hat{y}_{T+h T} + 1 ^{\frac{1}{\lambda}} & \text{if } \lambda \neq 0 \end{cases}$	$\hat{x}_{T+h T}^* = \begin{cases} \hat{x}_{T+h T} (1 + \sigma_h^2/2) & \text{if } \lambda = 0 \\ \hat{x}_{T+h T} \left(1 + \frac{\sigma_h^2 (1 - \lambda)}{2 (\lambda \hat{y}_{T+h T} + 1)^2} \right) & \text{if } \lambda \neq 0 \end{cases}$

Exercise

- Review notebook from Time Series Forecasting in Python chapter 11
- Review sktime forecasting notebook.
- Forecasting real-world time series:
 - Split data in training, validation and testing sets
 - Use cross-validation to generate different validation folds
 - Compute baseline rolling forecast performance
 - Compare with the ARIMA model
 - Compare with the Auto-ARIMA model
 - Test different forecasting approaches